Fundamentals of Technological Processes

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Contents

Part I

Micro and Nano Technologies

Chapter 1

Wafer Production and Monocrystals Growth

To create micro-things we need a special technology: semiconductive materials. They're useful because tipically they ca modify electrical and optical properties.

1.1 Monocrystals growth for miniatursed devices technology

. **Semiconductor materials used** include:

- **Silicon**: the most used, even if the 90% of what we'll se is valid as semiconductor.
- **III-V compounds**: *GaAs (gallium arsenide)* (mainly for high frequency applications) and InP (mainly optoelectronic applications).
- SiC ($\rightarrow$ high T & harsh environments applications).

The **aims of monocrystals growth** process are:

- Obtaining **extremely pure silicon** to be able to control doping. $\rightarrow$ To control electrical properties. The usual accepted level of impurities is < 1 over 10^9 atoms of silicon.
- Obtaining **very regular silicon crystals** (defects in the crystallographic building worsen conduction characteristics of the semiconductor). Note: Perfection does not exist, we try to minimize the defect. In quantum technologies we voluntarly introduce defect.
- Obtain a **monocrystalline structure** rather than a **polycrystalline one** (a set of different crystallographic buildings with different orientations $\rightarrow$ polysilicon). In a polycrystalline structure, the carriers mobility is reduced because of discontinuities between different crystals.

The fabrication process moves sequentially from raw materials to finished wafers:

Raw materials	(Sand, coke, coal, wood chips)
Metallurgical grade silicon	(used in making steel)
Electronic grade silicon	(polysilicon)
Single crystal boule	
Single crystal wafers	
Finished wafers	

1.2 Starting Material Purification

The starting material for silicon is relatively pure quartz based sand (SiO_2 or silica).

Silicon Refining

Sand is reduced in an arc furnace with coal, coke, and wood chips at $2000°C$ for ~ 8 days. The reaction is:

$$SiO_2 + 2\ C \rightarrow Si(solid) + 2CO(gas)$$

This forms **metallurgical grade silicon (MGS)**. Its purity is $\sim 98\%$. Major impurities (metals) include: Fe 0.8%, Al 0.3%, Cr 0.04%, Ti 0.03%, Mn & V 0.02%. **MGS** is used in Steel, Semiconductor, Silicones, and Aluminium alloys.

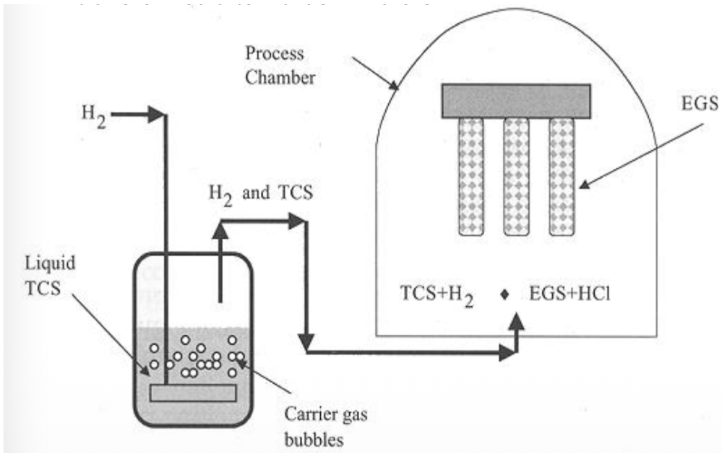


Figure 1.1: Siemens process

1.2.1 Trichlorosilane (TCS) Production

MGS is crushed and reacted with HCl to obtain trichlorosilane (TCS). The reaction is:

$$3HCl(gas) + Si(solid) \rightarrow HSiCl_3(gas) + H_2(gas)$$

This occurs at $325°C$. The reaction is exothermic and must be cooled to keep at $325°C$ to minimize byproduct formation.

TCS Purification

$HSiCl_3$ is a liquid at room temperature (boiling point at $32°C$). B and P impurities are present in $HSiCl_3$. Impurities are eliminated through **fractioned distillation** of the liquid.

1.2.2 Fractioned distillation

Fractioned distillation is the separation of a mixture into its component parts, or fractions. Chemical compounds are separated by heating them to a temperature at which one or more fractions of the mixture will vaporize. It uses distillation to fractionate.

In the distillation process, the crude oil mixture is heated in a boiler into a super-hot mixture of liquid and vapour called the feed. The mixture is fed into a distillation tower. Compounds with a lower boiling point rise up as vapours, while compounds with a higher boiling point fall downwards as liquids. The tower contains trays that allow the vapour to bubble upward through the liquid, helping to exchange heat and resulting in more effective separation. The distilled products (fractions or distillates) are piped off from the different levels. This may take place along multiple distillation towers.

Petrochemical outputs resulting from distillation can include Propane & Butane, Petrol (Gasoline), Kerosene / Jet Fuel, Diesel Fuel, Heavy Fuel Oil, Waxes & Lubricants, and Bitumen (Asphalt). Light Ends have a Boiling point $< 0°C$ and Residuals have a Boiling point $> 500°C$.

After purification, TCS is 99.9999999% pure.

1.3 Electronic-Grade Silicon (EGS) Production

Purified TCS is reduced with H_2 to obtain **electronic-grade silicon (EGS)**. The **Siemens process**(Figure 1.1) is used:

$$HSiCl_3(gas) + H_2(gas) \rightarrow 3HCl(gas) + Si(solid)$$

This reaction (Chemical Vapor Deposition, **CVD**) occurs in a cold chamber of a reactor containing an heated Si rod which acts as a deposition seed at $1100°C$.

After a few deposition days, the result is an **extremely pure** (~ 1 impurities over 10^9 atoms) **polycrystalline silicon**, which will be the raw material for the production of the monocrystalline one with the Czochralski (CZ) or with the Float-Zone (FZ) process.

The fabrication of slices from pure silicon consists first of fabricating a crystal, precisely an **ingot**, which is then cut into slices and transformed into wafers.

A comparison of impurity levels (ppm) between MGS and EGS is provided:

Impurity	MGS (ppm)	EGS (ppm)
A1	1570	-
B	44	<0,001
Fe	2070	4
P	28	<0,002
Sb	-	0,001
Au	-	0,00007

1.4 Czochralski Process (CZ)

The **Czochralski Process** was discovered, more than invented, in 1916. **Process Conditions**:

- Hyper-pure Si (EGS Si) is placed in a quartz crucible.
- Temperature: $\geq 1415°C$ (Si m.p.).
- Pressure: 20 Torr (Ar or He).
- Ar Flow Rate: 20 to 50 $liters/min$.
- Time: 18 to 24 hours (for initial setup/melting).
- Typical growth speed: some mm/min.
- A monocrystalline Si seed is introduced.

The seed is rotated about its axis to mix the melt and produce a circular cross-section crystal. The rotation inhibits the natural tendency of the crystal to grow along certain orientations to produce a faceted crystal. The pulling time takes 3 days.

1.4.1 Ingot Characteristics and Parts

- Length: Up to 2 meters.
- Diameter: Up to 300mm (12").
- Weight: Up to 250 kg.

The parts of the ingot definition are:

- **Seed**: determines crystal orientation.
- **Neck**: removes the dislocations (like Edge Dislocation or "extra" plane of atoms) generated by thermal shock when the seed is dipped into the melt.
- **Shoulder or taper**: transitions to required diameter.
- **Body**: Part of the crystal actually used for wafers.
- **Endcone or bottom**: prevents shockback into the body when the crystal is removed from the melt.

1.4.2 Doping from the Melt

Dopants are added to the melt to pull controlled doped crystals. Dopants are typically added in the form of doped polysilicon for concentration control.

The concentration of dopant in the crystal (C_S) is generally not the same as that in the melt (C_L). This difference is governed by the **segregation coefficient** (k):

$$k = \frac{C_S}{C_L}$$

Where C_S = Concentration of dopant in solid (wt dopant/wt solid) and C_L = Concentration of dopant in melt (wt dopant/wt melt).

Since k in general is less than 1, the dopant becomes increasingly more concentrated in the melt. Dopant concentration thus changes along the length of the crystal.

Dopant Segregation Coefficients (k)

Dopant	P	As	B
k	0.35	0.3	0.8
atomic weight	30.97	74.92	10.81

Since k in general is less than 1, the dopant becomes increasingly more concentrated in the melt.
Boron produces a relatively flat profile because its k is close to 1. Dopants with $k \ll 1$ (like Antimony, $k = 0.023$) produce much more doping variation along the crystal.

1.4.3 III-V Monocrystals Growth: Liquid Encapsulated CZ (LEC)

The **Liquid Encapsulated CZ (LEC)** is a CZ technique modification used for the growth of $GaAs$. $GaAs$ has a low boiling point, and evaporation could result in toxic fumes and non-uniform crystal growth. Furthermore, the components Ga and As have differing vapor pressures, which complicates a stoichiometric precipitation over a large crystal volume.
Solution: The solution is to pressurize the chamber and melt B_2O_3 (about 1 cm thick) on top to seal and suppress evaporation.

- B_2O_3 does not react with GaAs during growth.
- Ga melts at $30°C$, B_2O_3 melts at $500°C$ and seals the crucible, while Ga and As start reacting at $800°C$ to produce GaAs.
- A high pressure As atmosphere is maintained in the reactor to avoid As evaporation (a very volatile element).
- Process pressure is approximately 200 $atm\ Ar + AsH_3$.
- Pyrolitic BN crucibles are used, which do not react with $GaAs$.

After crystalline seed introduction and monocrystal growth, slow cooling down ($30 - 80°C/h$) is performed. The melt GaAs required temperature is $T_F\ GaAs = 1238°C$.

1.5 From Boule to Wafers

The process steps after monocrystal growth are:

1.5.1 Ingot cropping

This consists of cutting, or cropping, the extremities of the ingot, which are regions with high defect concentration and variable diameter.

1.5.2 Ingot inspection

This involves:

- Control of the size of the ingot: if undersized, is trashed.
- Control of the resistivities on the top and bottom faces of the ingot: due to the variation of the doping concentration during the pulling, the final resistivity varies as a function of location. A check of resistivity and an agreement with specifications are needed. A four probe method equipment makes these measurements.
- Check of the crystallographic orientation of the ingot (XRD).

Approximately 50% of the material can be rejected at this stage.

1.5.3 Shaping of ingot (Outer Diameter Grinding)

Make it round and of the correct diameter: during the pulling, due to the very large set of physical parameters to control, the diameter of the ingot slightly varies. That creates some waves at the ingot surface. To get slices of calibrated diameter suitable for automatic equipment, a cylindrical polishing is needed.

1.5.4 Flat(s) grinding

One or two flat zones on the edge of the ingot are processed to get a crystallographic orientation reference for the wafer fabrication. This reference will be used during the wafer process (orientation of conducting zones, crystallographic axes for the die cutting...). Orientation flat grinding references: 45° for (111) type n, 180° for (111) type p, 190° for (100) type n, and a primary flat for (100) type p.

1.5.5 Ingot sawing (Wafer Slicing)

This sawing is proceeded with a diamond tooth saw. The creation of a thickness of powder equivalent to the saw thickness leads to loss. Typical thickness range is 300 $\mu m - 1\ mm$.

The saw blade itself is about 400 μm thick. Together with the loss at the seed and tail end of the crystal, only 50% of the ingot ends up in wafer form. The dominant state of the art slicing technology is **Multi-Wire Sawing (MWS)**. In MWS, a thin wire is arranged over cylindrical spools so that hundreds of parallel wire segments simultaneously travel through the ingot. The sawing effect is achieved by SiC or other grinding agents that run along the rotating wire.

1.5.6 Edge grinding

After sawing, "peaks of matter" remain on the peripheral zones of the slices, which must be removed. A circle shaped edge is created to make manipulation easier during IC fabrication, avoiding degradation of transfer equipment and preventing cracks or dislocations that lead to definitive breaking. This step can improve cleaning results and reduce breakage up to 400%.

1.5.7 Wafer lapping and grinding

The thicknesses and section shapes after sawing can be significantly different. To decrease the cost, the quantity of matter to grind has to be minimized. Usually, the wafers are sorted by thickness range of ten micrometers (10 μm). In order to improve their surface quality, the slices are polished using a mixture that contains alumina or diamond grains for which the size is about several micrometers (final roughness < 2 μm).

1.5.8 Wafer polishing

This polishing is **chemico-mechanical based**. The equipment is similar to the lapper, but the solution is less aggressive, containing smaller grains of alumina or diamond (grain diameters can be as low as 0.1 μm) and acid or basic chemical agents. Chemical agents used can include $NaClO$, $Br_2 + CH_3OH$, and $NaOH + SiO_2$ gel.

Mirror finishing is necessary for lithographic steps. The polishing process creates a very smooth surface with a roughness of 1 nm or better. Polishing results in regular reflection from a smooth surface, unlike diffuse reflection from a rough surface. Wafers can be **SSP** (Single side Polished) or **DSP** (Double side Polished).

1.5.9 Wafer cleaning

This step removes abrasive species and contaminants by ultra pure deionized water rinsing.

1.5.10 Laser marking for ID

The writing of the lots mentioning ingot number, date, etc., is performed through a laser beam scanning. These indications allow controlling each wafer during the fabrication process of circuits and devices. The identification code includes VENDOR IDENTIFICATION, RESISTIVITY IDENTIFICATION, DOPANT SPECIES, CRYSTAL GROWTH ORIENTATION, and CHECK CHARACTERS.

1.6 Wafer Evolution and Advantages of Larger Diameters

1.6.1 Wafer Evolution

The evolution of standard wafer diameters is:

- 1964: 25 mm.
- 1969: 50 mm.
- 1974: 75 mm.
- 1978: 100 mm.
- 1982: 125 mm.
- 1985: 150 mm.
- 1990: 200 mm. (Current MEMS standard).
- 1998: 300 mm. (Current ICs standard).
- 450 mm (18") diameter wafer was expected in production by 2018-20 through an alliance between Intel, Samsung, and TSMC, but was abandoned.

Note that 1 inch $= 2,54$ cm, and 1 $mil = 0,001$ $in = 25,4$ μm.

1.6.2 Advantages of Larger Diameter Wafers

Larger Diameter Wafer Advantages are summarized below, comparing surface area relative to a 50mm wafer, considering a 3mm exclusion zone (black ring):

	100mm	150mm	200mm
Gross surface area advantage over 50mm	4X	9X	16X
Surface area % lost to 3mm exclusion zone	22%	12%	8%
Net surface area advantage over 50mm	4.4X	10.3X	18.8X

Further comparison of surface area and chip count for various configurations is available:

	56x50mm (2")	8x150mm (6")	5x200mm (8")
Net surface area	88,274 mm^2	130,288 mm^2	$147,796\ mm^2$
Surface area gain vs. 50mm		48%	67%
Count of 45x45mil LED chips	62,944	97,600	111,280
LED chip count gain vs. 50mm		55%	77%

Chapter 2

Crystallography and crystalline defects

2.1 3 types of solids

- **Crystalline**: All atoms arranged on a common lattice.
- **Polycrystalline**: Different lattice orientation for each grain.
- **Amorphous**: Atoms are disordered (no lattice).

2.1.1 Microstructure of electronic materials

Examples of microstructure components include:

- Polycrystalline materials: Lines show lattice orientation, Grain boundary.
- Amorphous materials: SiO_2, nitride spacer, silicide, gate oxide.
- Single-Crystal Material: substrate Si.
- Features size example: 50nm.

2.2 Crystal structure

Atoms are arranged on a regular grid called a **lattice**. A Bravais lattice is an infinite array of points that appears the same from any lattice point, and there are **only 14 unique lattices possible in three dimensions**.(Figure 1.2)

These lattices correspond to 7 Crystal families/Lattice systems: triclinic, monoclinic ($\beta \neq 90°$), orthorhombic ($a \neq b \neq c$), tetragonal, rhombohedral, hexagonal ($\gamma = 120°$), and cubic ($a = b = c$, $\alpha = \beta = \gamma = 90°$).

2.2.1 Cubic Crystal Structure

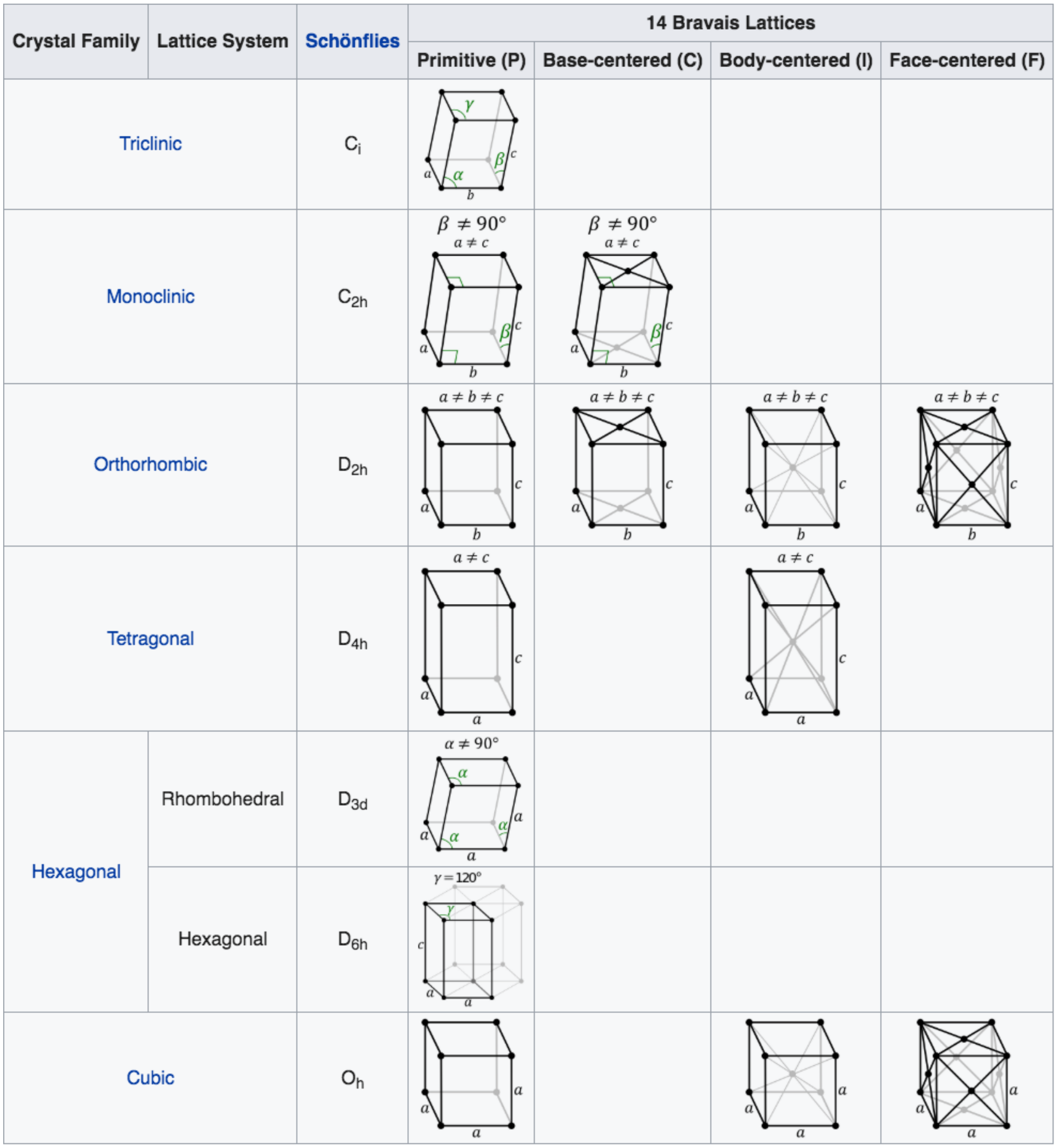
- **Simple Cubic** (only Polonium).
- **Body Centered Cubic (bcc)** (Na, W, Cr, Fe, K, Mo ...).
- **Face Centered Cubic (fcc)** (Al, Au, Cu, Pt, ...).

The **lattice parameter** ($= a_0$) is the length of a cube side.

2.2.2 Silicon lattice (Diamond Structure)

The crystalline structure of Si and Ge (also Sn) belongs to the family of cubic crystals.
It is formed by the **interpenetration of two fcc** sub-lattices where one sub-lattice is moved away from the other along the diagonal of the cube by an amount corresponding to $\frac{1}{4}$ of its length (a shift $= a_0\frac{\sqrt{3}}{4}$).
In diamond structure:

- All atoms are equal.
- Every atom is surrounded by 4 atoms placed at the same distance at the corners of a tetrahedron.(Figure 1.3)



Crystal Family	Lattice System	Schönflies	14 Bravais Lattices			
			Primitive (P)	Base-centered (C)	Body-centered (I)	Face-centered (F)
Triclinic		C_i				
Monoclinic		C_{2h}	$\beta \neq 90°$, $a \neq c$	$\beta \neq 90°$, $a \neq c$		
Orthorhombic		D_{2h}	$a \neq b \neq c$	$a \neq b \neq c$	$a \neq b \neq c$	$a \neq b \neq c$
Tetragonal		D_{4h}	$a \neq c$		$a \neq c$	
Hexagonal	Rhombohedral	D_{3d}	$\alpha \neq 90°$			
	Hexagonal	D_{6h}	$\gamma = 120°$			
Cubic		O_h				

Figure 2.1: 14 Bravais Lattices

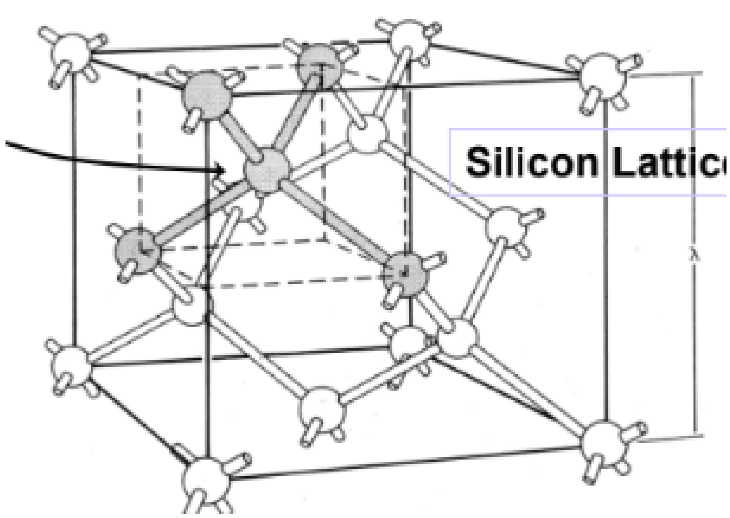


Figure 2.2: Silicon lattice. Gray spheres denote the corners of a tetrahedron

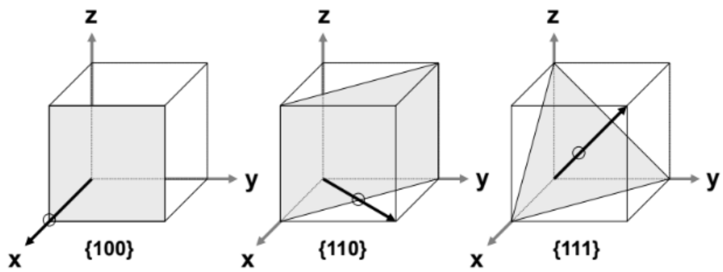


Figure 2.3: Common cubic crystal planes illustrated are {100}, {110}, and {111}.

Zincoblend Structure

This structure is typical of **III-V compounds** (GaAs, InP, ...).
It is **similar to diamond**, but one fcc lattice is made by III atoms and the other by V atoms.

2.3 Miller indices

Miller indices are used to describe surfaces (or planes) cut in crystals. Different cuts result in different physical/chemical properties.

General Procedure

1. **Cut** crystal along some plane.
2. **Determine x, y, z, intercepts** (x_o, y_o, z_o). If a plane does not intersect an axis, the intercept is infinity.
3. **Take reciprocals** $(\frac{1}{x_o}, \frac{1}{y_o}, \frac{1}{z_o})$ and **reduce to smallest integer**.

A bar is placed over negative intercepts.

2.3.1 Anisotropic Wet Etching and Surface Properties

Different surfaces have different arrangements of atoms, leading to different properties (mechanical, chemical, electrical, ...).

- **(100) Surface Orientation (for Silicon)**:
  - Gives a lesser number of dangling bonds on the surface.
  - Results in less impurity accumulation, leading to better control.
  - This characteristic is **required for a MOSFET**.
- **(111) Surface Orientation (for Silicon)**:
  - Gives an advantage of **fast oxidation** as more atoms are available on the surface to react with oxygen.
  - This is a characteristic **desired for BJTs**.

The angle between the (100) and (111) surfaces in silicon is 54.74°. (Figure 1.4)

2.4 Crystalline defects

A real crystal is different from ideality due to:

- **Finite dimensions** (surface atoms present "dangling bonds").
- **Presence of defects**.

Defects have a **strong influence** on electrical, chemical, mechanical, and optical properties of the semiconductor.

2.4.1 Defects Classification

- Point defects.
- Line defects.
- Surface defects.
- Volume defects.

2.4.2 Point Defects

Point defects often arise from too fast deposition or low substrate temperatures, leaving no time for atoms to move to crystal lattice sites.

- **Vacancy** (missing atom).
- **Self interstitial** (extra atom).
- **Substitutional impurity** (impurity atom in lattice).
- **Interstitial impurity** (impurity atom not in regular lattice site).
- **Antisite**.

These defects cause distortion of planes.

2.4.3 Line Defects: Edge Dislocation

An edge dislocation is the **insertion of an extra half-plane of atoms** which distorts the lattice. This defect is called a line defect because the locus of defective points produced in the lattice lies along a line, which runs along the edge of the extra half-plane. Inter-atomic bonds are significantly distorted only in the immediate vicinity of the dislocation line.

- The dislocation creates stresses (compression and tension).
- Typical densities are often $\mathbf{10^{10} - 10^{12}}$ dislocations/cm^2 in films.
- Dislocations form from the film growth process, dislocations in the substrate continuing into the film, or contamination on the substrate.

2.4.4 Surface Defects

Grain Boundaries

are interfaces between two single crystal regions of different orientation. Atoms at grain boundaries tend to be **loosely bound**. That implies that those boundaries are **more reactive** (leading to corrosion/etching/oxidation) and allow **accelerated diffusion** along boundaries.
The number of grain boundaries in a film (grain size) depends on deposition rate and substrate temperature. Generally, lower T yields smaller grains and many boundaries, while higher T yields larger grains and fewer boundaries.
Grain size is often proportional to film thickness (thinner films tend to have smaller grains).

Gemination

is an abrupt variation of crystal orientation.

2.4.5 Volume Defects

Impurities have an intrinsic **solid solubility** in the host crystal, which is variable with temperature. Usually, solid solubility decreases with decreasing temperature.
If impurities are introduced at maximum concentration for a certain temperature (T) and then T is lowered, the excess impurities (respect to the solid solubility at that temperature) can **precipitate in local concentration**. This local concentration change can drastically change the volume of the crystal, leading to volume dislocations.

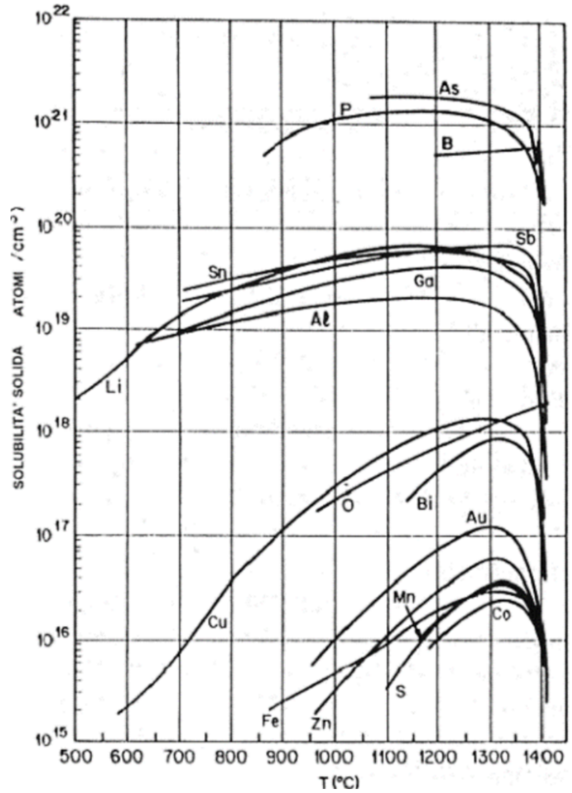


Figure 2.4: A graph detailing Solid Solubility (ATOMI/cm^3) versus Temperature ($T(^\circ C)$) shows solubility profiles for various elements (As, P, B, Sb, Sn, Ga, Al, Li, Bi, Au, Mn, Co, Cu, S, Fe, Zn).

Chapter 3

Cleanroom Technology

3.1 Why Cleanroom Technology is Important

A cleanroom is defined as **an environment where particulate concentration is reduced and kept under a specified level**. Typically, temperature and humidity are also controlled. Cleanroom Technology is crucial because contamination can cause several significant problems:

- **Ruin devices:** A single ruined device in a complex circuit can cause the whole chip to fail. This leads to a lower yield of good chips per wafer, resulting in higher costs and lower profits per chip.
- **"Poison" equipment:** Equipment must be removed from the manufacturing line, which reduces production throughput and revenue.
- **Pose a health risk:** Contamination can endanger employees, customers, and the environment, greatly increasing costs, and potentially leading to litigation, insurance issues, and compliance with safety issues.

3.2 Sources of Contamination

Humans cause most of the contamination. People continuously exfoliate skin and replace hair. Sources associated with people include dirt and oils tracked into labs on shoe soles, widespread use of make-up, perfume, and hair gels.

Particle shedding rates from people are substantial:

- Sitting quietly: $100,000$ particles shed per minute.
- Moving: 1 million particles shed per minute.
- Walking: 5 million particles shed per minute.

Other sources of contamination include:

- Ventilation and Room Structure.
- Equipment and Machines.
- Abrasion during automated wafer handling.
- Mechanical mechanism wear and lubrication.
- Aging plastic and rubber parts (valves, pumps, filters, etc.).
- Materials (chemicals, water, gas, targets).
- Processes (residues, depositions, slurry).

3.2.1 Examples of Common Air Contaminants and Size (in microns)

Contaminant	**Size (μm)**
Human Hair	70-100
Human Sneeze	10-100
Pollen	10-100
Pet Dander	5-100
Spores from Plants	6-100
Household Dust	0.05-100
Dust Mite Debris	0.5-50
Mold	2-20
Skin Flakes	0.4-10
Bacteria	0.35-10
Smoke	0.01-1

3.3 Contamination Induced Problems

Particles are defined as undesired objects (size between ~ 0.06 μm and a few microns) whose composition depends on the contamination source. Particles adhere to surfaces by van der Waals forces, condensation of vapour, or the formation of chemical bonds.

Contamination leads to various issues in semiconductor manufacturing:

- **Mobile ions in oxides** (e.g., Gate (+5V), Source (OV), Drain (OV)) can change electric fields and voltages at the semiconductor/dielectric interface.
- **Unintentional films between layers** can create open circuits or short circuits between layers. They can also impede adhesion between films.
- **Effect on photolithography:** Dust particles can create defects (cuts or protrusions) through shadowing.
- **Effect on movable microstructures:** Contamination can cause jamming.

3.4 Classification Standards and Measurement

The 1966 publication of Fed. Std. 209 defined a standard for clean rooms divided by classes. A clean room belongs to a certain class if its distribution of concentration/dimension of particles is under the line corresponding to the class. For example, **Class 100** must have less than 100 particles greater than 0.5 μm per cubic foot. The Normal environment value is $10,000$ particles per cubic foot.

Particulate concentration can be directly measured with methods of filtering and counting (usually too slow) or with a suitable apparatus exploiting **optical scattering of a laser beam** (just like looking at particulate through a sun beam).

The ISO standard (ISO 14644-1) requires results to be shown in cubic meters (1 cubic meter = 35.2 cubic feet).

3.4.1 Classification Details (Fed. Std. 209 E)

The classification table provides maximum particles per cubic foot and associated design requirements (Air changes per hour, ACH, and Capital Cost per ft^2):

- **Class 100,000**: $100,000$ particles per ft^3 at 0.5 μm. Air changes per hour: 12-18. Approx. capital cost: \$10 per ft^2.
- **Class 10,000**: $10,000$ particles per ft^3 at 0.5 μm. Air changes per hour: 18-30. Approx. capital cost: \$50 per ft^2.
- **Class 1,000**: $1,000$ particles per ft^3 at 0.5 μm. Air changes per hour: 150-300. Approx. capital cost: $\$350-400$ per ft^2.
- **Class 100**: 100 particles per ft^3 at 0.5 μm. Air changes per hour: 400-540. Approx. capital cost: $\sim$ \$1200 per ft^2.
- **Class 10**: 10 particles per ft^3 at 0.5 μm. Air changes per hour: 400-540. Approx. capital cost: $\sim$ \$3500 per ft^2.
- **Class 1**: 1 particle per ft^3 at 0.5 μm. Air changes per hour: 540-600. Approx. capital cost: $\sim$ \$10,000+ per ft^2.

Class limits (maximum allowable particles cubic foot)

ISO	SI	CLASS	0.1	0.2	0.3	0.5	5
CLASS 3	M 1	1	100 35			35 1	
	M 1.5	1	35	7	3	1	0
	M 2	10	99.1			2.83	
CLASS 4	M 2.5	10	10,000 345	75	30	352 10	0
	M 3	100	991			28.1	
CLASS 5	M 3.5	100	100,000 3,450	750	300	3520 100	0
CLASS 6	M 4.5	1,000	1,000,000 34,500	N/A	N/A	35,200 1,000	7
CLASS 7	M 5.5	10,000	345,000	N/A	N/A	352,000 10,000	70
CLASS 8	M 6.5	100,000	3,450,00	N/A	N/A	3,520,000 100,000	700

Figure 3.1: ISO 14644-1 Fed. Std. 209 E(SI) International
ISO standard requires results to be shown in cubic meters (1 cubic meter=35.2 cubic feet)

3.5 Environmental Control and Airflow

3.5.1 Controlled Parameters

- **Temperature** is controlled, typically 20 to 22°C. Tolerances range from ±3.0°F (Class 10,000) to ±0.1°F (Class 5).
- **Humidity (RH)** is controlled, usually around 50% RH, or 40 ÷ 60%. Tolerances range from ±5% (Class 10,000/1,000/100) down to ±1% (Class 5).
- **Pressure:** The room is held at **positive pressure** to blow dust OUT.
  - Pressure is typically ∼ 25 Pa for Class 100, 1000, and 10,000.
  - Pressure is typically 75 to 100 Pa for Class 1 and 10.
  - Usually, doors open inward, so room pressure closes them shut.
  - Note: Biohazard rooms operate at negative pressure to keep bugs IN.

3.5.2 Airflow and Design

Laminar flow is a key factor to reduce environmental particle contamination. Factors affecting laminar flow include the number of operators, cleanroom design, equipment layout, and operator behavior.

Design elements include:

- Supply air often flows through HEPA Filters located in the ceiling.
- Return air can utilize low-wall returns or return air plenums below a raised floor.
- A **raised floor** assures optimal performance for Class 5. Raised floors are very common, though low wall returns also work.
- Class 1 requires Gel/Flush grid ceiling systems with raised floors. It also mandates 540 to 600+ air changes per hour and $98+\%$ ceiling coverage.
- Class 4 requires 540 to 600 air changes per hour and $85-90\%$ ceiling coverage.

3.5.3 Cleanroom Air Filters

- **High Efficiency Particulate Air (HEPA) Filters:** This is the most common type of cleanroom air filter. They offer high efficiency, low pressure drop, and good loading characteristics. They use glass fibers in a paper-like medium. By definition, a true HEPA-rated filter will retain **99.97%** of incident particles with a diameter of 0.3 μm or larger.
- **ULPA Filters:** Class of air filters which meet a minimum performance level of **99.999%** on 0.3 μm efficiency. In the cleanroom market, ULPA can also be rated to 99.9995% on $0.12-0.16$ μm. Class 1 requires ULPA filters.

3.6 Cleanroom Clothing (Gowning)

Cleanroom clothing requirements escalate with the cleanliness class.

- **Highest Classes (Class 1, Class 10):** Require Hood, Hair Cover, Coverall, Intersuit, Boots, Facial Cover, and Gloves. The frequency of change is **Each Entry**.

- **Class 100:** Requires Hood, Hair Cover, Coverall, Boots, Facial Cover, and Gloves. The Intersuit is Optional. Frequency of change is Daily.
- **Lower Classes (Class 1,000, Class 10,000, Class 100,000):** Require less coverage, typically Hood/Cap/Hair Cover and Coverall/Frock, Boots/Footwear, and optional facial cover/gloves. Frequency of change ranges from 3 Times/Week to 2 Times/Week.

Gowning procedures include specific steps, such as putting on safety glasses, ensuring all hair is contained by the cap, checking for a snug fit and good face seal for the hood, bending the nose piece of the face mask for a snug fit, and pulling the hem of the glove over the coverall sleeve. Shoe Covers may be needed before entering the gowning area.

3.7 Deionized Water (DI Water)

Deionized water must be particulate- (and contaminant-) free. A large amount of water is used in washing steps.

- **Salt Removal:** Salts are removed through ion-exchange resins, which are both cationic (steal metals substituted by H^+) and anionic (steal anions substituted by OH^-). These resins finally capture salts, producing pure water.
- **Organic Removal:** Organic compounds (non-ionic) are adsorbed by active carbon. Water is then filtered through an absolute filter ($< 0.2\ \mu$m).
- **Quality Control:** Electric conductance is a good parameter for evaluating water quality. Resistance must always be in the $10 - 18\ \mathrm{M}\Omega \cdot \mathrm{cm}$ range.
- **Sterilization:** Water is sterilized with UV light to avoid bacteria and algae proliferation.
- **Drying:** After washing, substrates are dried with a **filtered hyperpure nitrogen flow under pressure** to avoid the slow natural drying (evaporation) of water, which would leave a small amount of contamination.

Chapter 4

Introduction to Lithography

Lithography derives from the Greek words $\lambda\acute{\iota}\theta o\varsigma$ (lithos, 'stone') and $\gamma\rho\acute{\alpha}\varphi\varepsilon\iota\nu$ (graphein, 'to write'). It is the fundamental process used to **transfer circuit or device patterns from a layout to substrates**.
Micro and Nano Technology often employs **batch fabrication**. Because selective processing (directly growing, depositing, or etching films or introducing dopants only in desired areas) is typically impossible, processes involve blanket deposition followed by removal from areas where the material is not needed. Technologies are categorized as:

- **SELECTION:** Lithography.
- **ADDITIVE:** Deposition (Epitaxy, CVD, PVD, ALD, Electroplating, ...) and Thermal Oxidation.
- **SUBTRACTIVE:** Etching.
- **MODIFICATION:** Doping, Annealing.

Photolithography is the specific technique used to **select**, transferring a desired pattern from a master copy (or mask) into a photo-sensitive polymer layer called **photoresist** on the wafer surface. The patterned polymer layer is then used to protect areas of the wafer during subsequent processes, such as etching or ion implantation. For the realization of a single device, **many photolithographic steps are needed** (many mask layers required), and alignment between these steps is crucial. During every step, similar regions of different devices are realized in parallel across the wafer surface.

4.1 Masks and Alignment

4.1.1 Mask Generation

Masks are fabricated using high-resolution lithographic techniques.

- **Substrates:** Pure and amorphous $\mathbf{SiO_2}$ slabs that are perfectly planar (rectified). Quartz glass maintains a transmission of approximately 90%, even at wavelengths lower than soda lime glass.
- **Absorbing Layer:** A thin absorbing metal layer is deposited, typically **Cr** (Chromium), chosen for optimal adhesion to the substrate.
- **Features:** Obtained through laser or e-beam lithography, depending on the required resolution.

4.1.2 Alignment

Alignment (or Overlay) ensures that circuit elements from the current step are precisely positioned relative to features already on the wafer. It is accomplished by superimposing previously printed alignment marks on the wafer with corresponding marks on the mask. Verification is achieved through a **split-field microscope**, which allows simultaneous viewing of opposite sides of the wafer, ensuring precise alignment. Sources of misalignment include $\mathbf{X} - \mathbf{Y}$ displacement and θ rotation. For alignment using front-backside exposure tools:

- **IR transparent samples (e.g., Cr):** Infrared (IR) alignment is utilized, providing high resolution of $1-5\ \mu$m. The IR transparency of materials like Cr enables precise alignment through the wafer.
- **Not IR transparent samples:** Alignment marks on the front side of the wafer are used, achieving a resolution of $5-20\ \mu$m.

4.2 Photoresist Components and Properties

4.2.1 Components

Photoresists (resists) are photosensitive organic mixtures that contain:

- **Inactive polymer resins:** A binder that provides mechanical and chemical properties, such as adhesion, chemical resistance, rigidity, and thermal stability.
- **PhotoActive Compounds (PAC).**
- **Solvent:** Controls the viscosity of the base, keeping it liquid for spinning (affecting film thickness).
- Other components: Surfactants, leveling agents, sensitizers, and dyes (to absorb λ).

4.2.2 Properties: Polarity (Positive vs. Negative)

- **Positive Resist**:
  - Light (photons) breaks the polymer chains (CHAIN SCISSION), making them more soluble in the developing solution.
  - Areas exposed to light are dissolved.
  - Resulting pattern replicates the mask pattern (holes in the resist correspond to holes in the mask).
  - Positive resist generally has **better resolution** due to the smaller size of the polymer.
- **Negative Resist**:
  - Light induces **cross-linking** of the polymer chains, making them less soluble in the developing solution.
  - Areas exposed to light become cross-linked and resist the developer chemical.
  - Resulting pattern replicates the inverse (negative) pattern of the mask (resist remains where the mask had holes).

4.2.3 Properties: Resolution and Thickness

Resolution is the smallest opening or space that can be produced in the photoresist layer. It is related to the exposure source and developing process. A **thinner layer has better resolution**. However, thicker layers may be required to serve as effective etch and implantation barriers or to ensure a pinhole-free film.

4.2.4 Properties: Sensitivity and Exposure Energy

Sensitivity is also known as the **Dose-to-Clear** (D_0 or E_i), which is the amount of exposure energy required to just clear (or consolidate) the resist in a large area for a given process.

- Typical exposure energies (Dose) range from $50 - 150$ mJ/cm^2 for Positive (UV) resist and $20 - 30$ mJ/cm^2 for Negative (UV) resist.
- Energy (E) is related to intensity (I) by $E := I \cdot t$.

4.2.5 Properties: Contrast (γ)

The photoresist contrast (γ) is the **maximum slope of the development rate curve** (H-D curve).

$$\gamma := \frac{1}{log_{10}(E_f) - log_{10}(E_i)} = \frac{1}{log_{10}\frac{|E_f|}{|E_i|}}$$

Typically, γ is ~ 2 to 4 for g- and i-line resists, and ~ 5 to 10 for DUV resists.

- **Good Resist Contrast** yields sharp walls and no swelling.
- **Poor Resist Contrast** yields sloped walls and swelling.

Sensitivity and contrast are not purely intrinsic properties; they also depend on process conditions (developer, development time, baking time, wavelength, substrate).

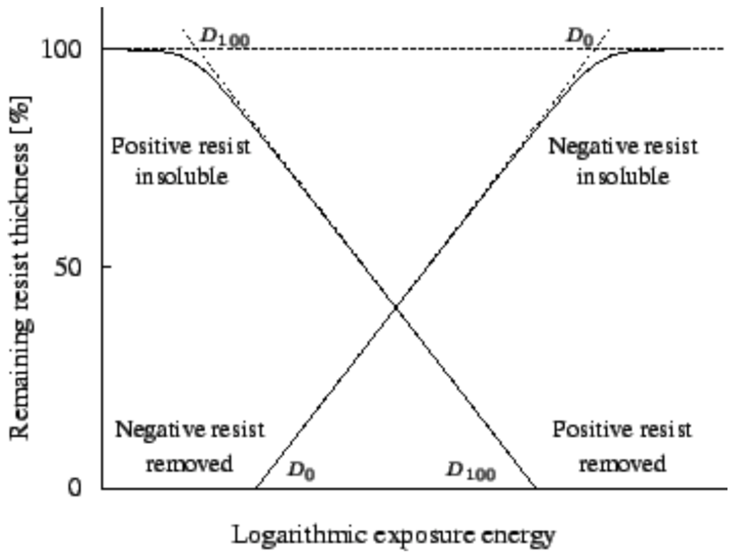


Figure 4.1: This figure presents the relation between Sensitivity and Contrast.

4.3 The Photolithography Process Flow

A typical Photolithography process recipe flow involves several sequential steps:

1. Priming (Wafer Preparation)
2. Resist Spin-On Coating
3. Resist SoftBake
4. Resist Exposure
5. Resist Post Exposure Bake (PEB)
6. Resist Image Developing
7. Resist HardBake

4.3.1 Wafer Preparation and Priming

Wafer surfaces are prepared prior to resist coating.

- **Dehydration Bake:** Wafer surfaces are baked to drive off adsorbed water, typically at **130°C** to **200°C** in a convection oven.
- (If needed) **Adhesion Promoter (Primer):** An adhesion promoter is applied. **HMDS** (hexamethyl disilazane, $[(CH_3)3Si]2NH$) is commonly used and can be applied as a vapor in an oven or spun on.

4.3.2 Resist Spin-On Coating

The resist is deposited onto the wafer by the spin-on method.

1. **Deposition:** An excess of the liquid polymer (photoresist) is delivered onto the substrate.
2. **Spin-up:** Under centrifugal force generated by the spinning disk, the liquid flows radially.
3. **Spin-off:** The liquid flies off the edge, and a uniform thick film tends to form and thins slowly to an equilibrium thickness.
4. **Evaporation:** The liquid solidifies due to solvent evaporation.

Resist thickness (t) is dependent on variables like viscosity (ν) and angular velocity (ω):

$$t = K \cdot S \cdot \left(\frac{\nu}{\omega^2 \cdot R^2}\right)^{1/3}$$

Where K is a constant, S is the fraction of solids, and R is the radius. The thickness decreases as the spin speed (rpm) increases. Quality measures during this step include thickness, uniformity, and the absence of particles and defects.

4.3.3 Soft Bake

The Soft Bake (or Pre-Bake) step is typically performed on a hot plate at **90°C** to **115°C** for 60 up to 90 seconds. **Purpose of Soft Bake**:

- Partial evaporation of photoresist solvents.
- Improves adhesion, uniformity, etch resistance, and linewidth control.
- Optimizes the light absorbance characteristics of the photoresist.

To leave the resist film, a solvent molecule must diffuse on the resist surface, evaporate, diffuse through the diffusion boundary layer above the surface, and ultimately be carried away by the air stream. The time and temperature of the soft bake significantly impact the average remaining solvent content (e.g., PGMEA).

4.3.4 Resist Exposure

The exposure step transfers the mask image to the resist-coated wafer and activates the photo-sensitive components. Quality measures include linewidth resolution, overlay accuracy, particles, and defects. Note: it's always a fight against diffraction.

Exposure Sources

- **Mercury (Hg) Lamps:** Traditionally used, has some spectral lines from a high-intensity plasma. Key lines include **G-line** (436 nm), **H-Line** (405 nm), and **I-Line** (365 nm). Optical filters can limit exposure wavelengths.
  The use of filters depends on the photoresistor: if we're using a positive one, we should use a filter, because its peak absorption frequency is the same as **I-line**(we'll limit diffraction); instead for a negative we don't use a filter.
- **UV LEDs:** An attractive alternative to energy-consuming mercury lamps due to technological advancements. Benefits include long lifespan, low power consumption, limited heating, instant-on functionality (no mechanical shutter), and no daily calibration or maintenance costs. High-power LEDs can cover the UV spectrum between 360 and 410 nm, mimicking Hg lines.
- **DUV Lasers (Excimer Lasers):** Used to increase resolution for **deep ultraviolet (DUV)** lithography systems. **Excimer pulsed lasers** are typically used. These are powerful, but extremely expensive to purchase and maintain.
  In excimer lasers, two elements, e.g. a noble gas and a halogen (from a halogen containing compound), which can react and "bind" together only in the excited state but not in their ground states, are present.
  Providing energy will therefore drive the reaction, creating the excimer.
  When the excitation energy is removed, the excimer dissociates and releases the energy at the characteristic wavelength.
  A pulsed excitation is used to repeat the process.
  $Kr + NF_3 \rightarrow energy \rightarrow KrF \rightarrow photon\ emission$
  - KrF (Krypton Fluoride) at **248** nm is used for 0.25 μm resolution.(lower wavelength than Hg, brighter).
  - ArF (Argon Fluoride) at **193** nm is used for 0.12 μm resolution.
  - F_2 at **157nm** was theorized but never used due to technical issues: finding suitable resists and transparent optical components at these wavelengths.
- **EUV (Extreme Ultraviolet):** Generates light at **13.5**nm by firing a high-energy IR pulsed laser on a droplet of molten tin. It's hit two times: First time is melt and the second vaporized →emits radiation. EUV lithography must operate in a **vacuum** and exclusively use nearly 11 **reflective optics** (including reflective masks) because EUV is absorbed by glass, oxygen, and water. Mirrors typically use about 50 pairs of $\lambda/4$ multilayer layers. It's important to remember that those mirrors reflect only 70% of total light: this means that the wafer receives only 2% of the initial light.
  ASML is the leader company in this industry.
- **Electron-Beam Lithography (EBL):** Uses a focused beam of electrons instead of light, allowing for **direct writing** of nm-scale features without a mask. Electrons have wave-like properties with wavelengths on the order of $0.2 - 0.5$Å, making diffraction negligible. E-beam lithography is primarily used to produce photomasks or in Nanotech/Quantum Research. It requires a vacuum environment.

PMMA: Poly(methylmethacrylate) is a positive photoresist used for short-wavelength lithography (deep UV, EUV, e-beam, X-ray). It is not sensitive to $\lambda > 240$ nm and has low sensitivity, often requiring sensitizers to improve speed. This ensures and high resolution.

Exposure Tools

The three main optical exposure systems are Contact, Proximity, and Projection.

- **Contact Exposure**:
  - *Advantages:* Fast (you expose the whole wafer at once), simple, inexpensive.
  - *Disadvantages:* The mask **contacts** the wafer, causing mask wear and contamination, leading to unacceptable defect densities and a shorter lifetime for the mask. No magnification is possible.
  - Max resolution in given by **Minimum Line/Gap Period** ($2 \cdot b_{min}$): $2 \cdot b_{min} = 3\sqrt{\lambda \cdot \frac{z}{2}}$; being λ the wavelength of the radiation and z the thickness of the photoresist. The definition is about $0.5 - 1\mu m$.
- **Proximity Exposure**:
  - *Advantages:* Fast (wafer exposed at once) and the mask **does NOT contact** the wafer (gap $s = 10 - 25$ μm), preventing mask wear or contamination.
  - *Disadvantages:* The separation gap causes greater diffraction, leading to **less resolution**. Cannot easily print features below a few μm (except for X-ray systems). It's also more expensive because the mask is usually the same size of the wafer.
  - Minimum Line/Gap Period ($2 \cdot b_{min}$): $2 \cdot b_{min} = 3\sqrt{\lambda \cdot (s + \frac{z}{2})}$; s being the distance between the Mask and the Photoresist.

  Note: Contact and Proximity are performed by the same machine, that costs about half a million, for Projection we need another one that costs about 50 millions.
- **Projection Exposure (Steppers/Scanners)**:
  - *Characteristics:* Projection printing **dominates today** due to high resolution and low defect densities. It uses a complex projection lens (e.g., 25-30 glass elements, 500 Kg).
  - *Advantages:* Mask **does NOT contact** the wafer. Offers de-magnification (usually 4X or 5X reduction), which makes masks cheaper to create because they have larger features. High resolution.
  - *Disadvantages:* An high de-magnification means each die exposed separately, this means it takes longer to expose the entire wafer. It's also very complex and expensive, requires precision stepper motor. This all implies a lower throughput.

Projection Exposure: Steppers and Scanners

Note: now the mask is called ***reticle***.

- **Stepper:** The step and repeat method exposes one die (or part of one die) at a time. Only the wafers move to expose different dice. Steppers rely on a precision system to automatically expose all dice after an initial alignment. Extra lenses allow reduction of the mask dimensions onto the wafer (4:1 to 10:1): it's easier to create defect-free masks at larger geometries.
- **Scanner:** All state-of-the-art tools use this approach today. **Both wafers and reticle move** to expose different dyes (reticle's speed depends on magnification). By using a fixed exposure slit, scanners achieve the required exposure field size with a **much smaller lens**, providing better control for aberrations. Nowadays it' used with immersion optical tool.

4.3.5 Post-Exposure Processing and Pattern Transfer

Reflections and Standing Waves

Standing waves occur due to **interference between the incident and reflected light** waves within the photoresist film. This creates a sinusoidal intensity variation along the thickness of the photoresist, leading to high and low exposed regions separated by $\lambda/4n$ (n=refractive index). This non-uniform exposure causes faster and slower development rates.
It'll show photoresistor's areas that are Over and Under exposed, creating a knurling.

Fixing: Post Exposure Bake (PEB)

PEB is a crucial (and more used) step used to **remove standing waves**.; though is not always necessary.

- **Mechanism:** PEB causes thermal re-distribution (diffusion) of the exposure products, smoothing the concentration pattern created by the sinusoidal intensity distribution.
- **Benefits:** It increases focus latitude and resolution, and improves thermal stability.
- **Conditions:** Hot plate bake $110°C$ to $125°C$ for 60 up to 90 seconds.

Prevention: Antireflective Coatings (ARC)

The use of ARCs, dyes, and filters can also help prevent interference. ARC materials common for 365 nm are Titanium Nitride, while Silicon nitride and silicon oxynitride are used for $365 - 248$ nm.

4.3.6 Developing

Developing involves dissolving the soluble areas of the photoresist using a developer chemical. The final visible patterns (windows or islands) appear on the wafer.

- **Developer Solution:** Typically an aqueous-alkaline solution (normality: 0.1N – 0.35N). NaOH, KOH, or TMAH based solutions are used for PR, while xylene is used for NR.
- **Methods:** Puddle development (100-200cc dispensed), Immersion Develop, or Spray-On Develop.
- **Timing:** The timing is critical. **Overdevelopment** (too long) or **underdevelopment** (too short) both negatively affect linewidth. Underdeveloped resist can prevent access to the underlying layer.

4.3.7 Hard Bake

The Hard Bake step involves evaporating all solvents remaining in the PR. It is not always mandatory but is needed for acid wet etching.

- **Purpose:** Improves etch and implantation resistance, improves PR adhesion, polymerizes and stabilizes photoresist, and allows PR flow to fill pinholes.
- **Conditions:** Hot plate bake $110°C$ to $125°C$ for 60 up to 120 seconds.
- **Caution:** Over-baking can cause excessive PR flow, negatively affecting photolithography resolution.

4.3.8 Pattern Transfer Methods

- **Etching Process:** The PR is spun, patterned, and used as a mask while the underlying layer is etched, before the PR is removed. Will always introduce impurities, and the process is too long and difficult in the case we have multiple metals.
- **Lift-Off Process:** The PR is spun and patterned. Metal (or other material) is deposited over the patterned PR, and then the excess metal is lifted off with the PR layer.
  - It's used a lot in the Quantum world.
  - This process allows features to be obtained without chemical wet etching: a solvent is used (often the same of the photoresistor) instead of acid.
  - It requires relatively high thickness resist (> 2 μm).
  - The deposited film thickness must be less than the resist thickness (typically a maximum of 1/3).
  - **Image Reversal (IR) resists** are beneficial as they have a particular behavior: thay act like a positive resistor under exposure; thanks to a **reversal bake** the soluble area becomes cross-linked, creating a negative profile slope (undercut) which facilitates the lift-off of deposited layers right after a **flood exposure**.
    In this way we have strong adhesion, a negative slope with resolution of a positive resistor.

4.4 Models and Simulation

Lithography simulation relies on models from two fields: Optics (to model aerial image formation) and Chemistry (to model latent image formation in the resist).

- **Aerial Image Simulation:** Commercial tools (PROLITH, DEPICT, ATHENA) calculate the aerial image based on mask design and optical system characteristics. These simulators are powerful, as the math is well understood and fast algorithms are implemented. Simulation can show how variables like numerical aperture (NA), feature size, and wavelength (λ) affect image quality.
- **Latent Image Simulation:** This step calculates the concentration of the PhotoActive Compound (PAC) after exposure. Simulation can demonstrate how PEB "smears out" standing wave effects, producing a more uniform PAC distribution.

Chapter 5

Silicon Oxidation

5.1 Thermal Oxidation of Silicon

Thermal Oxidation of Silicon is a process used to grow a layer of silicon dioxide (SiO_2) on the surface of a silicon wafer. This is achieved by exposing the silicon to an oxidizing ambient, such as oxygen (O_2) or water vapor (H_2O), at elevated temperatures typically ranging from 700°C to 1200°C. Alternatively, the process can also start from a polysilicon film.

The thermal oxidation process is highly controlled, allowing the growth of SiO_2 layers with thicknesses ranging from as thin as 30 Å (angstroms) to several micrometers ($\sim 2\ \mu$m). These layers are characterized by their excellent interface properties, which are critical for the performance of silicon-based devices.

Thermal oxidation is a cornerstone in the manufacturing of silicon devices due to the versatile and beneficial properties of silicon dioxide. Some of its key applications include:

- **Electrical Insulation:** Silicon dioxide serves as a gate oxide in MOSFETs (Metal-Oxide-Semiconductor Field-Effect Transistors) and as an isolation layer between different devices on the same wafer. Its high dielectric strength and insulating properties make it ideal for these purposes.
- **Masking Layer:** During processes such as doping or etching, SiO_2 acts as a protective layer to shield selected regions of the silicon wafer. This selective masking is essential for creating intricate device structures.
- **Thermal Insulation:** Silicon dioxide is thermally insulating, which is advantageous in certain applications where heat management is critical.
- **Sacrificial Layer:** In Surface Micromachining Technology, SiO_2 is often used as a sacrificial layer. After serving its purpose, it can be selectively removed to create microstructures or cavities.

While silicon dioxide can also be deposited using other techniques, such as Chemical Vapor Deposition (CVD), the SiO_2 layers produced by thermal oxidation generally exhibit superior electrical characteristics. This makes thermal oxidation the preferred method for applications requiring high-quality oxide layers.

The ability to form stable and high-quality silicon dioxide layers was one of the key reasons silicon became the dominant material in the semiconductor industry, replacing germanium (Ge). Unlike silicon, germanium does not form a stable native oxide, which limits its utility in modern device fabrication. Similarly, while gallium arsenide (GaAs) offers excellent electrical properties, its thermal oxidation results in non-stoichiometric films with poor electrical insulation and surface protection. This has limited the widespread adoption of GaAs in favor of silicon.

It is worth noting that silicon naturally forms a very thin layer of silicon dioxide, known as **native oxide**, when exposed to air. This native oxide layer is typically about 1 nm thick and forms spontaneously due to the reaction of silicon with oxygen in the environment. While this layer is not sufficient for most device applications, it highlights the natural tendency of silicon to form a protective oxide layer, which has been leveraged and optimized in the thermal oxidation process.

5.2 Properties of Silicon Dioxide

Thermal silicon dioxide, often referred to as fused silica, is an amorphous material with a short-range structure. This structure consists of tetrahedrons where oxygen ions (O^{2-}) are positioned at each corner, and a silicon atom (Si^{4+}) is located at the center. This arrangement gives silicon dioxide its unique properties, making it an essential material in semiconductor manufacturing.

- **Amorphous Nature:** Unlike crystalline materials, thermal silicon dioxide does not have a long-range periodic atomic arrangement. Instead, it exhibits a disordered, glass-like structure, which is highly stable and reproducible during growth.

- **Melting Point:** The melting point of silicon dioxide is extremely high, approximately 1700°C. This makes it suitable for high-temperature applications.
- **Density:** The density of thermal silicon dioxide is 2.21 g/cm^3, which is slightly lower than that of silicon (2.33 g/cm^3). This lower density is due to its amorphous structure, which is less tightly packed compared to crystalline materials.
- **Crystalline Quartz Comparison:** Crystalline silicon dioxide, known as quartz, has a higher density of 2.65 g/cm^3. The difference in density between thermal silicon dioxide and quartz arises from the more open structure of the amorphous form. In thermal silicon dioxide, only about 43% of the available volume is occupied by silicon dioxide molecules, leaving space for interstitial diffusion of impurities such as sodium (Na).
- **Atomic Density:** The atomic density of thermal silicon dioxide is 2.3×10^{22} molecules/cm^3. For comparison, silicon has an atomic density of 5×10^{22} atoms/cm^3, which reflects the more compact nature of the silicon lattice.
- **Refractive Index:** The refractive index of silicon dioxide is $n = 1.46$, which is an important property for optical applications, including waveguides and photonic devices. It's relatively low refractive index compared to silicon ($n \approx 3.5$) allows for effective light confinement in silicon-based optical structures.
- **Dielectric Constant:** The dielectric constant of silicon dioxide is $\epsilon = 3.9$. This relatively low value makes it an excellent insulator, suitable for use in capacitors and as a gate dielectric in MOSFETs.
- **Electrical Insulation:** Silicon dioxide is an outstanding electrical insulator, with a resistivity greater than 10^{20} $\Omega \cdot$cm. Its wide energy gap of $8-9$ eV contributes to its insulating properties, preventing the flow of charge carriers under normal operating conditions.
- **Breakdown Electric Field:** Silicon dioxide can withstand very high electric fields, with a breakdown strength exceeding 10^7 V/cm. This property is critical for its use in high-voltage and high-reliability applications.

In summary, thermal silicon dioxide is a versatile material with properties that make it indispensable in the semiconductor industry. Its amorphous structure, high thermal stability, excellent electrical insulation, and compatibility with silicon processing have established it as a cornerstone of modern electronics.

5.3 How Does Silicon Oxidize?

The process of silicon oxidation involves heating silicon wafers in an atmosphere containing an oxidizing agent, typically molecular oxygen (O_2) or steam (H_2O). This process results in the formation of a silicon dioxide (SiO_2) layer on the surface of the silicon wafer. The two primary reactions that occur during this process are:

- **Dry Oxidation:** This process involves the reaction of silicon with molecular oxygen to form silicon dioxide:

$$\mathrm{Si} + \mathrm{O_2} \rightarrow \mathrm{SiO_2}$$

- **Wet Oxidation:** In this process, silicon reacts with water vapor (steam) to produce silicon dioxide and hydrogen gas:

$$\mathrm{Si} + 2\mathrm{H_2O} \rightarrow \mathrm{SiO_2} + 2\mathrm{H_2}$$

It is important to note that during the oxidation process, the silicon substrate is **consumed** as it reacts with the oxidizing species to form the SiO_2 layer.

5.3.1 Oxidation Occurs at the $\mathrm{Si}-\mathrm{SiO_2}$ Interface

The growth of the silicon dioxide layer does not occur on the outer surface of the oxide layer. Instead, it takes place at the interface between the silicon substrate and the existing silicon dioxide layer. This means that the oxidizing species (O_2 or H_2O) must diffuse through the already-formed oxide layer to reach the silicon surface where the reaction occurs. As a result, the rate of oxidation decreases over time as the oxide layer becomes thicker, due to the increased diffusion distance.

5.3.2 Molecular Density Ratio of Oxide to Silicon

The molecular density of silicon dioxide is lower than that of silicon. Specifically, the ratio of the molecular density of silicon dioxide to the atomic density of silicon is approximately 0.44. This can be calculated as follows:

$$\frac{\text{Molecular Density of Oxide}}{\text{Atomic Density of Silicon}} = \frac{2.2 \times 10^{22}\ \text{molecules/cm}^3}{5.0 \times 10^{22}\ \text{atoms/cm}^3} = 0.44$$

This difference in density means that the silicon dioxide layer occupies more volume than the silicon consumed during the oxidation process.

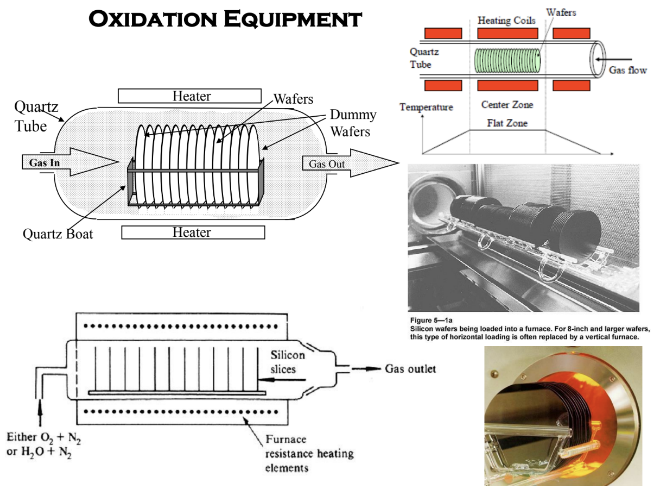


Figure 5.1: Oxidation Equipment

5.3.3 Oxidation Equipment

The equipment used for silicon oxidation is typically a high-temperature furnace designed to provide a controlled environment for the oxidation process. The main components of an oxidation furnace include:

- **Quartz Tube:** A cylindrical quartz tube serves as the reaction chamber. It is resistant to high temperatures and chemical reactions, ensuring a clean environment for oxidation.
- **Heating Elements:** The furnace is equipped with heating elements, often made of silicon carbide or other high-temperature materials, to maintain the required temperature (typically 700°C to 1200°C).
- **Gas Flow System:** The system introduces oxidizing gases such as oxygen (O_2) or water vapor (H_2O) into the chamber. The flow rate and composition of the gases are precisely controlled to ensure uniform oxidation.
- **Temperature Control:** A precise temperature control system, including thermocouples and a feedback loop, ensures uniform heating throughout the chamber.
- **Wafer Boat:** Silicon wafers are placed in a quartz or silicon carbide boat, which holds them in a vertical or horizontal orientation inside the furnace.
- **Exhaust System:** The exhaust system removes byproducts such as hydrogen (H_2) or hydrogen chloride (HCl) from the chamber, maintaining a clean environment.
- **Load/Unload Mechanism:** A mechanism for loading and unloading wafers ensures minimal contamination and efficient handling.

Oxidation furnaces can be classified into two types based on the wafer orientation:

- **Horizontal Furnaces:** Wafers are placed horizontally in the furnace. These are commonly used for batch processing.
- **Vertical Furnaces:** Wafers are placed vertically, allowing better uniformity in gas flow and temperature distribution. These are often used for advanced processes.

Modern oxidation equipment also includes automation features for wafer handling, recipe control, and real-time monitoring of process parameters to ensure high-quality oxide layers.

5.4 Key Variables in Oxidation

The oxidation process and the characteristics of the resulting oxide layer are influenced by several critical factors. These variables determine the growth rate, uniformity, and quality of the silicon dioxide layer. Below is a detailed discussion of the key variables:

- **Time:** The duration of the oxidation process is a primary factor in determining the thickness of the oxide layer. Longer oxidation times allow for more silicon to react with the oxidizing species, resulting in a thicker oxide layer. However, the growth rate decreases over time due to the increasing diffusion distance as the oxide layer thickens.

- **Temperature:** The temperature at which the oxidation process is carried out plays a crucial role in determining the reaction rate and the rate of solid-state diffusion. Higher temperatures accelerate the chemical reaction between silicon and the oxidizing species, as well as the diffusion of oxidizing molecules through the growing oxide layer. This leads to faster oxidation and thicker oxide layers within a given time frame.

- **Oxidizing Species:** The choice of oxidizing agent significantly impacts the oxidation rate. Wet oxidation, which uses steam (H_2O), is much faster than dry oxidation, which uses molecular oxygen (O_2). This difference arises because water molecules diffuse more readily through the silicon dioxide layer and react more efficiently with silicon compared to oxygen molecules.

- **Crystallographic Orientation:** The crystallographic orientation of the silicon substrate affects the oxidation rate. For monocrystalline silicon, the (111) surface oxidizes approximately 1.7 times faster than the (100) surface. This is due to the higher density of silicon atoms on the (111) planes, which provides more sites for the oxidation reaction. In the case of polycrystalline silicon, the oxidation rate may vary depending on the grain size and orientation of the individual crystals.

- **Second-Order Variables:** Several secondary factors can also influence the oxidation process, including:
  - **Surface Cleanliness:** The cleanliness of the silicon surface is critical. Metallic contamination, such as residues from previous processing steps, can act as a catalyst for the oxidation reaction. This can lead to localized variations in the oxidation rate and affect the uniformity of the oxide layer.
  - **Furnace Pressure:** The pressure inside the oxidation furnace can impact the oxidation rate. Higher pressures increase the concentration of oxidizing species in the gas phase, thereby enhancing the rate of oxidation.
  - **Silicon Doping:** The type and concentration of dopants in the silicon substrate can influence the oxidation behavior. For example, p-type and n-type dopants can have different effects on the oxidation rate due to their impact on the electronic properties of the silicon surface and the diffusion of oxidizing species.

5.5 Modeling Thermal Oxidation of Silicon

The modeling of thermal oxidation of silicon is a critical aspect of understanding and predicting the growth of silicon dioxide layers during the oxidation process. This model provides a simplified yet effective tool to calculate and predict the following:

- The oxidation rate, represented as $\frac{\partial x_{\text{ox}}(t)}{\partial t}$, which describes the rate at which the oxide layer grows over time.

- The final oxide thickness at a given time t, denoted as $x_{\text{ox}}(t)$, which is the total thickness of the silicon dioxide layer formed after a specific duration of oxidation.

The most widely accepted and successful model for this purpose was developed by B.E. Deal and A.S. Grove in 1965. Their work, titled *"General Relationship for the Thermal Oxidation of Silicon"*, was published in the *Journal of Applied Physics* (Vol. 16, No. 12, pg. 3770, 1965). This model has since become a cornerstone in the field of semiconductor manufacturing.

5.5.1 Diffusion-Limited Process

The thermal oxidation process involves three primary fluxes:

- F_1: The flux of oxidant molecules (e.g., O_2 or H_2O) diffusing through the gas phase to the surface of the growing oxide film.

- F_2: The flux of oxidant molecules diffusing through the silicon dioxide layer to the SiO_2/Si interface.

- F_3: The reaction flux, which represents the rate at which oxidant molecules react with silicon at the interface to form silicon dioxide.

For the system to be in a steady state, the fluxes must satisfy the condition:

$$F_1 = F_2 = F_3$$

Here, N_0 and N_i represent the concentrations of oxidant species, measured in $\frac{\text{number of particles}}{\text{cm}^3}$, at the outer surface of the oxide and at the SiO_2/Si interface, respectively.

5.5.2 Fick's First Law of Diffusion

The diffusion flux J through the oxide layer can be described using Fick's first law of diffusion:

$$J = F_2 = -D \cdot \frac{dN}{dx} = D \cdot \left(\frac{N_0 - N_i}{x_{\text{ox}}} \right)$$

where:

- D is the diffusion coefficient, measured in $\frac{\text{cm}^2}{\text{sec}}$.
- x_{ox} is the thickness of the oxide layer.
- N_0 and N_i are the oxidant concentrations at the outer surface and the interface, respectively.

The reaction flux at the interface is given by:

$$J = F_3 = k_s \cdot N_i$$

where k_s is the surface reaction rate constant.

By combining these equations and eliminating N_i using the steady-state condition ($F_2 = F_3$), the flux J can be expressed as:

$$J = \frac{D \cdot N_0}{x_{\text{ox}} + \frac{D}{k_s}}$$

5.5.3 Oxidation Growth Rate

The growth rate of the oxide layer, $\frac{dx_{\text{ox}}}{dt}$, can be derived by considering the relationship between the flux J and the number of oxidant molecules required to form the oxide:

$$\frac{dx_{\text{ox}}}{dt} = \frac{J}{N_{\text{ox}}}$$

where N_{ox} is the concentration of oxidant molecules in the oxide required to form the silicon dioxide layer. For dry oxidation, $N_{\text{ox}} = 2.2 \times 10^{22}$ molecules/cm^3, while for wet oxidation, $N_{\text{ox}} = 4.4 \times 10^{22}$ molecules/cm^3.

Substituting J into the equation, the growth rate becomes:

$$\frac{dx_{\text{ox}}}{dt} = \frac{\left(\frac{D \cdot N_0}{N_{\text{ox}}} \right)}{\left(x_{\text{ox}} + \frac{D}{k_s} \right)}$$

5.5.4 General Relationship for Oxidation

The general relationship for the thickness of the oxide layer as a function of time is given by:

$$\frac{x_{\text{ox}}^2}{B} + \frac{x_{\text{ox}}}{\left(\frac{B}{A} \right)} - (t + \tau) = 0$$

where:

- $A = \frac{2D}{k_s}$ is the linear growth parameter.
- $B = \frac{2D \cdot N_0}{N_{\text{ox}}}$ is the parabolic growth parameter.
- $\tau = \frac{x_i^2}{B} + \frac{x_i}{\left(\frac{B}{A} \right)}$ is the time shift to account for the initial oxide thickness x_i.

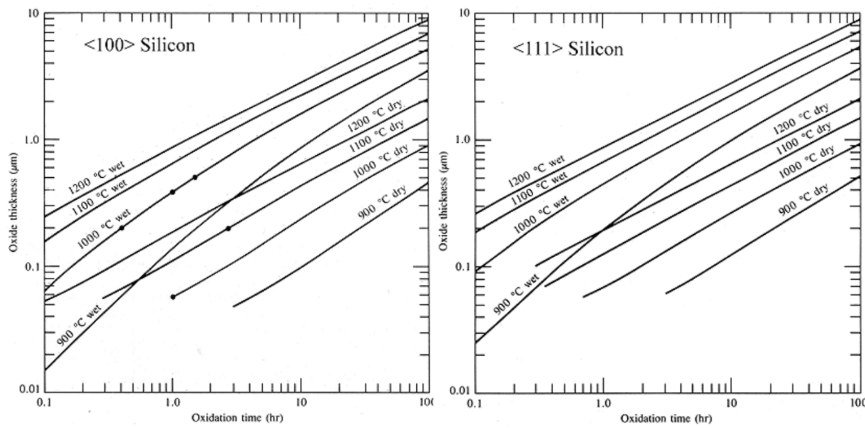
The solution for the oxide thickness at time t is:

$$x_{\text{ox}}(t) = \frac{1}{2} A \left[\sqrt{1 + \frac{4B}{A^2}(t + \tau)} - 1 \right]$$

This equation provides a comprehensive description of the oxidation process, accounting for both diffusion-limited and reaction-limited regimes.

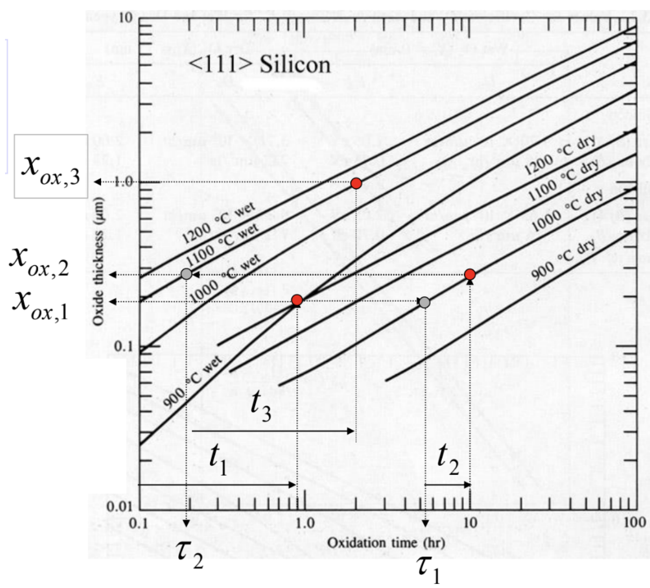
5.6 Oxidation Graphs

Graphical representations of the oxidation process are invaluable for visualizing the relationship between oxide thickness, time, and the governing parameters. These graphs provide insights into the behavior of the oxidation process under varying conditions, such as temperature, oxidizing species, and initial oxide thickness. They are essential tools for process engineers to optimize and predict the outcomes of thermal oxidation.



5.6.1 Multiple Oxidations (Tracking τ)

In practical scenarios, the oxidation process often involves multiple steps, each with different conditions such as temperature and oxidizing species. To account for these changes, the concept of a time shift, τ, is introduced. This shift allows the current oxidation step to be treated as an extension of the previous step, simplifying the calculation of the overall oxide thickness.



The relationship for time and thickness in such cases is given by:

$$t = \frac{x_{\text{ar}}^2}{B} + \frac{x_{\text{ar}}}{\left(\frac{B}{A}\right)} - \tau$$

where τ is the time shift, calculated as:

$$\tau = \frac{x_i^2}{B} + \frac{x_i}{\left(\frac{B}{A}\right)}$$

To illustrate this, consider the following table of oxidation steps:

Step	Time	Temp	Type	Overall Thickness
0	0.0 hr			0 nm
1	0.9 hr	900°C	Wet	190 nm
2	4.5 hr	1000°C	Dry	280 nm
3	1.8 hr	1100°C	Wet	

The challenge is to determine the time required to achieve the current oxide thickness under new oxidation conditions (temperature and type of oxidation). By shifting the time origin using τ, the current step can be analyzed as a continuation of the previous step, simplifying the calculations.

5.7 Growth Rate Regimes

The growth of the oxide layer can be described by the general equation:

$$x_{\text{ox}}(t) = \frac{1}{2}A\left[\sqrt{1+\frac{4\cdot B}{A^2}(t+\tau)}-1\right]$$

Depending on the time scale, the growth rate can be categorized into two distinct regimes:

5.7.1 Short-Time Regime (Reaction-Limited Growth)

For short times, where $(t+\tau) \ll \frac{A^2}{4\cdot B}$, the growth rate is limited by the reaction of the oxidizing species with silicon. Using a Taylor series expansion, the equation simplifies to:

$$x_{\text{ox}}(t) = \left(\frac{B}{A}\right)\cdot(t+\tau)$$

In this regime, the growth rate is linear, and the parameter $\frac{B}{A}$ is referred to as the **linear growth parameter**.

5.7.2 Long-Time Regime (Diffusion-Limited Growth)

For long times, where $(t+\tau) \gg \frac{A^2}{4\cdot B}$, the growth rate is limited by the diffusion of oxidizing species through the growing oxide layer. The equation simplifies to:

$$x_{\text{ox}}(t) = \sqrt{B\cdot(t+\tau)}$$

In this regime, the growth rate follows a parabolic trend, and the parameter B is known as the **parabolic growth parameter**.

The relationship between the linear and parabolic growth parameters is given by:

$$\frac{B}{A} = k_s\frac{N_0'}{N_{\text{ox}}}$$

where k_s is the surface reaction rate constant, N_0' is the oxidant concentration at the surface, and N_{ox} is the oxidant concentration in the oxide.

The parabolic growth parameter is expressed as:

$$B = \frac{2DN_0}{N_{\text{ox}}}$$

5.8 Rapid Thermal Oxidation (RTO)

Rapid Thermal Oxidation (RTO) is a modern technique increasingly employed for the growth of thin, high-quality dielectric layers. It is particularly advantageous for reducing the thermal budget in semiconductor manufacturing.

RTO differs from conventional thermal oxidation in several key aspects:

- **Chamber Design:** RTO systems feature more complex chamber designs to ensure uniform heating and rapid temperature changes.
- **Radiation Source:** High-intensity radiation sources, such as halogen lamps, are used to achieve rapid heating.
- **Temperature Monitoring:** Advanced temperature monitoring systems ensure precise control of the oxidation process.

From a kinetic perspective, RTO is influenced by both thermally activated processes and non-thermal, photon-induced processes. The latter involves the generation of monatomic oxygen atoms by ultraviolet (UV) radiation, which creates a parallel oxidation reaction. This photon-induced process is particularly significant at lower temperatures, where it can dominate the oxidation kinetics.

RTO is widely used for applications requiring thin oxide layers with excellent electrical properties, such as gate dielectrics in advanced MOSFETs.

Chapter 6

Epitaxy, CVD, PVD, electroplating

6.1 Epitaxy

Epitaxy is a growth technique involving monocrystalline layers grown on a monocrystalline substrate, ensuring structural continuity. The substrate acts as a crystal seed, and the film is grown by progressively adding atoms.

6.1.1 Conditions and Types

For successful epitaxial growth, **the crystalline structure and lattice parameter should be similar**, generally having a maximum mismatch of approximately 2%. Epitaxial techniques are categorized based on the substrate (A) and the grown layer (S):

- **Homoepitaxy**: Layer S and Substrate A are the same material ($S = A$).
- **Heteroepitaxy**: Layer S and Substrate A are different materials ($S \neq A$), for example, 3C-SiC on Si.

Epitaxy is used to control the doping profile, such as adding a lightly doped or undoped layer on a heavily doped substrate to increase the breakdown voltage of the junction, or creating isolation junctions with opposing doping. Epi-layers often have a smoother surface than the substrate and are generally oxygen and carbon-free, unlike semiconductive substrates. Epi-layers can also be deposited over heavily doped areas called **buried layers**, which serve as low-resistance contacts.

6.1.2 Classification by Mother Phase

Epitaxial techniques are divided according to the characteristic of the Mother Phase (This is the source of the atoms that will form the new crystal layer):

1. **Vapour Phase Epitaxy (VPE)** or **Chemical Vapor Deposition (CVD)**: The mother phase is a mix of gases or vapors at room or reduced pressure.
2. **Liquid Phase Epitaxy (LPE)**: The mother phase is a solution of the active material with a suitable diluent.
3. **Molecular Beam Epitaxy (MBE)**: The mother phase is a gas at a very reduced pressure, where collisions between molecules are practically absent.

6.1.3 Why is Epitaxy important?

Epitaxy plays a crucial role in the fabrication of advanced semiconductor devices and integrated circuits. Its importance stems from the ability to precisely control the crystalline structure, doping profile, and physical properties of the deposited layers. Below are the key reasons why epitaxy is significant:

- **Enhanced Breakdown Voltage:** Epitaxy allows the addition of a lightly doped or completely undoped layer on top of a heavily doped substrate. This configuration increases the breakdown voltage of the junction, which is essential for high-power and high-voltage applications. The lightly doped layer reduces the electric field intensity at the junction, thereby improving the device's ability to withstand higher voltages without breakdown.
- **Electrical Isolation:** By depositing an epitaxial layer with doping opposite to that of the substrate, it is possible to create isolation junctions. These reverse-polarized junctions act as barriers, electrically isolating different regions of the device. This is particularly useful in complex integrated circuits where multiple components need to operate independently.

- **Buried Layers for Low-Resistance Contacts:** Epitaxial layers can be deposited over patterned, heavily doped regions known as **buried layers**. These buried layers serve as low-resistance contacts, improving the electrical performance of the device. The epitaxial layer ensures that the buried layer is encapsulated, providing a smooth surface for further processing.
- **Smoother Surface for Subsequent Processing:** Epitaxial layers typically have a smoother surface compared to the substrate. This smoothness is advantageous for subsequent fabrication steps, such as lithography and etching, as it reduces defects and improves the overall yield of the manufacturing process.
- **Precise Doping Profile Control:** Epitaxy provides an additional means of controlling the doping profile in a device structure, beyond what is achievable through doping by diffusion or ion implantation. By carefully controlling the doping concentration and distribution during the epitaxial growth process, device designers can achieve highly tailored electrical characteristics.
- **Superior Material Quality:** The physical properties of epitaxial layers often differ from those of bulk materials. For instance, epitaxial layers are generally free from oxygen and carbon impurities, which are commonly present in semiconductive substrates. This high purity enhances the electrical and optical properties of the material, making it suitable for high-performance applications.
- **Flexibility in Device Design:** The ability to grow one or more epitaxial layers on a substrate provides device designers with greater flexibility in creating complex structures. This flexibility is critical for the development of advanced devices such as heterojunction bipolar transistors (HBTs), high-electron-mobility transistors (HEMTs), and multi-junction solar cells.
- **Compatibility with Advanced Materials:** Epitaxy enables the integration of materials with different properties, such as III-V compounds on silicon, through heteroepitaxy. This compatibility opens up new possibilities for combining the advantages of different materials in a single device, such as high mobility and low-cost manufacturing.

6.1.4 Vapor Phase Epitaxy (VPE) / Chemical Vapor Deposition (CVD)

VPE Details

VPE consists of growing the crystal from **gaseous precursors** (gas sources) including the material and its associated doping. It is the *most used* epitaxial growth technique, especially for Si.

For VPE, **gases are dissociated** in the chamber to produce atoms (e.g., silicon atoms) that move towards and bond with the pre-existing lattice. Substrates must be heated, typically at **high temperatures** (e.g., around 1100°C) to ensure epitaxial growth. An example reaction is $SiCl_4 + 2H_2 \Leftrightarrow Si + 4HCl$.

CVD Hazards

Many gases utilized in CVD systems present significant hazards.

Table 6.1: Properties and Hazards of Common CVD Gases

Hazard	**Description**
Toxic	Hazardous to humans. E.g., Arsine (AsH_3) has an exposure limit of 0.05 ppm.
Corrosive	Causes corrosion to stainless steel and other metals. E.g., Hydrogen chloride (HCl).
Flammable	Burns when exposed to an ignition source and oxygen source. E.g., Hydrogen (H_2) (4-74% vol limits).
Pyrophoric	Spontaneously burns or explodes in air, moisture, or when exposed to oxygen. E.g., Phosphine (PH_3) and Silane (SiH_4).

VPE Reactors

VPE reactors consist of a gas inlet/control system, a deposition chamber, and an exhaust (e.g., scrubber). Wafers are typically loaded onto graphite susceptors that are heated via RF induction. This RF heating warms the susceptor while the chamber walls remain cold, limiting outgassing and contamination. Common reactor shapes include **Horizontal, Vertical, Barrel, Pancake,** and **Rotating Disk (Planetary)** reactors.

6.1.5 Liquid Phase Epitaxy (LPE)

Liquid Phase Epitaxy (LPE) is a technique that involves the growth of crystalline layers from a liquid phase, typically a **metallic solution**, onto a **crystalline substrate**. This process relies on the principle of **oversaturation**, where the solution becomes supersaturated, leading to the precipitation of the solute onto the substrate in an epitaxial manner. The substrate acts as a **template**, ensuring that the deposited layer maintains the same crystalline structure.

One of the key aspects of LPE is the **absence of wetting** between the liquid solution and the crucible, which ensures that the growth occurs exclusively on the substrate. The growth mechanism involves the direct precipitation of the solute from the liquid phase as the solution is cooled. During this cooling process, the oversaturated solution deposits the excess solute onto the crystalline substrate, forming an epitaxial layer.

The growth conditions in LPE are carefully controlled to maintain **near-equilibrium conditions**. This is achieved by ensuring that the relative temperature difference ($\Delta T/T$) is kept within the range of 10^{-3} to 10^{-2}. Such precise control minimizes **defects** and promotes high-quality epitaxial growth.

LPE is particularly suited for the growth of III-V compound semiconductors, silicon carbide (SiC), and multilayered epitaxial structures with varying properties, such as different **doping levels**. The choice of solvent plays a crucial role in the process. Typically, a low melting point metal component of the compound is used as the solvent. For example:

- **Gallium (Ga)**, with a melting point of 30°C, is commonly used for the growth of GaAs and GaAlAs on GaAs substrates.
- **Indium (In)**, with a melting point of 156°C, is used for the growth of In, InGaAs, and InGaAsP on InP substrates.

The **growth temperature** (T_{growth}) is another critical parameter and is chosen to balance the **solubility** of the solute in the solvent and the risk of **thermal degradation** of the substrate. For instance:

- Epitaxy on **GaAs substrates** is typically performed at around 800°C.
- Epitaxy on **InP substrates** is carried out at a lower temperature, around 600°C.

Overall, LPE is a versatile and precise technique for producing high-quality epitaxial layers, making it invaluable in the fabrication of advanced semiconductor devices.

LPE Reactor

Growth takes place under a flux of purified H_2 (passed through a Pd barrier) at atmospheric pressure to avoid oxidation of the solution and the substrate.
The apparatus utilizes a **Graphite crucible** which features fixed wells for the liquid growth solutions and a sliding part that hosts the substrate.
The substrate is pushed under the oversaturated solutions. The temperature (T) is then slowly lowered ($\approx$ 1°C/min), and the substrate is kept in position for the time necessary to achieve the desired layer thickness.
Growth stop is obtained by removing the substrate from the solution. This is possible because molten metallic solutions do not wet graphite and substrates $\rightarrow$ **sliding off** of the substrate is enough to remove the solution and immediately stop the growth.

6.1.6 Molecular Beam Epitaxy (MBE)

General Conditions: Molecular Beam Epitaxy (MBE) is a highly controlled epitaxial growth technique that operates under ultra-high vacuum (UHV) conditions. The mother phase consists of a low-pressure gaseous molecular flux, which ensures precise deposition of materials. The process takes place in a specialized UHV deposition chamber, where the substrate is heated and positioned to face multiple furnaces. Achieving and maintaining UHV conditions is critical for the success of MBE, as it minimizes contamination and ensures high-quality epitaxial growth.

Key Components:

- **Furnaces (Knudsen Cells):** These are thermally controlled sources that contain the elements and dopants to be deposited. The furnaces generate fluxes of atoms (for metals) or molecules (for non-metals) that are directed toward the substrate. The precise thermal control of these cells allows for accurate regulation of the flux rates.

- **Substrate:** The substrate is heated to a specific temperature suitable for epitaxial growth and is kept in constant rotation to ensure uniform deposition across its surface. This rotation is essential for achieving consistent film thickness and composition.
- **Sliding Shutters:** Each furnace is equipped with a sliding shutter that can abruptly stop the flux of material. This feature allows for rapid changes in the flux composition, enabling the growth of layered structures with sharp interfaces. The shutters are crucial for creating complex multilayered epitaxial structures.
- **Liquid N_2 Panels and Pumps:** Liquid nitrogen-cooled panels and high-performance pumps are used to achieve and maintain the UHV conditions required for MBE. These components work together to minimize the presence of unwanted gases and contaminants in the chamber.
- **Loading Chamber (LOAD-LOCK):** The load-lock chamber is a secondary chamber used to introduce substrates into the system without breaking the UHV conditions of the main deposition chamber. This feature is particularly useful for maintaining the vacuum integrity of the system while allowing for the addition of new substrates or materials.

Importance of UHV:

- **Molecular Flux Conditions:** UHV is essential for maintaining molecular flux conditions, which are characterized by the free movement of atoms and molecules without significant collisions. This ensures that the deposition process is highly controlled and precise.
- **Minimized Contamination:** The UHV environment significantly reduces the incorporation of unwanted species, such as residual gases, that could unintentionally dope or contaminate the material. This is critical for achieving high-purity epitaxial layers.

Advantages and Control:

- **Thickness Control:** MBE is renowned for its ability to control the thickness of deposited layers with atomic-scale precision. This makes it ideal for applications requiring ultrathin films and quantum structures.
- **Chemical Composition Control:** The technique allows for precise control of the chemical composition of the deposited layers, with variations as small as $\pm 1\%$. This level of control is essential for tailoring the electronic and optical properties of the material.
- **High Resolution:** The small molecular flux used in MBE enables the growth of highly complex compounds with nanometer-scale resolution. However, this also means that the process is relatively slow and less productive compared to other deposition techniques.

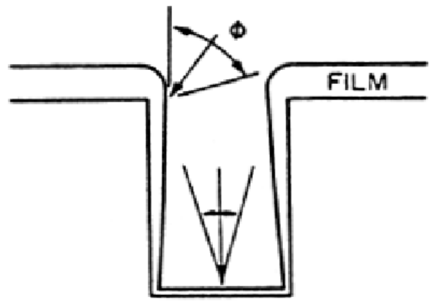
Applications: MBE is widely used in research and development for the fabrication of advanced semiconductor devices, quantum wells, superlattices, and heterostructures. Its ability to produce high-quality materials with precise control over thickness and composition makes it invaluable for cutting-edge technologies.

Reference Atmosphere Conversion: 1 atm = 760 torr (mm Hg) = 101323.2 Pa (N/m^2) = 1.013232 bars = 1013.232 mbar = 14.7 psi

6.2 Chemical Vapor Deposition (CVD)

CVD is the formation of thin films from vapor phase reactants (gaseous precursors). High temperatures and low pressures are common conditions, but not necessary. CVD films include metals (W, Al, Cu), semiconductors (Si, Ge, GaAsP), insulators (Si_3N_4, SiO_2), and silicides ($TiSi_2$).
All CVD involves using an energy source to break reactant gases into reactive species for deposition.
CVD and epitaxy are two families of deposition techniques: VPE is both CVD and Epitaxy. CVD is not strictly monocrystalline.

6.2.1 Acceptance Angle and Conformality (Non-Conformal Deposition)



The growth rate is dependent on the flux density of gas molecules incident on the surface, which is a function of the **acceptance angle**.
Not uniform coverage → NOT CONFORMAL. This occurs because the reagent is adsorbed and reacts without significant surface migration.
Deposition Rate and Incidence Angle: The deposition rate is proportional to the incidence angle of molecules:

- 180° for the upper surface (maximum deposition rate).
- 270° for the upper edge of the trench.
- An angle proportional to $\arctan(w/d)$ for the bottom surface of the trench (where w is the width and d is the depth).

This angular dependence leads to a **reduced thickness** on the side walls of the trench and on its bottom. A non-continuous film (due to **auto-shadowing**) is even possible.
This characteristic is typical of **PVD (Physical Vapor Deposition)** methods.

Step coverage is defined using ratios like Conformity $= b/c$, Sidewall Step Coverage $= b/a$, and Bottom Step Coverage $= d/a$, where a, b, c, d are thickness measurements at specific points. High aspect ratio (h/w) structures are more difficult to fill without forming **voids** or **keyholes**.
Sometimes you want your film to be not conformal <3.

6.2.2 Types of Chemical Vapor Deposition (CVD) Reactors

Chemical Vapor Deposition (CVD) is a foundational technique in semiconductor fabrication, with different reactor types offering various trade-offs in performance, dictated primarily by operating pressure and temperature. The three major types are Atmospheric Pressure CVD (APCVD), Low Pressure CVD (LPCVD), and Plasma Enhanced CVD (PECVD).

Atmospheric Pressure CVD (APCVD)

APCVD operates at high temperatures, typically between 350°C and 1200°C, and near atmospheric pressure (10 to 100 kPa). Its main appeal lies in its operational simplicity, resulting in **very high deposition rates** and high throughput. However, the high pressure environment leads to severe limitations, namely **poor uniformity** and **poor conformality** (step coverage), and the resulting film purity is lower compared to other methods. APCVD is primarily used for less demanding applications, such as depositing low-temperature oxides.

Low Pressure CVD (LPCVD)

LPCVD operates at high temperatures, usually between 550°C and 600°C, but at a much lower pressure of approximately 100 Pa. This low-pressure environment significantly improves the quality of the deposited film. LPCVD is highly valued for its **excellent uniformity**, high film **purity**, and **good conformity** across complex topographies. The primary drawback compared to APCVD is its **lower deposition rate**. It is the workhorse for depositing crucial films like **polysilicon**, **nitride**, and high-quality **oxide** layers.

Plasma Enhanced CVD (PECVD)

PECVD is distinguished by its use of plasma (that is a complex mixture resulting from the ionization and excitation of gas atoms) to supply the energy needed for the chemical reactions, which allows it to operate at much **lower temperatures** (150°C to 450°C). This low-temperature capability is critical for depositing films over substrates that cannot withstand high heat, such as those with pre-deposited metallization. PECVD also offers **good step coverage** (conformality) and a deposition rate that is higher than LPCVD. However, the use of plasma introduces potential drawbacks, including **plasma damage** to the underlying device structures and possible **chemical contamination**. The tool itself is also more complex. PECVD is often used for creating low-temperature oxides and for **passivation** layers.

Collision Processes: In a plasma, several key collision processes occur that are fundamental to its behavior and applications:

- **Ionization**: $e^- + \mathrm{Ar} \rightarrow 2e^- + \mathrm{Ar}^+$ (sustains the plasma).
- **Dissociation**: $e^- + \mathrm{SiH}_4 \rightarrow e^- + \mathrm{SiH}_3^\bullet + \mathrm{H}^\bullet$ (generates highly reactive free radicals).
- **Excitation**: $e^- + \mathrm{Ar} \rightarrow e^- + \mathrm{Ar}^*$ (transfer of energy to higher level).
- **Relaxation**: $\mathrm{Ar}^* \rightarrow Ar + hv$
  - As the excited element/molecule falls back to its original energy level, it must release energy. In this case a photon is released. This gives the plasma its "glow".
  - Color of the glow is characteristic of the gas: $hv = \Delta E$ oxygen plasmas are blue, nitrogen purple. A stable colored glow is the proof that everything is stable.

Plasma Generation Plasma is created by applying voltage to a neutral gas, which accelerates free electrons. As the voltage increases, these electrons gain enough energy to ionize or excite gas atoms through cascading collisions, sustaining the plasma. The visible glow in a plasma is emitted when excited gas atoms return to their ground state. The process involves the following steps:

- **Initial State:** The gas consists of neutral atoms with no applied voltage.
- **Voltage Application:** Free electrons are accelerated, causing elastic collisions with gas atoms.
- **Increased Voltage and Field:**
  - Energetic electrons ionize or excite gas atoms, producing additional free electrons (cascading collisions), which sustain the plasma.
  - Excited gas atoms emit photons as they return to their ground state, creating the characteristic glow (e.g., oxygen plasmas appear blue, nitrogen plasmas purple).
- **Chamber Preparation:** The chamber is evacuated and then filled with the desired gas(es).
- **RF Energy Application:** RF energy (typically 13.56 MHz) is applied to electrodes, switching the electric field more than 13 million times per second.
- **Electron Acceleration:** The applied energy accelerates electrons, increasing their kinetic energy.
- **Ionization:** Electrons collide with neutral gas molecules, forming ions and additional electrons. The alternating electric field causes electrons to oscillate, gaining enough energy to sustain ionization. Gas ions, being much heavier, do not respond to the high-frequency field.
- **Steady State:** A steady-state plasma is achieved when the rates of ionization and recombination balance.

Plasmas are used to force reactions that would not be possible at low temperature (plasma is the energy source generating reactive gases).

- **Advantages:** uses lower temperatures ($150-450°\mathrm{C}$) necessary for near end processing or deposition on T sensitive materials, more degrees of freedom to achieve film properties (good step coverage, density, composition), higher deposition rate than LPCVD.
- **Disadvantages:** plasma damage or chemical contamination typically results.

PECVD Process Control Parameters The Plasma Enhanced Chemical Vapor Deposition (PECVD) process involves several critical parameters that must be carefully controlled to achieve the desired film properties and deposition characteristics. These parameters include:

1. **Temperature:** The substrate temperature plays a crucial role in determining the film's quality, density, and adhesion. Lower temperatures are typically used in PECVD compared to other CVD methods, making it suitable for temperature-sensitive substrates.
2. **Deposition Time:** The duration of the deposition process directly affects the thickness of the deposited film. Longer deposition times result in thicker films, but excessive durations may lead to non-uniformity or other defects.

3. **Gas Composition (Precursors) and Flow Rate:** The choice of precursor gases and their flow rates determine the chemical composition of the deposited film. Precise control of the gas mixture is essential for achieving the desired stoichiometry and film properties.

4. **System Pressure and Reactant Partial Pressure:** The overall pressure in the deposition chamber, as well as the partial pressures of the reactant gases, influence the reaction kinetics and the uniformity of the film. Lower pressures generally improve film quality by reducing contamination.

5. **RF Power, Frequency, and Bias:** The radio frequency (RF) power and frequency control the plasma density and energy, which in turn affect the deposition rate and film properties. The RF bias applied to the substrate can influence the film's conformality and stress.

PECVD Advantages and Disadvantages The PECVD technique offers several advantages and disadvantages, which must be considered when selecting it for specific applications.

Advantages:

- **Lower Processing Temperatures:** PECVD operates at significantly lower temperatures compared to Low Pressure CVD (LPCVD), making it ideal for back-end processing and for substrates that cannot withstand high temperatures, such as those with pre-deposited metallization.
- **High Deposition Rates:** PECVD achieves faster deposition rates compared to LPCVD, which can improve throughput and reduce manufacturing time.
- **Good Step Coverage:** The plasma environment enables better conformality over complex topographies, making PECVD suitable for depositing films on high-aspect-ratio structures.

Disadvantages:

- **Complexity of Equipment:** PECVD systems are more complex to operate and maintain due to the additional components required for plasma generation and control.
- **Lower Film Quality:** The films deposited by PECVD often have lower quality compared to LPCVD films. This is due to potential contamination, hydrogen incorporation, and other impurities introduced during the plasma process.
- **Chemical Byproducts:** The plasma environment generates a "soup" of reactive species, which may include undesirable or harmful byproducts. These byproducts can pose challenges for process control and environmental safety.

6.2.3 Atomic Layer Deposition(ALD)

Atomic Layer Deposition (ALD) is a modified CVD process used to manufacture conformal thin films. The process involves introducing several gases into the process chamber alternately. Each gas reacts in a self-limiting manner at the surface with the previously introduced gas, causing the reaction to stop once the surface is fully saturated. Between these reactions, the chamber is purged with an inert gas, such as nitrogen or argon.
A specific example of an ALD process is the deposition of aluminum oxide, which can be achieved using trimethylaluminum (TMA, C_3H_9Al) and water (H_2O).

ALD Process Steps (Example: Al_2O_3)

1. **First step**: elimination of hydrogen atoms bound to oxygen (e.g., due to environment humidity) at the wafer surface. The methyl groups (CH_3) of TMA react with the hydrogen to form methane (CH_4). The remaining molecules bond with the unsaturated oxygen. If these atoms are saturated, no more TMA molecules can react at the surface.

2. **Second step**: The chamber is purged and subsequent water steam (H_2O) is led into the chamber. Every hydrogen atom of the H_2O molecules reacts with the former deposited surface atoms to form methane, while the hydroxyl anion is bond to the aluminum atoms. This leaves new hydrogen atoms at the surface which can react in a subsequent step with TMA like in the beginning.

ALD Characteristics

- Each cycle, including gas injection and pumping, takes few seconds.
- With ALD, even conformal 3D structures can be deposited very uniformly.
- Insulating films are possible as well as conductive ones, on different substrates (semiconductors, polymers, ...).
- Film thickness can be controlled very precisely by the number of cycles.

6.2.4 Silicon Dioxide (CVD)

CVD silicon dioxide has worst electrical properties with respect to thermally grown oxide, but complementary applications. The reaction is:

$$SiH_4 + O_2 \Rightarrow SiO_2 + 2H_2$$

- **Undoped oxide**: used for separation and insulation among different levels of metals (IMO: InterMetalOxide; where oxidation is not feasible), as a masking layer for implantation and diffusion, and to increase the thickness of thermal oxides.
- **Phosphorous doped oxides (PSG)**: used for separation and insulation among different levels of metals, and as the final passivation layer for devices.
- **P, As or B doped oxides**: occasionally used as diffusion sources.

Silicon Dioxide Deposition Methods Comparison

Method	Temp (°C)	Composition	Thermal Stability	Dielectric Strength (V/cm)	Etch Rate (Å/min)	Step Coverage
PECVD $SiH_4 + O_2$	200	SiO_2(H)	Loses H	5×10^6	400	Non-conformal
LPCVD $SiH_4 + O_2$	450	SiO_2(H)	Densifies	8×10^6	60	Non-conformal
LPCVD TEOS	700	SiO_2(C...)	Stable	10×10^6	30	Conformal
LPCVD $SiCl_2H_2 + N_2O$	900	SiO_2(Cl)	Loses Cl	10×10^6	30	Conformal
Thermal Oxidation	1000	SiO_2	Stable	11×10^6	25	Conformal

Table 6.2: Comparison of Silicon Dioxide Deposition Methods

A lower HF etch rate means better film quality (denser film).

6.2.5 Polycrystalline Silicon ("Poly")

- Used as gate electrode in MOS devices (more affordable than Al, particularly for thin gate oxides). Also used for the fabrication of high resistivity resistors.
- Process: LPCVD process at $600 - 650°C$ →pyrolysis of silane reaction:

$$SiH_4 \rightarrow Si + 2H_2 (600°C)$$

- Can be doped by diffusion, implantation or in-situ doping (doping gases in the reaction mixture). Implantation is the usual choice for its low process temperature.
- **Grain size depends on deposition temperature; hotter deposition leads to larger grain structure.**
- Polysilicon can exhibit columnar or random small grain structures.

6.3 PVD (Physical Vapor Deposition)

Physical Vapor Deposition (PVD) refers to a family of deposition techniques where film growth is achieved through the physical condensation of vaporized material onto a cooler substrate. The term "physical" emphasizes that the bonding and deposition processes are driven by physical phenomena, such as vapor condensation, rather than chemical reactions. This process is analogous to the condensation of water vapor on a cold window during winter.

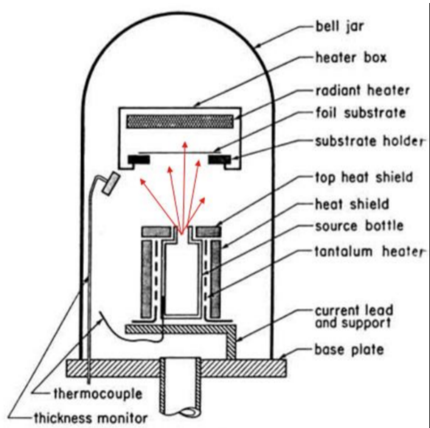
PVD is widely used for depositing thin films of metals, alloys, and other materials. The two primary methods of PVD are:

1. **Evaporation:** Material is vaporized by heating and then condenses on the substrate.
2. **Sputtering:** Material is ejected from a target by ion bombardment and then deposits on the substrate.

PVD is particularly suitable for depositing metal layers or alloys due to its ability to achieve high purity and precise control over film thickness.

6.3.1 PVD-Thermal Evaporation

Thermal evaporation is one of the simplest and most widely used PVD techniques. It involves heating a source material to its evaporation point in a high-vacuum environment, allowing the vaporized atoms to travel to and condense on the substrate.



Key characteristics of thermal evaporation include:

- **High Vacuum Environment:** The process requires a low working pressure (5×10^{-5} Pa $\approx 5 \times 10^{-7}$ Torr) to increase the **mean free path** of the vaporized atoms. This ensures that the deposition is directional and follows a line-of-sight trajectory.
- **Deposition Rate:** The deposition rate is determined by the emitted flux of vaporized material and the geometry of the target and wafer holder.
- **Material Limitations:** Thermal evaporation is limited to materials with relatively low melting points, as high-melting-point materials are difficult to vaporize efficiently.
- **Orientation of Wafers:** Wafers are typically placed upside down to minimize particulate contamination during the process.
- **Shutter Control:** A shutter is used to control the timing of deposition, ensuring precise film thickness.

Conformality

- **Line-of-Sight Deposition:** Thermal evaporation is inherently a line-of-sight process, meaning that the vaporized material travels in straight paths from the source to the substrate. This results in **shadowing effects**, where steps and complex topographies are not uniformly covered.
- **Thickness Variation:** The thickness of the deposited film depends on the angle between the surface and the incoming source flux. Surfaces perpendicular to the flux receive the maximum deposition, while inclined or recessed surfaces receive less material.
- **Comparison with CVD:** Unlike Chemical Vapor Deposition (CVD), which involves rapid surface migration of atoms, PVD relies on a long mean free path and exhibits minimal surface migration. This makes PVD less conformal than CVD.

6.3.2 PVD-Thermal Evaporation Techniques

Thermal evaporation can be achieved using two primary techniques: **Thermal Evaporation** and **E-beam Evaporation**.

Thermal Evaporation

In thermal evaporation, the source material is placed in a **crucible**, which is heated to vaporize the material. The heating current flows through the crucible, producing the vapor. Crucibles are made from materials such as tungsten (W), tantalum (Ta), molybdenum (Mo), or carbon (C) to avoid contamination. Alternatively, non-conductive refractory materials like alumina or boron nitride (BN) can be used as crucibles, heated by a spiral filament.

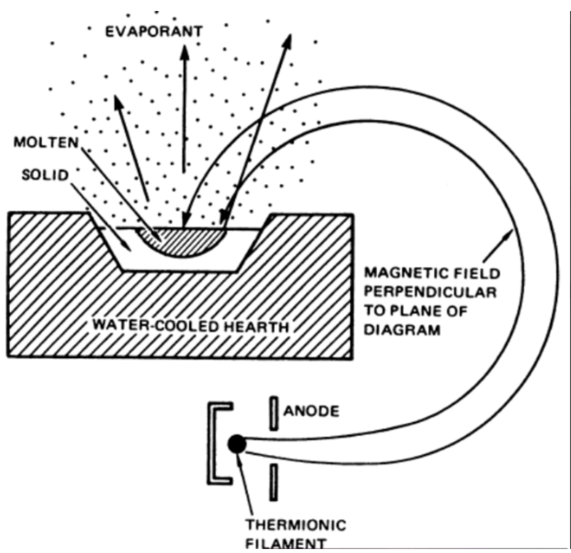
- **High Deposition Rate:** Thermal evaporation can achieve high deposition rates, making it suitable for applications requiring thick films.

- **Material Limitations:** Not all materials can be evaporated using this method. For example, tungsten (W) is nearly impossible to evaporate, while molybdenum (Mo) and platinum (Pt) are very difficult.
- **Rotating Wafer Holder:** For high throughput and uniform thickness, a rotating planetary wafer holder can be used to ensure even deposition across multiple wafers.

Material	Symbol	MP (°C)	S/D	g/cm^3	Temp.(°C) for Given Vap. Press. (Torr)			Evaporation Techniques					Sputter	Comments
									Thermal Sources					
					10^{-8}	10^{-6}	10^{-4}	E-Beam	Boat	Coil	Basket	Crucible		
Aluminum	Al	660		2.70	677	821	1010	Ex	TiB_2,W	W	W	TiB_2-BN, ZrB_2, BN	RF, DC	Alloys and wets. Stranded W is best.
Aluminum Oxide	Al_2O_3	2072		3.97	-	-	1550	Ex	W	-	W	-	RF-R	Sapphire excellent in E-beam; forms smooth, hard films. n = 1.66
Chromium	Cr	1857	S	7.20	837	977	1157	G	**	W	W	VC	RF, DC	Films very adherent. High rates possible.
Copper	Cu	1083		8.92	727	857	1017	Ex	Mo	W	W	Al_2O_3, Mo, Ta	DC, RF	Adhesion poor. Use interlayer (Cr). Evaporates using any source material.
Silicon	Si	1410		2.32	992	1147	1337	F	W, Ta	-	-	BeO, Ta, VC	DC, RF	Alloys with tungsten; use heavy tungsten boat. SiO produced above 4 x 10^{-6} Torr. E-beam best.
Titanium	Ti	1660		4.5	1067	1235	1453	Ex	W	-	-	TiC	DC, RF	Alloys with refractory metals; evolves gas on first heating.

E-beam Evaporation

E-beam evaporation uses an **electron beam** to vaporize the source material. A heated filament emits electrons, which are accelerated and directed by a magnetic field to strike the material in a **liner** (crucible), causing it to evaporate.



Key features of E-beam evaporation include:

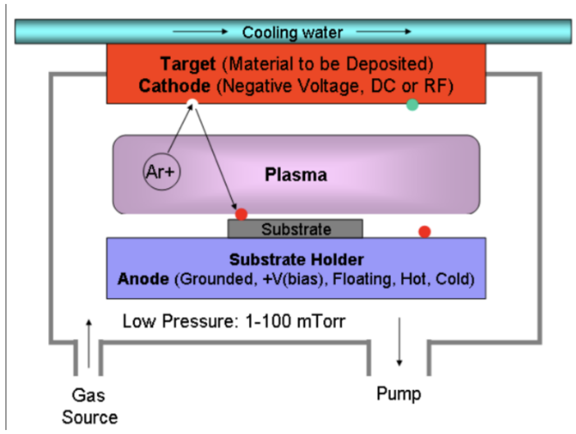
- **Water-Cooled Liners:** Liners are water-cooled to prevent contamination from the liner and filament. This ensures high-purity deposition.
- **Sequential Evaporation:** Multiple materials can be evaporated in sequence without breaking vacuum, allowing for the deposition of multilayer films or alloys.
- **High Deposition Rate:** E-beam evaporation can achieve deposition rates up to 0.5 μm/min, depending on the distance between the source and substrate.
- **Material Versatility:** High-melting-point materials like tungsten (W), molybdenum (Mo), and platinum (Pt) can be evaporated using this method.

Example Application: Ti/Al/Au Ohmic Contacts

- A typical Ohmic contact structure consists of a thin adhesion layer (e.g., $h_{\text{Ti}} = 20$ nm) beneath the metal bulk ($h_{\text{Al}} = 500$ nm) and a passivation layer ($h_{\text{Au}} = 20$ nm).
- Titanium (Ti) provides adhesion, aluminum (Al) offers excellent conductivity, and gold (Au) prevents oxidation.
- The entire process is performed under vacuum to prevent oxidation and contamination.

Uniform Deposition Across Multiple Wafers: To achieve uniform deposition across multiple wafers, a **Planetary Substrate Holder** is used. This holder rotates the wafers in a spherical configuration, ensuring even exposure to the vapor flux. The double rotation mechanism ensures uniform thickness across all wafers.

6.3.3 PVD- Sputtering

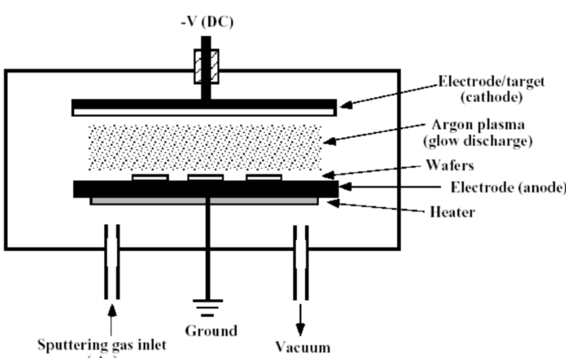


Sputtering is a physical vapor deposition (PVD) technique that relies on ion bombardment to dislodge atoms from a target material, which then travel through a vacuum chamber and condense as a thin film on the substrate. Typically, a plasma (usually Ar^+ ions) is used to bombard the target, causing atoms to be ejected. These atoms then travel in a straight-line trajectory to the substrate, where they deposit and form a thin film.

Key characteristics of sputtering include:

- **Pressure Requirements:** Sputtering requires higher pressures (1-100 mtorr) compared to evaporation, which can increase the risk of contamination.
- **Material Versatility:** Sputtering can deposit a wide range of materials, including metals, alloys, and compounds. It is particularly effective for depositing alloys and compounds because the composition of the target is replicated in the deposited film.
- **Adhesion Quality:** Sputtered films exhibit excellent adhesion to the substrate due to the higher energy of the adatoms compared to evaporation.
- **Cost:** Sputtering is generally more expensive than evaporation due to the higher voltages and more complex equipment required.
- **Plasma Generation:** Glow discharges for sustaining the plasma can be created using either DC voltages or AC voltages (e.g., RF systems).

DC Sputtering



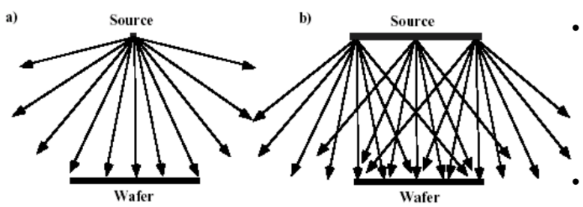
DC sputtering is the simplest form of sputtering. In this method:

- The target material (cathode) is connected to a negative potential, while the substrate (anode) is grounded or connected to a positive potential.
- A high voltage (several hundred volts) is applied between the electrodes, creating an electric field that ionizes the sputtering gas (e.g., Argon).
- Positive ions from the plasma are accelerated toward the negatively charged target, causing atoms to be ejected from the target surface.
- These ejected atoms travel through the vacuum and deposit on the substrate.

Key features:

- **Neutral Atoms:** The sputtered atoms are mostly neutral when they arrive at the substrate.
- **Plasma Maintenance:** The discharge is sustained as accelerated electrons continuously ionize new ions by colliding with the sputtering gas.
- **Limitations:** DC sputtering is limited to conductive target materials. Non-conductive materials cannot be sputtered effectively using DC sputtering due to charge buildup on the target surface.

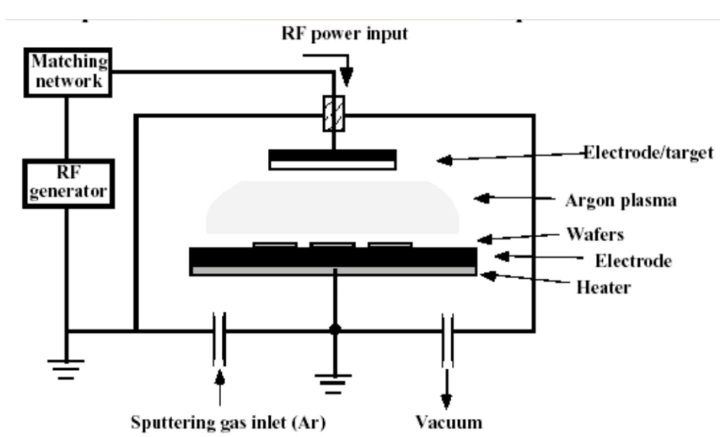
Sputtering Conformality



The conformality (step coverage) of sputtered films is influenced by the geometry of the target and the line-of-sight nature of the deposition process:

- **Line-of-Sight Deposition:** Sputtered atoms travel in straight paths from the target to the substrate, leading to shadowing effects in high-aspect-ratio structures.
- **Wide Arrival Angles:** Unlike evaporation, where atoms are emitted from a single point source, sputtering involves a large target area, resulting in a wider range of arrival angles. This makes sputtering more conformal than evaporation.
- **Sticking Factor:** The sticking factor of incoming species is close to 1, meaning that atoms adhere to the surface upon impact without significant surface diffusion. This limits the ability to achieve uniform coverage in complex geometries.
- **Improving Conformality:** To improve step coverage, techniques such as heating the substrate to enhance surface diffusion or using chemical vapor deposition (CVD) may be employed.

RF Sputtering



RF sputtering is used to deposit dielectric or non-conductive materials, which cannot be sputtered using DC sputtering. In RF sputtering:

- An RF power source (typically 13.56 MHz) is capacitively coupled to the target.
- The alternating electric field prevents charge buildup on the target surface by allowing electrons to neutralize the positive charge during each half-cycle.
- At frequencies above 50 kHz, heavy ions cannot follow the polarity switching, while electrons can, ensuring a stable plasma.

Key advantages:

- **Dielectric Materials:** RF sputtering is ideal for depositing insulating materials such as SiO_2 and Al_2O_3.
- **Plasma Stability:** The alternating electric field ensures a stable plasma, even for non-conductive targets.

Magnetron Sputtering

Magnetron sputtering enhances the efficiency of the sputtering process by using magnetic fields to confine electrons near the target surface:

- Magnetic fields are applied perpendicular to the electric field, forcing electrons into helical paths.
- This increases the probability of ionizing collisions, leading to a higher ionization rate and increased sputtering efficiency.
- Magnetron sputtering allows for higher deposition rates at lower pressures, reducing contamination risks.

Key considerations:

- **Higher Deposition Rates:** Deposition rates can be 10-100 times higher than standard sputtering.
- **Lower Pressure:** Operating at lower pressures improves film quality.
- **Target Erosion:** The magnetic field can cause uneven target erosion, requiring careful design of the target geometry.

Reactive Sputtering

Reactive sputtering is used to deposit compound films by introducing a reactive gas into the sputtering chamber:

- A metallic or semiconductor target is used, and a reactive gas (e.g., O_2, N_2, or NH_3) is introduced into the plasma.
- The reactive gas reacts with the sputtered atoms to form a compound film (e.g., SiO_2, SiN_x, or Al_2O_3) on the substrate.

Types of sputtering:

- **Non-Reactive Sputtering:** The sputtering gas (e.g., Ar) does not participate in the film formation.
- **Reactive Sputtering:** The sputtering gas reacts with the target material to form a compound film.

Ionized Sputtering (HDP Sputtering)

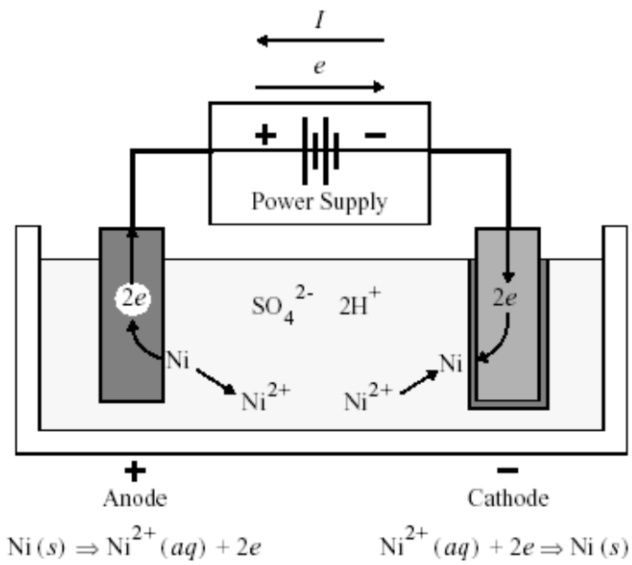
Ionized sputtering, also known as High-Density Plasma (HDP) sputtering, involves ionizing the sputtered atoms themselves:

- An RF coil around the plasma induces additional collisions, ionizing the sputtered atoms.
- The ionized atoms are directed toward the substrate using an electric field, ensuring a narrow distribution of arrival angles.
- This technique is particularly useful for filling deep contact holes or vias with metal.

Example application:

- Filling vias (vertical interconnections between metal layers) with metal to achieve low electrical resistivity and high-quality interconnects.

6.4 Electroplating - Electrodeposition-Galvanic Deposition



Electrodeposition, also known as electroplating or galvanic deposition, is one of the oldest and most widely used deposition technologies. It involves the deposition of a metal layer onto a conductive substrate through the use of an electrochemical process. This technique has been utilized for centuries and remains a cornerstone in various industrial and technological applications.

The basic setup for electrodeposition consists of:

- A bath containing an electrolyte solution with dissolved metal ions.
- A cathode, which is the electrode or substrate where the metal deposition occurs.
- An anode, which serves as the counter electrode.
- An external power supply to drive the electrochemical reactions.

When an electric current is applied, cations (positively charged ions) in the electrolyte migrate toward the cathode, where they are reduced and deposited as a solid metal layer. Simultaneously, anions (negatively charged ions) move toward the anode, where oxidation reactions occur.

6.4.1 Processes

Electrodeposition involves two primary electrochemical processes:

- **Cathodic Processes:** At the cathode, metal ions in the electrolyte gain electrons (reduction) and are deposited as a solid metal layer. For example:
$$Ni^{2+}(aq) + 2e^- \rightarrow Ni(s)$$
Here, nickel ions (Ni^{2+}) in the aqueous solution are reduced to form solid nickel (Ni) on the cathode surface.
- **Anodic Processes:** At the anode, the metal is oxidized, releasing metal ions into the electrolyte. For example:
$$Ni(s) \rightarrow Ni^{2+}(aq) + 2e^-$$
In this case, solid nickel from the anode is oxidized to form nickel ions (Ni^{2+}), which dissolve into the electrolyte.

6.4.2 Characteristics and Applications

Electrodeposition offers several unique characteristics and advantages, making it suitable for a wide range of applications:

- **Versatility:** A variety of metals can be deposited, including gold (Au), nickel (Ni), copper (Cu), and others. However, not all metals are suitable for electrodeposition due to factors such as cost, toxicity, or chemical reactivity.
- **High Deposition Rate:** Electrodeposition is a fast process, with deposition rates typically in the range of micrometers per minute (μm/min).

- **Thick Layers:** It is possible to deposit layers with thicknesses exceeding 100 μm, making it ideal for applications requiring substantial material buildup.
- **Seed Layer Requirement:** A conductive seed layer is essential for the process, as the entire electrochemical circuit must be electrically conductive. The seed layer provides the initial surface for metal deposition.
- **Electrical Connectivity:** All parts of the substrate must be electrically connected to ensure uniform deposition. Isolated segments cannot be plated unless they are connected to the cathode.
- **3D Structures:** Electrodeposition can be used to create complex three-dimensional structures, as the deposited material can conform to the shape of the substrate.
- **Patterned Deposition:** The process relies on a patterned seed layer, which can be defined using photolithography or other masking techniques. The typical steps include:
  1. **Deposit Seed Layer:** A thin conductive layer (e.g., copper) is deposited on the substrate using a technique such as PVD (Physical Vapor Deposition).
  2. **Define Seed Layer and Contact:** The seed layer is patterned to define the areas where deposition will occur.
  3. **Electroplate Material:** The desired metal is deposited onto the patterned seed layer through the electrodeposition process.

6.4.3 Key Considerations

For successful electrodeposition, the following conditions must be met:

- The surface of the substrate must be **electrically conductive.** This is a mandatory requirement to allow the flow of current and the reduction of metal ions at the cathode.
- The conductive areas of the substrate must be **electrically connected.** Isolated regions will not receive deposition unless they are part of the electrical circuit.
- The electrolyte composition, temperature, and current density must be carefully controlled to achieve the desired deposition rate, uniformity, and material properties.

Electrodeposition is widely used in industries such as electronics, automotive, and jewelry for applications ranging from the fabrication of microelectronic interconnects to the decorative plating of consumer goods.

Chapter 7

Doping-Implantation

7.1 Silicon Doping

Doping is the process of introducing specific atoms, called dopants, into a semiconductor material to modify its electrical properties. This process is the cornerstone of modern semiconductor technology, enabling precise control over the local electronic properties of a semiconductor crystal. By altering the doping concentration, the resistivity of silicon can be changed by as much as 8 orders of magnitude.

7.1.1 Doping Levels

Doping levels are categorized as follows:

- **N^- or P^- (Lightly Doped):** 10^{14} cm^{-3} $< N_D$ or $N_A < 10^{16}$ cm^{-3}. Results in relatively high resistivity.
- **N or P (Moderately Doped):** 10^{16} cm^{-3} $< N_D$ or $N_A < 10^{18}$ cm^{-3}. Provides a balance between conductivity and resistivity.
- **N^+ or P^+ (Heavily Doped):** 10^{18} cm^{-3} $< N_D$ or $N_A < 10^{20}$ cm^{-3}. Results in significantly enhanced conductivity.
- **N^{++} or P^{++} (Degenerately Doped):** N_D or $N_A > 10^{20}$ cm^{-3}. The material behaves almost like a metal.

7.1.2 Dopant Species: P-Type and N-Type

Dopants are specific atoms introduced into a semiconductor to create n-type or p-type regions. For dopants to be electrically active, they must occupy substitutional sites in the crystal lattice. Typical doping levels range from 10^{15} to 10^{20} atoms/cm^3, compared to silicon's atomic density of 5.2×10^{22} atoms/cm^3.

- **Donor Atoms (n-type):** Donate free electrons to the conduction band. Examples:
  - Phosphorus (P)
  - Arsenic (As)
  - Antimony (Sb)
- **Acceptor Atoms (p-type):** Accept electrons, creating holes in the valence band. Examples:
  - Boron (B)
  - Gallium (Ga)
  - Aluminum (Al)
- **Preferred Dopants:** Phosphorus, arsenic, and boron are the most commonly used dopants due to their high solubility in silicon, which exceeds $5 \cdot 10^{20}$ cm^{-3}.
- **Unwanted Dopants:** Certain elements, such as gold (Au), iron (Fe), copper (Cu), and nickel (Ni), are considered contaminants in semiconductor manufacturing. These elements can:
  - Degrade the performance of solid-state electronic devices.
  - Diffuse rapidly through the silicon lattice, making them difficult to control.

Doping is a critical process not only for altering the electrical properties of semiconductors but also for influencing their physical and chemical characteristics.

7.1.3 Piezoresistive Pressure Sensor

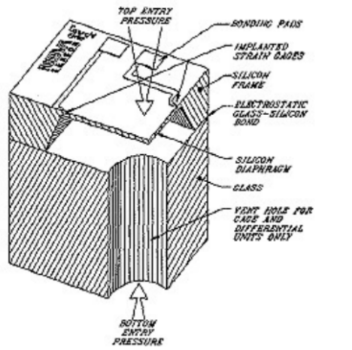


Figure 1: Simplified piezoresistive pressure sensor design (quarter of the structure shown).

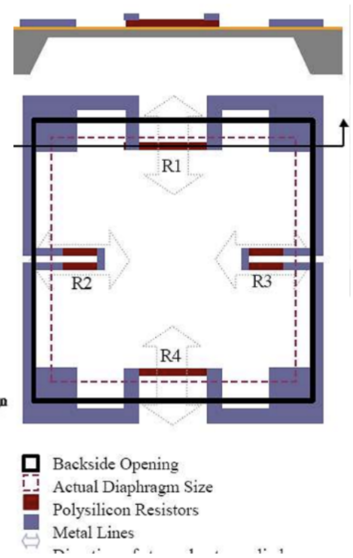


Fig. 1. Piezoresistive Pressure Sensor

A piezoresistive pressure sensor is a device that converts pressure into an electrical signal. At the core of the sensor is a thin silicon membrane that deforms under differential pressure. This deformation alters the resistance of a piezoresistor, which is a doped silicon region. The change in resistance is proportional to the applied pressure, allowing the sensor to generate a voltage signal that corresponds to the pressure differential.

Doped silicon is used as the piezoresistor because its electrical resistance changes predictably under mechanical stress. This property, known as the piezoresistive effect, is a direct result of the interaction between the mechanical and electrical properties of the doped material. By carefully controlling the doping concentration and distribution, the sensor's sensitivity and performance can be optimized.

7.1.4 Doping Parameters

The characterization of doping processes involves defining key parameters that allow for the prediction and control of the dopant distribution within the semiconductor. These parameters are essential for ensuring the desired electrical properties of the device.

1. **Dopant Distribution:** The dopant concentration as a function of depth, denoted as $N(x)$ or $C(x)$ (depending on the literature), is measured in [atoms/cm^3].

2. **Total Dose:** The total dose, $Q(t)$, is expressed in [atoms/cm^2]. It represents the cumulative number of dopant atoms introduced into the material and is a critical parameter for controlling the electrical characteristics of the doped region.

$$Q(t) = \int_0^\infty N(x,t) \cdot dx$$

3. **Junction Depth:** The junction depth, x_j, is the depth at which the dopant concentration equals the background doping concentration, N_B (or C_B). Mathematically, this is defined as the point where:

$$N(x,t) = N_B$$

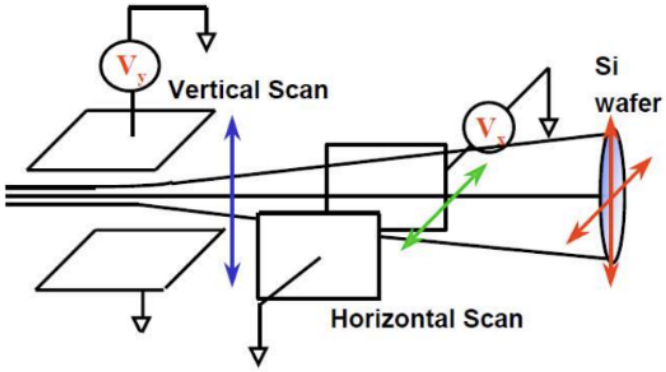
   The junction depth is typically measured in nanometers (nm) or micrometers (μm) and is a key parameter for determining the electrical behavior of the device.

7.2 Ion Implantation

- **Purpose:** The primary goal of ion implantation is to introduce doping atoms or impurities into substrates such as silicon, GaAs, InP, SiC, or diamond. This process is fundamental in semiconductor device fabrication, as it allows precise control over the electrical properties of the material.

- **Process Overview:** An ion implanter generates an ionic beam of controlled species and energy. This beam is directed toward the substrate, scanning its surface to introduce the desired quantity of dopant atoms. The process is highly controlled to ensure uniformity and precision.

In essence, ion implantation is akin to "spraying" dopant atoms onto the substrate, where they penetrate and modify the material's properties in non-masked areas.

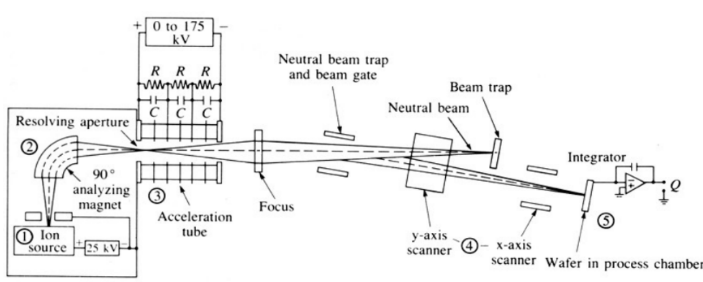


The concept of ion implantation was first proposed by Shockley in 1954. However, it was not until the late 1970s that the technique was widely adopted for mass production in the semiconductor industry.

7.2.1 Key Features of Ion Implantation

- **Energy Control:** Ions are electrostatically accelerated using high potentials (ranging from 3 kV to 3 MV). This allows precise control over the velocity and energy of the ions, enabling them to penetrate the substrate to specific depths.
- **Low-Temperature Process:** The process is performed at low temperatures (room temperature to approximately $200°C$), making it compatible with photoresist masks, which are cost-effective and easy to use.
- **Instant Control:** The process offers instant-on and instant-off control, allowing for precise adjustments during implantation.
- **Dose Control:** Implanting current and charge can be precisely controlled, enabling accurate monitoring of the implanted dose (Q_0) in real time.
- **Penetration Capability:** High-energy implants can penetrate thin films of materials, making the technique versatile for various applications.
- **Buried Profiles:** The peak concentration of implanted dopants is always located below the surface, creating buried profiles that are advantageous for certain device structures.
- **Mass Separation:** The implanter's mass separator allows it to handle multiple dopant species (e.g., B, P, BF_2, As) in a single tool.
- **Post-Implantation Annealing:** A critical step following implantation is annealing, which serves two purposes:
  - **Crystal Damage Repair:** The implantation process causes damage to the crystal lattice, which must be repaired to restore the material's properties.
  - **Dopant Activation:** Annealing moves dopant atoms from interstitial sites to substitutional sites in the lattice, enabling them to contribute to the material's electrical properties.

7.2.2 Implanter Components



An ion implanter consists of several key components, each playing a critical role in the implantation process:

- **Ion Source:** Generates dopant ions through electron cascading from a tungsten filament. A magnetic field forces electrons into spiral paths, enhancing ionization efficiency.
- **Extraction Region:** Extracts ions from the source and focuses them toward the mass analyzer.
- **Mass Analyzer:** Filters out undesired ions by tuning magnetic fields to allow only ions with a specific mass-to-charge ratio (m/q) to pass through.
- **Accelerator:** Uses an electric field to accelerate ions toward the substrate. The energy imparted to the ions is determined by the applied voltage ($E = qV$).
- **Vacuum System:** Maintains a clean vacuum environment to minimize ion loss due to neutralization or scattering.
- **Deflector Plate:** Bends the ion beam to remove neutralized species, which are trapped by a beam trap.
- **Scanning System:** Uses x and y-axis deflection plates to scan the beam across the wafer, ensuring uniform implantation of the desired dose.

Why Eliminate Neutral Species? Neutral species are uncountable because they do not carry charge. Since each ion carries a single charge, the number of ions can be determined by measuring the current. This ensures precise dose control.

7.2.3 Dose Calculation

The dose, which represents the total number of ions implanted per unit area, is a critical parameter in ion implantation. It is calculated using the following formula:

$$Q = \int_0^T \frac{I \cdot dt}{n \cdot q \cdot A}$$

where:

- I: The beam current, which indicates the number of ions per second in the beam.
- T: The total implantation time, representing the duration over which the ions are implanted.
- n: The number of charges per ion, which is typically 1 for singly charged ions.
- q: The elementary charge, approximately 1.602×10^{-19} C.
- A: The area of the substrate being implanted, measured in cm^2 or μm^2.

This formula ensures precise control over the number of dopant ions introduced into the substrate, which is essential for achieving the desired electrical properties.

7.2.4 Gaussian Approximation of Dopant Profiles

Ion implantation is inherently a stochastic process, leading to dopant distributions that can be approximated by a Gaussian profile. This approximation is widely used due to its simplicity and accuracy in describing the distribution of dopants within the substrate.

Key parameters of the Gaussian profile include:

- x: The depth from the wafer surface, typically measured in micrometers (μm).
- $N(x)$ or $C(x)$: The dopant concentration at a specific depth x, expressed in atoms/cm^3.
- N_p or C_p: The peak dopant concentration, which occurs at the depth of the projected range (R_p).
- Q: The total dose, representing the total number of implanted dopant ions per unit area, measured in atoms/μm^2.
- R_p: The projected range, which is the depth at which the dopant concentration is highest or the average depth of the dopants.
- ΔR_p: The vertical straggle, representing the standard deviation of the dopant concentration in the vertical direction.
- $\Delta R_\perp$: The lateral straggle, representing the lateral spread of the dopants due to scattering.
- Skewness: A measure of the asymmetry of the Gaussian distribution, which can arise from non-ideal implantation conditions.

The Gaussian distribution is mathematically expressed as:

$$N(x) = N_p \cdot \exp\left(-\frac{(x - R_p)^2}{2\Delta R_p^2}\right) = \frac{Q}{\sqrt{2\pi}\Delta R_p} \cdot \exp\left(-\frac{(x - R_p)^2}{2\Delta R_p^2}\right)$$

This equation describes how the dopant concentration varies with depth, with the highest concentration occurring at R_p and tapering off symmetrically on either side.

Dose and Range Calculations

The following calculations are essential for characterizing the dopant distribution:

- **Dose:** The total dose is the integral of the dopant concentration over the entire depth:

  $$Q = \int_0^\infty N(x) \cdot dx$$

  This represents the total number of dopant atoms implanted per unit area.

- **Projected Range:** The projected range is the average depth of the implanted dopants:

  $$R_p = \frac{\sum_i x_i}{N}$$

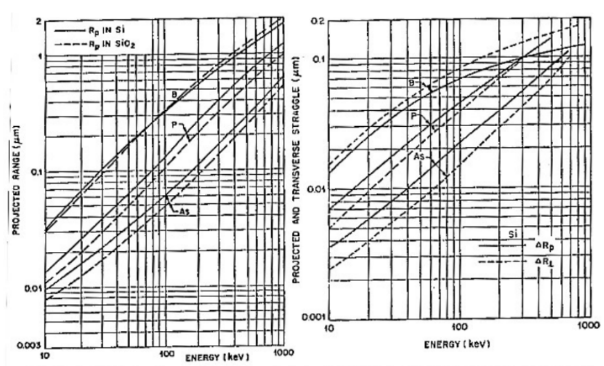
  where x_i represents the depth of individual dopant atoms, and N is the total number of dopant atoms.

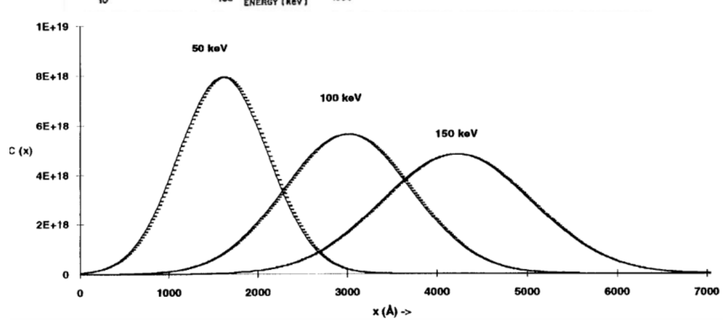
- **Straggle:** The vertical straggle quantifies the spread of the dopant distribution around the projected range:

  $$\Delta R_p = \sigma_p = \sqrt{\frac{\sum_i (x_i - R_p)^2}{N}}$$

  This parameter is critical for understanding the uniformity of the dopant distribution.

Projected Range and Straggle





Key Observations:

- **Heavy Ions:** Heavy ions, such as arsenic (As), result in shallower implantation depths and more vertically concentrated Gaussian profiles. This is due to their higher mass, which reduces their penetration depth.

- **Light Ions:** Light ions, such as boron (B), penetrate deeper into the substrate and exhibit wider distributions. Their lower mass allows them to travel further before coming to rest.

These observations are crucial for selecting the appropriate dopant species and implantation parameters for specific device requirements.

7.3 Implantation Control Parameters

- **Ion Energy:**
  - Controls the depth and shape of the implanted dopant profile. Higher ion energy results in deeper penetration of ions into the substrate, while lower energy leads to shallower profiles.
  - Affects the amount of crystal damage created during implantation. Higher energy ions cause more significant lattice disruption, which may require more extensive annealing to repair.
- **Dose:**
  - Determines the total number of dopant atoms introduced per unit area. The dose directly controls the dopant concentration, which in turn influences many device parameters such as threshold voltage and sheet resistance.
- **Ion Species:**
  - Determines the type of doping (n-type or p-type) based on the choice of dopant atoms. For example, boron is used for p-type doping, while phosphorus or arsenic is used for n-type doping.
  - Impacts the depth and shape of the dopant profile due to differences in atomic mass and scattering behavior. Lighter ions like boron penetrate deeper, while heavier ions like arsenic create shallower profiles.
- **Mask Shape (Layout) and Thickness:**
  - Defines the in-plane geometry of the implanted regions. The mask layout determines where the dopants are introduced and where they are blocked.
  - The thickness of the mask material must be sufficient to prevent ion penetration in masked areas, ensuring precise patterning.
- **Annealing Technique:**
  - The choice of annealing method, along with the time and temperature parameters, is critical for activating the dopants and repairing crystal damage caused by implantation.
  - Annealing redistributes dopants to achieve the desired profile and ensures that dopant atoms occupy substitutional lattice sites, making them electrically active.

7.3.1 Junction Depth (As Implanted)

The junction depth is the depth at which the concentration of implanted dopant atoms equals the substrate's background doping concentration. This parameter is critical for defining the electrical characteristics of the device.

$$C(x_j) = C_b \quad \rightarrow \quad C_b = \text{background doping concentration}$$

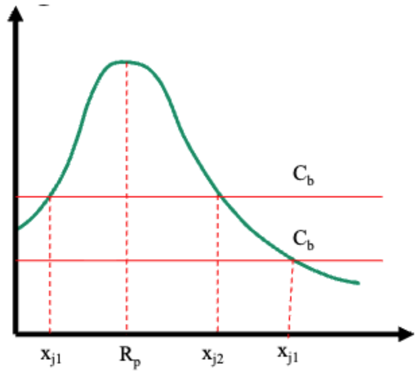
The junction depth, X_j, can be calculated using the following formula:

$$X_j = R_p \pm \sqrt{2} \cdot \Delta R_p \cdot \sqrt{\ln\left(\frac{C_p}{C_b}\right)}$$

Where:

- R_p: Projected range, representing the average depth of the implanted dopants.
- ΔR_p: Standard deviation of the dopant distribution (vertical straggle).
- C_p: Peak dopant concentration, given by:
$$C_p = \frac{Q_0}{\sqrt{2\pi} \cdot \Delta R_p}$$
- Q_0: Total implanted dose.
- C_b: Background doping concentration of the substrate.

Note: The values of R_p and ΔR_p are tabulated for various dopant species and substrate materials, making it easier to calculate the junction depth for specific implantation conditions.



7.4 Annealing - Activation

Annealing is a critical thermal process used to repair damage caused by ion implantation and to activate dopants, enabling them to contribute to the electrical properties of the semiconductor. The process involves heating the substrate to specific temperatures to achieve the desired effects. Below are the key effects and classifications of annealing:

7.4.1 Effects of Annealing

- **Electric Activation:** After ion implantation, dopant atoms are randomly distributed within the crystalline lattice, often occupying interstitial or other non-substitutional positions. To activate their dopant properties, these atoms must be moved to substitutional lattice sites, where they can effectively donate or accept charge carriers.
- **Reorganization of Atomic Scale Order:** Ion implantation disrupts the crystalline structure of the substrate, creating defects and disorder. Annealing helps restore the atomic-scale order of the lattice, repairing damage and improving the material's structural integrity.
- **Side Effect - Dopant and Contaminant Diffusion:** A side effect of annealing is the diffusion of dopants and contaminants within the substrate. This diffusion can alter the dopant profile, potentially impacting the device's performance. The extent of diffusion depends on the annealing temperature and duration.

7.4.2 Classification of Thermal Treatments

Annealing processes are classified based on the thermal equilibrium and heat transfer mechanisms involved. The four main types of thermal treatments are:

Isothermal Annealing

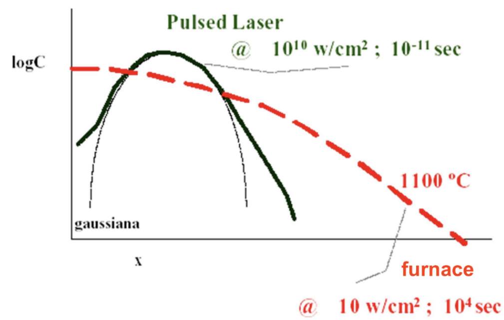
In isothermal annealing, the entire substrate maintains thermal equilibrium at every instant. Heat transfer is slow relative to the thermal constants of the substrate material. This category includes:

- **Furnace-Based Annealing:**
  - Heat transfer is dominated by conduction through the surrounding gas environment.
  - The process typically occurs on a time scale of minutes to hours.
  - Due to the extended duration, dopant redistribution is more pronounced, leading to wider dopant profiles.
- **Rapid Thermal Processing (RTP) or Rapid Thermal Annealing (RTA):**
  - Heat transfer is dominated by electromagnetic radiation, typically from high-intensity lamps.
  - The thermal gradient is rapid, with heating rates of approximately 150°C/s.
  - The process occurs on a time scale of seconds, limiting dopant redistribution and preserving sharper dopant profiles.

Adiabatic Annealing

In adiabatic annealing, thermal equilibrium is not maintained. Heat transfer occurs faster than the substrate's ability to dissipate energy. This category includes:

- **E-Beam Annealing:**
  - Heat transfer is achieved through radiation from an electron beam.
  - The process occurs on a time scale of a few seconds to milliseconds.
- **Laser Annealing:**
  - Heat transfer is achieved through radiation from a laser.
  - The process occurs on a time scale of milliseconds to nanoseconds.
  - Laser annealing is particularly useful for achieving localized heating with minimal impact on surrounding areas.



7.4.3 Comparison of Annealing Techniques

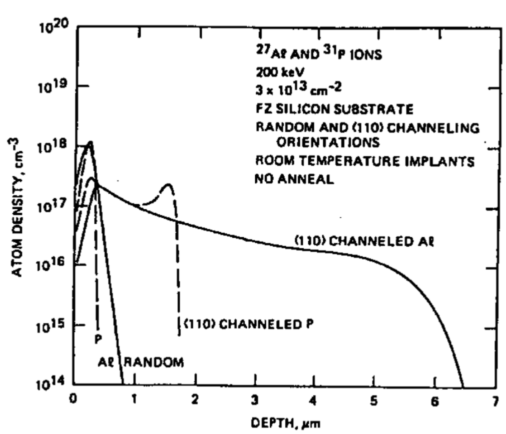
The choice of annealing technique depends on the specific requirements of the application, such as the desired dopant profile, the extent of lattice damage, and the thermal budget of the process. Furnace annealing is suitable for processes requiring uniform heating over extended durations, while RTP and laser annealing are preferred for applications requiring rapid, localized heating with minimal dopant diffusion.

7.5 Channeling Ions Through Lattice

In a monocrystalline material, there is a significant probability that the trajectory of implanted ions aligns with crystallographic planes of low Miller indices (planes with larger interatomic spacing). These planes act as channels where the ions are less likely to undergo nuclear scattering, and the primary energy loss mechanism becomes electronic interaction.

Under these conditions, the interatomic potential itself contributes to keeping the ion aligned with the channel, a phenomenon known as **channeling**.

As a result of channeling, the distance an ion travels before stopping can be significantly larger than in the case of an amorphous layer. This leads to an approximately 2× increase in the implant depth. The dopant profile deviates from a standard Gaussian distribution, often exhibiting an exponential tail due to the extended penetration of ions.



7.5.1 Minimizing Channeling

Channeling can lead to undesirable variations in the dopant profile, which may affect the performance and reliability of semiconductor devices. Several techniques can be employed to minimize channeling effects:

- **Implant into a damaged target:** Pre-damage the surface layers of the substrate using a pre-implantation step with neutral ions (e.g., Ar^+) at low energy and high dose. The induced damage disrupts the crystalline structure, reducing the likelihood of channeling.
- **Implant through an amorphous layer:** Introduce an amorphous layer, such as silicon oxide (SiO_2), on the substrate surface. The amorphous layer prevents ions from aligning with the crystalline channels.
  - While effective, this method is challenging to integrate into a standard integrated circuit (I.C.) process flow due to additional complexity and potential compatibility issues.
- **Implant at an angle not aligned with crystal channels:** Adjust the implantation angle to avoid alignment with the crystal channels. This can be achieved by introducing a **twist** (rotation about the wafer's normal axis) and/or a **tilt** (rotation about an axis perpendicular to the wafer's normal axis).

Twist and/or Tilt:

- A tilt angle of approximately $7° - 8°$ has been found to be effective in minimizing channeling effects for a wide range of implantation conditions.
- However, **shadowing** effects must be carefully considered when setting the tilt angle. Shadowing occurs when the mask edges block the ion beam, leading to incomplete implantation in certain areas.
- Adjusting the mask size or position to mitigate shadowing is generally not a viable solution, as it may cause misalignment issues with horizontal and vertical features in the device layout.

7.6 Sheet (or Square) Resistance

The concept of sheet resistance is widely used in the characterization of thin films, particularly in semiconductor and microelectronics applications. It provides a convenient way to describe the resistance of a thin film without requiring knowledge of its specific dimensions.

$$R = \frac{\rho \cdot L}{A} = (\frac{\rho}{t})(\frac{L}{W}) = R_s(\frac{L}{W}) \rightarrow R_s = (\frac{\rho}{t}) = R_{\square}$$

Where:

- R: Resistance of the film, measured in Ohms [Ω].
- ρ: Resistivity of the material, measured in Ohm-meters [$\Omega \cdot$ m].
- L: Length of the resistor, measured in meters [m].
- W: Width of the resistor, measured in meters [m].
- t: Thickness of the film, measured in meters [m].
- R_s: Sheet resistance, measured in Ohms per square [$\Omega/\square$].
  - The term "square" is unitless and refers to the aspect ratio of the resistor. It simplifies the resistance calculation for films of uniform thickness.

7.6.1 Sheet Resistance of an Implant

The sheet resistance of an implanted layer is determined by integrating the conductivity over the depth of the layer. The formula is given as:

$$R_s = (\frac{\rho}{t}) = \frac{1}{\int_{x_\beta}^{x_j} \sigma(x) \cdot dx}$$

Where:

- $\sigma(x)$: Conductivity at depth x, measured in Siemens per meter [S/m].

- x_β: Starting depth of the conductive layer.
- x_j: Junction depth, where the dopant concentration equals the background doping concentration.

For n-type materials:

$$\sigma(x) = q \cdot \mu(x) \cdot n(x)$$

For p-type materials:

$$\sigma(x) = q \cdot \mu(x) \cdot p(x)$$

Where:

- q: Elementary charge (1.602×10^{-19} C).
- $\mu(x)$: Mobility of charge carriers at depth x, measured in $\mathrm{m^2/V \cdot s}$.
- $n(x)$ or $p(x)$: Electron or hole concentration at depth x, measured in $\mathrm{cm^{-3}}$.

The sheet resistance can be expressed as:

$$R_s = \frac{1}{\int_{x_{j1}}^{x_{j2}} q \cdot \mu(x) \cdot (N_{A,D}(x) - N_B) \cdot dx}$$

Where:

- $N_{A,D}(x)$: Acceptor or donor concentration at depth x.
- N_B: Background doping concentration.

Since the mobility $\mu(x)$ is concentration-dependent, numerical methods are typically required to calculate the sheet resistance.

7.7 Implanting Through a Mask

When ion implantation is performed through a mask, the fraction of dopants transmitted through the mask is given by:

$$\text{Fraction Transmitted} = \frac{\int_d^\infty N(x) \cdot dx}{\int_0^\infty N(x) \cdot dx}$$

Where:

- $N(x)$: Dopant concentration at depth x, modeled as a Gaussian distribution.

The Gaussian distribution is expressed as:

$$N(x) = N_p \cdot e^{-\left(\frac{x-R_p}{\Delta R_p \sqrt{2}}\right)^2}$$

Using the error function (erfc), the integral can be simplified:

$$\int_z^\infty e^{-a \cdot u^2} \cdot du = \frac{1}{2}\sqrt{\frac{\pi}{a}}\mathrm{erfc}(z\sqrt{a})$$

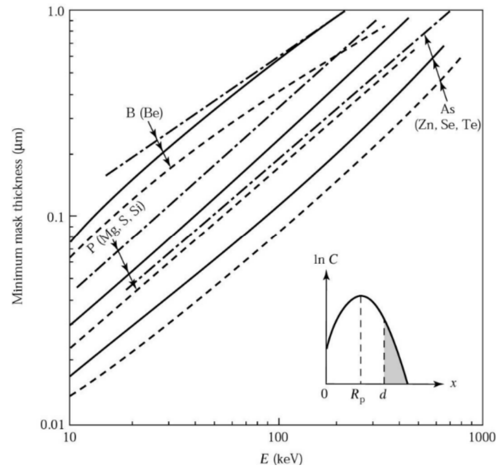
Substituting into the formula:

$$\int_z^\infty N_p \cdot e^{-\left(\frac{x-R_p}{\Delta R_p \sqrt{2}}\right)^2} \cdot dx = N_p \Delta R_p \sqrt{\frac{\pi}{2}}\mathrm{erfc}\left(\frac{z - R_p}{\Delta R_p \sqrt{2}}\right)$$

Resulting in:

$$\text{Fraction Transmitted} = \frac{1}{2}\mathrm{erfc}\left(\frac{d - R_p}{\Delta R_p \sqrt{2}}\right)$$

7.7.1 Masked Implantation

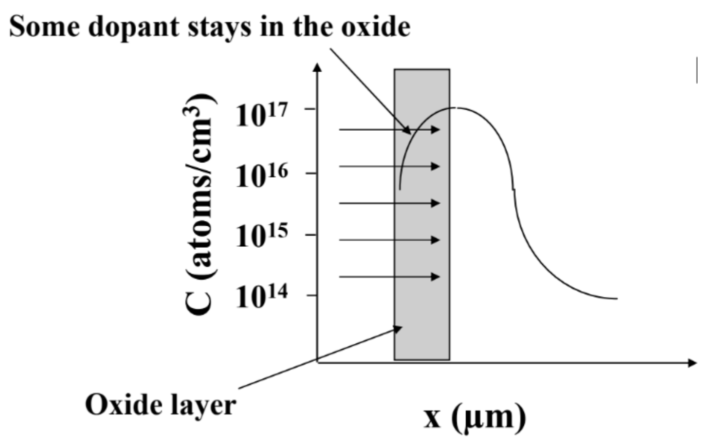


The figure illustrates the minimum thickness of SiO_2 (solid line), Si_3N_4 (dashed line), and photoresist (dotted line) required to achieve a masking effectiveness of 99.99%. The curves represent the effectiveness of these materials in blocking dopants during ion implantation.

7.8 Screen Oxides

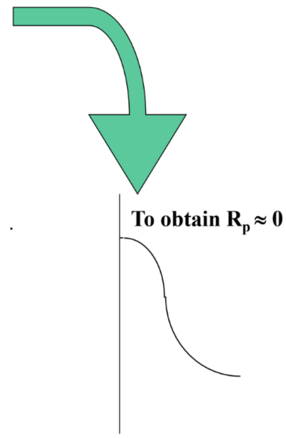
7.8.1 Implantation

Screen oxides are thin oxide layers used during ion implantation to protect the substrate and control the dopant profile. The oxide layer reduces channeling effects and minimizes surface damage.



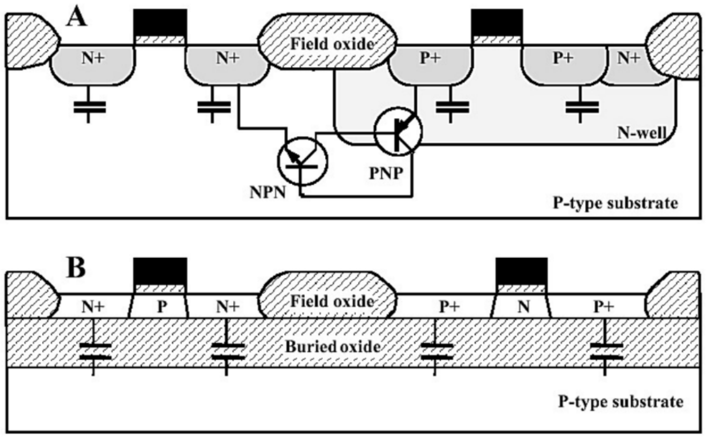
7.8.2 Oxide Etching

After implantation, the screen oxide is removed using an etching process to expose the substrate for subsequent processing steps.



7.9 Silicon on Insulators (SOI) Wafers

SOI wafers are used to reduce parasitic capacitance and improve device performance. They consist of a thin silicon layer (device layer) on top of an insulating layer, typically SiO_2.



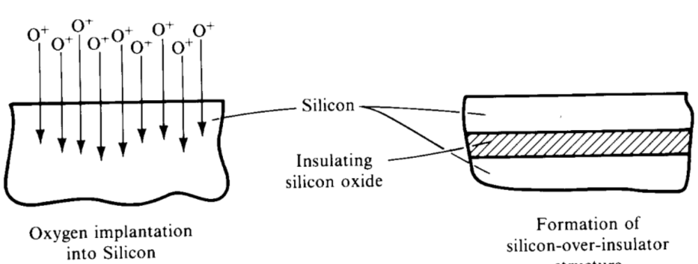
7.9.1 Key Features

- **Device Layer Thickness:** Typically ~ 2 μm for ICs but can be up to 100 μm for MEMS.
- **Oxide Thickness:** Typically ~ 0.5 to 2 μm.
- **Cost:** Expensive, ranging from \$100 to \$500 per wafer.

7.9.2 Fabrication Methods

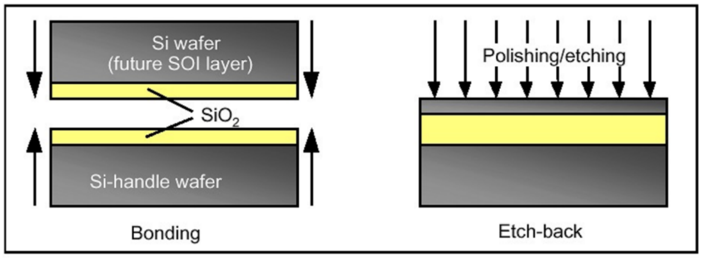
- **SIMOX (Separation by Implantation of Oxygen):** Involves implanting oxygen ions to create a buried oxide layer.
- **BESOI (Bond and Etchback Silicon On Insulator):** Two wafers are bonded, and one is polished to create a thin device layer.
- **Smart Cut:** Combines ion implantation and wafer bonding to transfer a thin silicon layer onto an insulating substrate.

SIMOX



SIMOX (Separation by IMplantation of OXygen) wafers are created by implanting oxygen ions into silicon, followed by annealing to form a buried oxide layer. By the early 1990s, SIMOX gained acceptance due to its compatibility with nearly-standard processing equipment and the availability of commercial suppliers. However, increased device performance requirements and costs have gradually reduced its market presence, though some users still rely on SIMOX wafers.

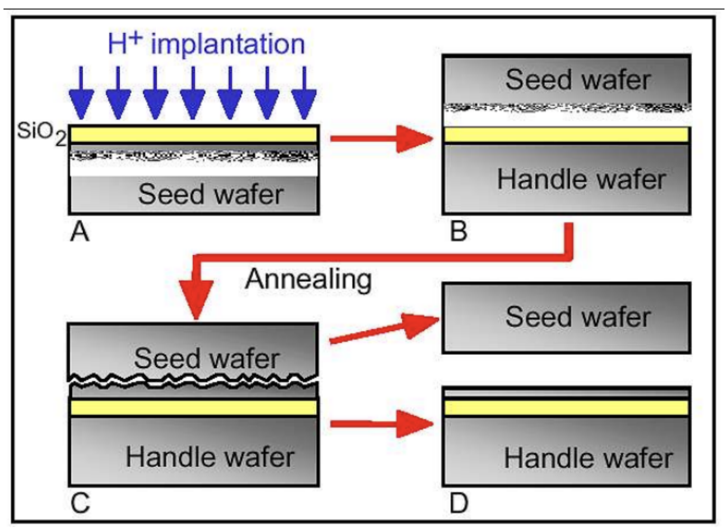
BESOI



Bonding of two oxidized silicon wafers (left) and polishing/etching back of one of the wafers

BESOI (Bond and Etchback Silicon On Insulator) involves bonding two wafers, one or both of which has an insulating oxide layer on the surface. After bonding, almost all of one wafer is polished away, leaving a structure with a thin top silicon layer, a buried oxide, and a full wafer below. This method was a contender for the SOI market alongside SIMOX.

Smart Cut



Smart Cut is Soitec's proprietary technology used to manufacture UNIBOND SOI wafers, based on two key techniques—ion implantation and wafer bonding. The process begins with two bulk silicon wafers, one of which is reused to create a SOI wafer later. An oxide layer is grown on one wafer (Wafer A), forming the buried oxide (BOX) layer of the SOI structure. High-dose hydrogen ion implantation through the oxide creates a damage layer at the ion's range.

Wafer A is bonded to Wafer B, and Wafer A is then cut across the damage plane by applying heat to break the layer off. A thin silicon layer is transferred to Wafer B, forming the SOI structure. Finally, a CMP polish finishes the SOI surface to the original bulk-silicon wafer specification. Hydrogen ion implantation acts as an atomic scalpel, enabling thin slices of monocrystalline film to be cut and transferred efficiently, optimizing material usage.

Chapter 8

Wet & Dry Etching

8.1 Etching

- Etching is a fundamental process in microfabrication where unwanted areas of films or bulk material are selectively removed to create desired patterns or structures.
  - This can be achieved by dissolving the material in a liquid chemical solution, a process known as **WET ETCHING**.
  - Alternatively, the material can be reacted with gases (either thermally or in a plasma) to form volatile products that are removed, a process referred to as **DRY ETCHING**.
- To protect areas that are not to be etched, a layer of **photoresist** is typically applied and patterned. This acts as a mask during the etching process.
- In some cases, a more durable **hard mask** is used instead of or in addition to the photoresist. Hard masks are usually made of materials like SiO_2, Si_3N_4, or metals such as chromium (Cr) or aluminum (Al). These are employed when:
  - The etch selectivity to photoresist is low, meaning the etching process would also attack the photoresist.
  - The etching environment is harsh and may cause the photoresist to delaminate or degrade.

8.1.1 Wet Etching

- Wet etching typically involves immersing the wafer in an acid or chemical bath that reacts with the exposed material to dissolve it.
- The reactions occur at the liquid-solid interface, and the process can be carried out in different ways:
  - **Immersion Etching:** The entire wafer is submerged in the etching solution.
  - **Spray Etching:** Chemicals are sprayed onto the wafer, which is often mounted on a spinning chuck to ensure uniform coverage.
  - **Chemical Mechanical Polishing (CMP):** Combines chemical etching with mechanical abrasion to achieve planarization or selective material removal.

8.1.2 Dry Etching

- Dry etching typically uses a plasma or reactive ion etching (RIE) process, where the material is removed by chemical reactions at the gas-solid interface.
- The etching process relies on the formation of gaseous by-products, which are then evacuated from the chamber. This ensures that the etched material is effectively removed.
- Dry etching is particularly useful for achieving high anisotropy (directional etching), making it ideal for creating vertical sidewalls in microstructures.

8.2 Etchant Properties

When selecting or evaluating an etchant for a specific application, several key properties must be considered to ensure the process is effective, controlled, and repeatable. These properties include etch rate, selectivity, anisotropy, surface roughness, control of parameters, safety, cost, and the capacity for etch-stops.

Etch Rate

The etch rate is a critical parameter that defines the speed at which material is removed, typically measured in micrometers per minute (μm/min). The etch rate depends on several factors:

- The specific etchant and the material being etched.
- The concentration and mixing of the etchant solution.
- The temperature of the etchant bath.
- The material layout and geometry.

The etch rate often follows an Arrhenius relationship, where the rate increases exponentially with temperature. It is essential to control and repeat the etch rate to ensure consistent results. Excessively fast etching can make the process difficult to control, leading to non-uniformity and potential damage to the underlying layers.

Selectivity to Masking Layers and Their Availability

Selectivity is the ability of the etchant to preferentially remove the target material while minimizing the etching of other layers, such as the masking layer or underlying films. Since etching also affects the mask and substrate, the etch time must be carefully calculated to avoid excessive damage.

- **Film-to-Mask Selectivity:** The ratio of the etch rate of the film to the etch rate of the masking layer. A high selectivity value is needed to minimize linewidth resolution loss.

$$S_{f:m} = \frac{R_{\text{film}}}{R_{\text{mask}}}$$

- **Film-to-Underlying Film Selectivity:** The ratio of the etch rate of the film to the etch rate of the underlying layer. A high selectivity value is required to minimize overetch material loss.

$$S_{f:u} = \frac{R_{\text{film}}}{R_{\text{under}}}$$

Anisotropy (Crystal Plane Selectivity)

Anisotropy refers to the directional dependence of the etch rate, which is particularly relevant when etching monocrystalline materials. For example, potassium hydroxide (KOH) has a relative etch rate of approximately 40:30:1 for the <110>, <100>, and <111> crystal planes, respectively. These relative rates can vary significantly with temperature and other process conditions. Anisotropy is a critical parameter for achieving precise geometries and well-defined sidewalls in microfabrication.

Surface Roughness

The surface roughness of the etched material is an important consideration, as it can affect the performance and reliability of the final device. The choice of etchant, etching conditions, and material properties all influence the resulting surface roughness.

Control of Etch Parameters

The etching process must be carefully controlled to achieve the desired results. Key parameters to monitor and adjust include:

- Temperature of the etchant solution.
- Agitation conditions to ensure uniform etching.
- Concentration and replenishment of the etchant.

Precise control of these parameters ensures repeatability and minimizes variability in the etching process.

Safety of Reactants and Products

The safety of the etching process is paramount. Both the reactants and the by-products of the etching reaction must be evaluated for toxicity, reactivity, and environmental impact. Proper handling, storage, and disposal procedures must be followed to ensure the safety of personnel and compliance with regulations.

Cost (Including Disposal and Fixed Costs)

The cost of the etching process includes both fixed costs (e.g., equipment) and variable costs (e.g., chemicals, disposal). Disposal costs are often the most significant expense, as they are calculated based on the volume of waste generated. Minimizing waste and optimizing the etching process can help reduce overall costs.

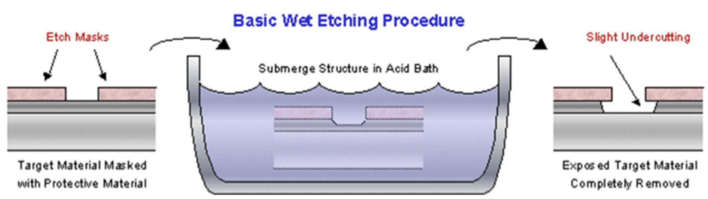
Capacity for Etch-Stops

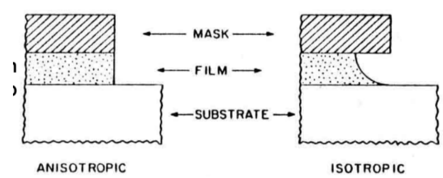
Etch-stops are mechanisms that halt the etching process at a specific point to prevent overetching. These can be achieved through:

- Time-controlled etching, where the process is stopped after a predetermined duration.
- Concentration-controlled etching, where the etchant is depleted to a point where etching slows or stops.
- Doping-dependent etch-stops, where the etch rate varies based on the doping level of the material.
- Electrochemical etch-stops, which use an applied voltage to selectively halt etching in specific regions.

The choice of etch-stop method depends on the material, etchant, and desired precision of the process.

8.3 Wet Etching





Wet etching is a process where the material is removed by immersing the substrate in a liquid chemical solution. The etchant reacts with the exposed material, dissolving it and leaving behind the desired pattern. Wet etching can be classified into two main types based on the etching profile: isotropic and anisotropic.

Isotropic etchants, such as HNA (a mixture of hydrofluoric acid, nitric acid, and acetic acid), etch the material uniformly in all directions, resulting in rounded profiles. This type of etching is:

- Best suited for applications with large geometries where the sidewall slope does not matter.
- Useful for undercutting the mask material.
- Quick, easy, and cost-effective.

Anisotropic etchants, such as KOH (potassium hydroxide), TMAH (tetramethylammonium hydroxide), and EDP (ethylene diamine pyrocatechol), preferentially etch certain crystallographic planes of silicon, such as the <100> planes, while etching much slower on the <111> planes. This type of etching is:

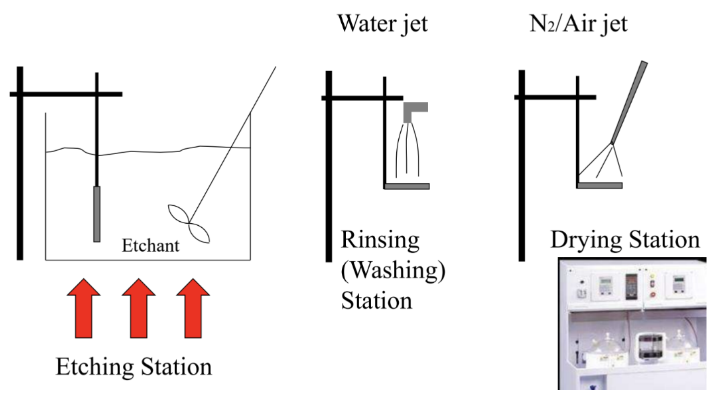
- Ideal for creating small gaps and vertical sidewalls or sidewalls with precise angulation.
- Typically more expensive and requires more precise control.

The resulting etching angle depends on the crystallographic orientation of the silicon substrate, as shown in the accompanying image.

8.3.1 Steps in Wet Etching

The wet etching process typically involves the following steps:

- **Etching Station:** The substrate is immersed in the etching solution, where the material is selectively removed.
- **Rinsing (Washing) Station:** The substrate is thoroughly rinsed to remove any residual etchant. This step is critical for ensuring the quality of certain devices.
- **Drying Station:** The substrate is dried to prevent watermarks or contamination.



Mechanical agitation is often required to ensure that fresh etching solution is in constant contact with the wafer surface. Surfactants are added to promote wetting and to remove or inhibit by-product gases, such as hydrogen (H_2), that may cling to the wafer surface.

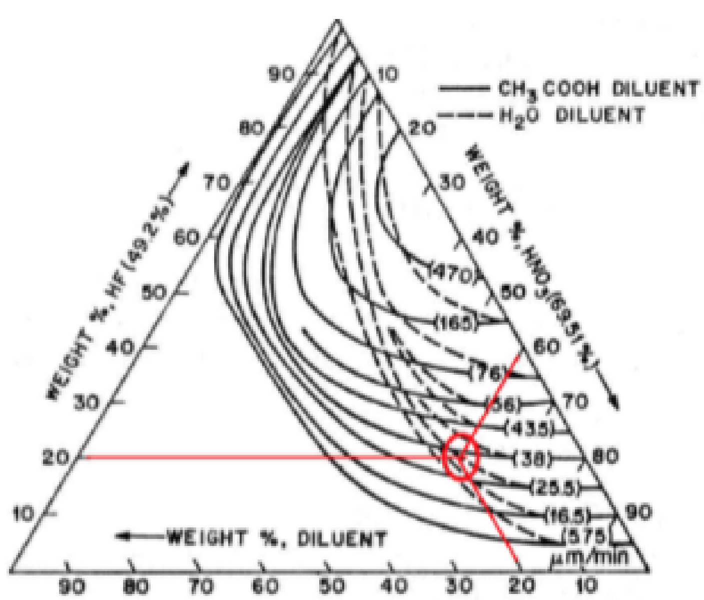
8.3.2 Wet Etching Silicon ("HNA")

The HNA etching solution is a mixture of nitric acid (HNO_3) and hydrofluoric acid (HF), often diluted with water or acetic acid (CH_3COOH). The etching process involves the following steps:

- HNO_3 oxidizes silicon, forming a silicon dioxide (SiO_2) layer.
- HF removes the SiO_2 layer, exposing fresh silicon for further oxidation.
- The process repeats, resulting in the removal of silicon.

Key characteristics of HNA etching:

- High HNO_3:HF ratio leads to etching limited by oxide removal.
- Low HNO_3:HF ratio leads to etching limited by oxide formation.
- Acetic acid is preferred over water for dilution, as it prevents HNO_3 dissociation.
- The etching process is fully isotropic.



Reference: H. Robbins and B. Schwartz, “Chemical Etching of Silicon II, the System HF, HNO_3, H_2O, and $HC_2H_3O_2$”, J. Electrochem. Soc., v. 107, p. 108 (1960).

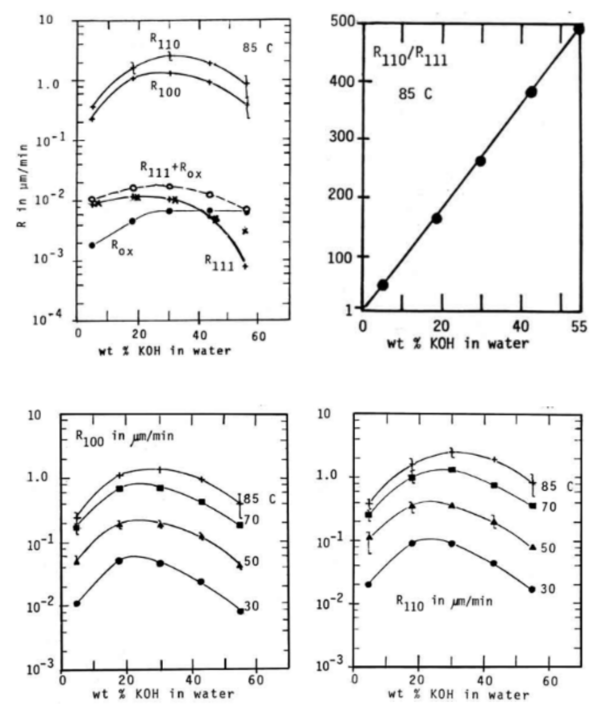
8.3.3 Anisotropically Wet Etching Silicon

Inorganic Alkaline Solutions (KOH)

Potassium hydroxide (KOH) is a commonly used anisotropic etchant for silicon. Key characteristics include:

- KOH pellets are dissolved in water, and the reaction produces hydrogen gas (H_2).
- It is the quickest silicon anisotropic etchant but is less controllable compared to other etchants.
- Not compatible with microelectronics (e.g., CMOS processes).
- SiO_2 and Si_3N_4 are used as masking materials, as photoresist will not survive, and oxide is attacked slowly.
- The setup is simple, typically involving a hot plate, Pyrex beaker, and stirrer. Stirring improves roughness by eliminating H_2 bubbles.
- High concentrations are needed to limit roughness.
- The etch rate is approximately 1 μm/min but strongly depends on the solution concentration and temperature (typically used around 80°C).

The etch rate of silicon in KOH depends on the crystallographic plane (anisotropy) and temperature:



KOH also etches silicon oxide slowly, providing good selectivity.

8.3.4 Hydrofluoric Acid (HF) for SiO_2 Etching

Hydrofluoric acid (HF) is a selective etchant for silicon dioxide (SiO_2) and does not etch silicon. Key characteristics include:

- **Selectivity:** HF etches SiO_2 but not silicon. However, it will also attack aluminum.
- **Etch Rate:** The rate depends strongly on concentration:
  - Maximum rate: 49% HF (“concentrated”) $\sim> 2$ μm/min.
  - Controlled rate: 5 to 50:1 (“timed”) $\sim< 0.1$ μm/min.
  - The etch rate is not linear at high concentrations.
- **Safety:** HF is extremely dangerous:

  - It is not a strong acid (check the pH of your HF etch).
  - It is deceptive in appearance, resembling water.
  - It penetrates the skin and attacks slowly, targeting bones.

- **Etch Geometry:** HF etching is completely isotropic and is often used to undercut or release structures.

- **Buffered HF (BOE):** Buffering with ammonium fluoride (NH_4F) keeps the pH constant, stabilizing the etch rate and lowering it if needed.

BOE (Buffered Oxide Etch) or **BHF (Buffered HF)** is commonly used for controlled etching of SiO_2.

8.4 What is a Dry Etch?

Dry etching is a process where material is removed from a substrate using gaseous or vapor-phase reactants. Unlike wet etching, which uses liquid chemicals, dry etching relies on gases that are often ionized to create reactive species. These reactive species chemically or physically interact with the exposed material, leading to its removal. Dry etching is broadly categorized into two types: plasma-based and non-plasma-based etching.

8.4.1 Why Use Plasmas in Dry Etching?

Plasmas are widely used in dry etching due to their unique advantages:

- **Lower Temperature Reactions:** Plasmas enable chemical reactions to occur at lower temperatures compared to thermally driven processes. This is particularly useful in Plasma-Enhanced Chemical Vapor Deposition (PECVD) and other low-temperature applications.

- **Directional Processing:** Plasmas can provide directionality to the etching process, allowing for anisotropic etching, which is essential for creating vertical sidewalls in microstructures.

- **Versatility:** The ability to control multiple parameters, such as gas composition, pressure, and RF power, provides higher versatility in achieving desired etch profiles and selectivity.

8.4.2 Vapor Etching

Vapor etching is a non-plasma, isotropic dry-etch process that uses vapor-phase reactants to remove material. One of the most common vapor etchants is xenon difluoride (XeF_2), which is particularly effective for silicon etching.

- **High Selectivity:** XeF_2 exhibits very high selectivity for silicon over materials like aluminum, silicon dioxide (SiO_2), phosphosilicate glass (PSG), silicon nitride (Si_3N_4), and photoresist.

- **Etch Rate:** The etch rate is generally fast, ranging from $1-3\ \mu m$/min, making it suitable for rapid material removal.

- **Reaction:** The chemical reaction for silicon etching is:

$$2XeF_2 + Si \rightarrow 2Xe + SiF_4$$

- **Surface Roughness:** While XeF_2 is highly selective, the etched surfaces can be quite rough, which may limit its use in applications requiring smooth finishes.

- **Safety Concerns:** XeF_2 reacts with water, including moisture in the air, to form xenon (Xe) and hydrofluoric acid (HF). HF is a safety hazard and can also unintentionally etch SiO_2.

HF Vapor Phase Etching

HF vapor etching is another non-plasma dry etching technique used for applications like surface micromachining and SOI-MEMS. The HF vapor etcher typically consists of a reaction cell, a heated wafer holder, and an electronic control unit. HF vapor is generated passively from a small liquid reservoir, requiring only about 60-80 ml of liquid HF. This method is particularly useful for dicing-free release and structure thinning. Safety is a critical concern when working with HF, and specialized equipment is designed to maximize operator safety.

8.4.3 Types of Dry Etching Processes

Dry etching processes can be categorized based on their mechanisms and applications. Below are the main types:

Species in Plasma	**Typical Density (cm^{-3})**
Neutral Molecules	10^{16}
Radicals	10^{14}
Electrons	10^{8}
Positive Ions	10^{8}

Table 8.1: Typical Plasma Composition

RF Plasma-Based Dry Etching

Plasma etching is an isotropic process that relies primarily on chemical reactions between the plasma-generated reactive species and the material being etched. It is highly selective, meaning it can etch the target material while leaving other materials relatively unaffected. However, due to its isotropic nature, it is less suitable for applications requiring high aspect ratios or vertical sidewalls. Plasma-based dry etching uses radio frequency (RF) energy to ionize gases, creating a plasma that facilitates the etching process. The steps involved are:

- The chamber is evacuated to create a vacuum.
- The chamber is filled with etching gases.
- RF energy is applied to electrodes, accelerating electrons and increasing their kinetic energy.
- High-energy electrons collide with neutral gas molecules, forming ions and additional electrons.
- A steady-state plasma is established, balancing ionization and recombination processes.

RF Plasma Characteristics

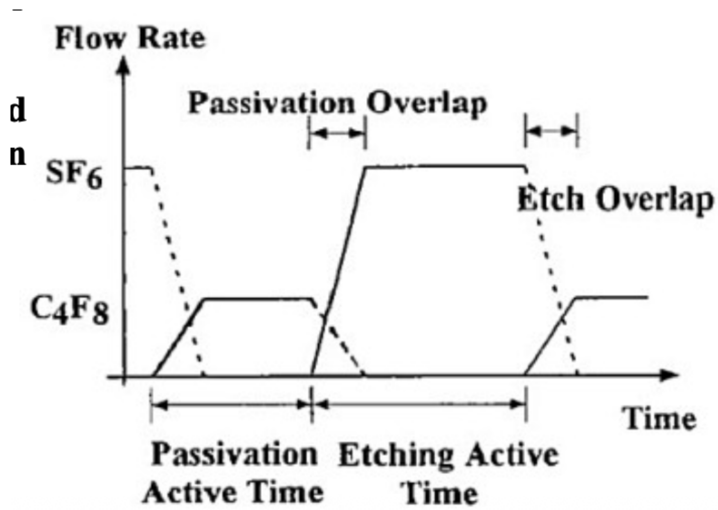
- **Frequency:** RF frequencies, typically 13.56 MHz, are used to sustain the plasma.
- **Electron Behavior:** Electrons oscillate in the glow region, gaining enough energy to cause ionization. The alternating potential polarity affects the electrons, sending them into oscillatory paths.
- **Ion Behavior:** Gas ions, being much heavier, do not respond to the high-frequency field, allowing for controlled ion bombardment of the substrate.
- **Etch Profile:** Plasma etching is generally isotropic and selective due to its strong chemical component.

Reactive Ion Etching (RIE)

RIE combines both physical and chemical etching mechanisms. It uses a plasma to generate reactive species for chemical etching while simultaneously applying an electric field to accelerate ions toward the substrate. This results in a directional etching process, making it suitable for creating features with vertical sidewalls. RIE is fairly selective and is widely used in microfabrication.

Deep Reactive Ion Etching (DRIE)

Deep Reactive Ion Etching (DRIE) is a highly anisotropic etching process used to create deep, high-aspect-ratio structures in silicon. The process alternates between an isotropic etching step (using gases like SF_6) and a passivation step (using gases like C_4F_8) to protect the sidewalls. This time-multiplexed process, also known as Time Multiplexed Deep Etching (TMDE) or the Bosch process, enables the creation of vertical sidewalls and is ideal for MEMS applications. DRIE achieves high aspect ratios while maintaining good selectivity and anisotropy.



Process Steps

- **Etching Step:** Sulfur hexafluoride (SF_6) is used as the etching gas. A substrate bias of 5 to 30 V accelerates cations generated in the plasma nearly vertically into the substrate, creating an anisotropic etch profile.
- **Passivation Step:** A mixture of octafluorocyclobutane (C_4F_8) and argon is introduced. This deposits a thin, Teflon-like polymer layer (polymerized CF_2) approximately 50 nm thick on all exposed surfaces, including sidewalls and horizontal surfaces.
- **Directional Bombardment:** During the passivation step, a small bias voltage prevents polymer formation on horizontal surfaces. Any polymer deposited on horizontal surfaces is removed by ion bombardment.
- **Repeat Etching:** The etching step is repeated, removing the polymer from horizontal surfaces and deepening the trench while preserving the passivated sidewalls.

Sequential Steps in TMDE

1. **Start:** The masking material (photoresist or silicon dioxide) is patterned on the substrate.
2. **Etching:** An isotropic SF_6 etch with anisotropic ion bombardment creates a shallow trench.
3. **Passivation:** A protective fluorocarbon film is deposited everywhere using C_4F_8.
4. **Etching Again:** The fluorocarbon film is removed from horizontal surfaces by directional ion bombardment, and another shallow trench is etched.

Illustration of TMDE Steps

- (a) The masking material is patterned.
- (b) During the etch step, a shallow, isotropic trench is formed.
- (c) In the passivation step, a protective fluorocarbon film is deposited everywhere.
- (d) During the next etch step, the fluorocarbon film is removed from horizontal surfaces, and another shallow trench is formed.

8.4.4 Dry Etch Chemistries

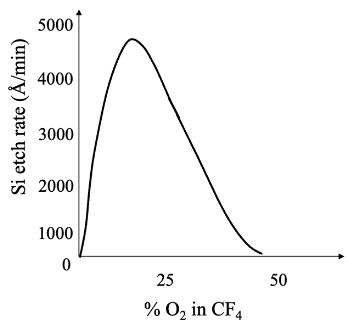
The choice of etch gases depends on the material being etched. Common chemistries include:

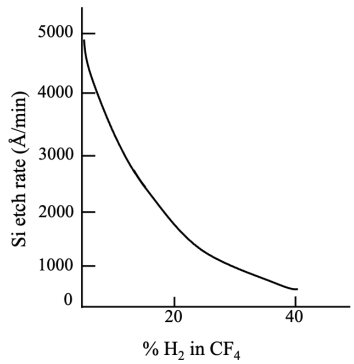
Material	**Etch Gases**
Silicon (Si), SiO_2, Si_3N_4	CF_4, SF_6, CHF_3, NF_3
Silicon (Si)	Cl_2, CCl_2F_2
Silicon Carbide (SiC)	CF_4, SF_6, CHF_3
Aluminum (Al)	BCl_3, CCl_4, $SiCl_4$, Cl_2
Organics	O_2, $O_2 + CF_4$
Other Metals (W, Ta, Mo, ...)	CF_4

8.4.5 Selectivity in Dry Etching

Selectivity is the ability to etch one material preferentially over another. For example:

- CF_4 plasmas can etch silicon selectively over SiO_2, and vice versa.
- Adding small amounts of O_2 to CF_4 enhances the etch rate of silicon more than SiO_2. Reactions between O_2 and CF_4 increase the fluorine (F) concentration in the plasma due to the generation of CO_2, which significantly increases the silicon etch rate.
- Adding H_2 to CF_4 decreases the etch rate of silicon more than SiO_2. Hydrogen reacts with fluorine to form HF in the gas phase, which scavenges fluorine atoms from the plasma. This reduces the silicon etch rate, while the SiO_2 etch rate remains nearly constant.
- At 40% H_2 in CF_4, the silicon etch rate is nearly zero, while the SiO_2 etch rate remains unaffected.





8.4.6 Anisotropy in Dry Etching

Anisotropy refers to the directionality of the etching process. It is quantified as:

$$\text{Anisotropy} = \frac{R_z}{R_x}$$

where R_z is the vertical etch rate and R_x is the lateral etch rate. By adjusting parameters like RF power and gas composition, a balance between isotropic and anisotropic etching can be achieved.

8.4.7 Aspect Ratio

The aspect ratio is defined as:

$$\text{Aspect Ratio} = \frac{\text{Height}}{\text{Width}}$$

High aspect ratios are desirable for creating deep, narrow features in microstructures.

sidewalls.

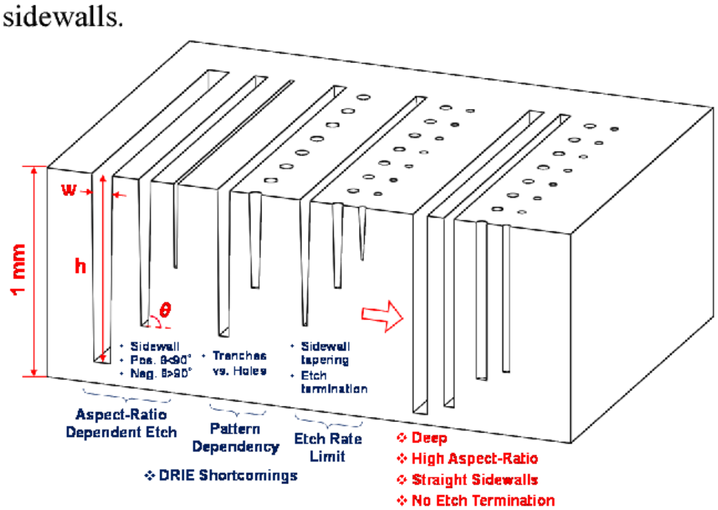


Figure 1: Ultra-deep ultra-high aspect ratio etching of

Chapter 9

Packaging

Packaging is a critical stage in the manufacturing of integrated circuits. The process starts after the wafer has been fabricated and tested.

9.1 Packaging Process Flow

The process generally follows these steps:

1. **Wafer Test**: The finished wafer undergoes testing. Defective dies are electrically identified.
2. **Defective die inked**: The defective dies are marked (inked) to distinguish them from functional dies, the eventually eliminated.
3. **Wafer Saw**: The wafer is cut into individual dies.
4. **Package only "good" die**: Only the dies that passed the test are prepared for packaging.
5. **Final Test**: The packaged integrated circuit undergoes a final test.

9.2 Steps in Packaging

The physical packaging process involves several key micro-assembly steps:

Die Separation and Handling

- **Saw wafers into dies**: The process of cutting the wafer into individual chips.
- **Wafers mounted on mylar tape in stainless steel ring**: This is done to hold the dies in place during the sawing and subsequent handling.

Die Bonding and Interconnection

- **Bond good dies into package**: The selected good die is attached to the package's heat sink or base. This is the **Die bond**.
- **Wire bond die pads to package leads**: This process, known as **Wire Bond**, connects the on-chip aluminum or gold bond pads to the package's external leads using Au or Al wires.
- This bonding is typically achieved through Heat or ultrasonic bonding.

Package Sealing (Encapsulation)

- **Seal package**: This step covers the die, providing protection and sealing the package.
- **Ceramic/metal packages** use a metal lid and solder for sealing.
- **Plastic packages** use encapsulant or injection molding, which must be cured in an oven.

Final Inspection

- **Inspect**: The final packaged components are inspected for quality.

9.3 Functions of Packaging

Packaging provides several critical functions necessary for a chip to operate reliably within an electronic system:

1. **Power Distribution**: Distributes power from the circuit board to the chip.
2. **Signal Distribution**: Distributes electrical signals to and from the chip.
3. **Dissipates Heat**: Manages thermal load. This involves structural and materials concerns to ensure the chip does not overheat.
4. **Houses and Protects Die**: Provides protection against mechanical stress, chemical attack, and electromagnetic interference.

9.4 Package Types

Packages are classified primarily by how they interface with the Printed Circuit Board (PCB).

9.4.1 Types of First-Level Packages

Through Hole Package

These packages have leads that are inserted through holes in the PCB.

- **DIP (Dual In-line Package)**: Two lines of leads electrically connect the die to the circuit board. Is a low performance package, but is simple and cheap
- **SH-DIP** (Shrink DIP)
- **SK-DIP, SL-DIP** (Skinny DIP, Slim DIP)
- **SIP** (Single In-line Package): The hole and the metal surface are usually attached to a heat dissipator. It is commonly used in power electronics.
- **ZIP** (Zig-zag In-line Package)

Surface Mounted Package

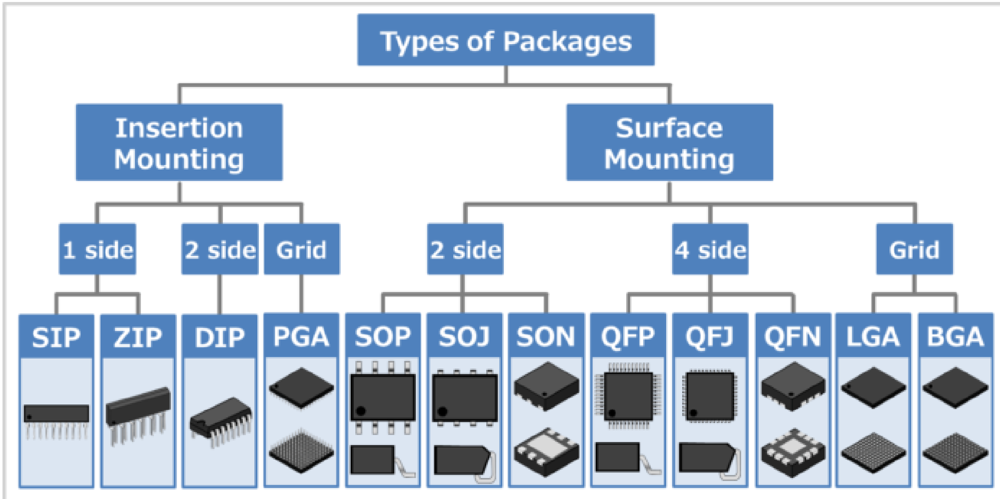
These packages are mounted directly onto the surface of the PCB.

- **SO or SOP (Small Out-line Package)**
- **QFP (Quad Flat Package)**
- **LCC (Leadless Chip Carrier)**: Chip carrier with no leads, metal lines run along package sides.
- **PLCC, SOJ** (Plastic Leaded Chip Carrier with Butt Leads)
- **TAB (Tape Automated Bonding)**
- **CSP (Chip Scale Package)**

Pin Array Packages

Electrical contact is made through an array of connections on the underside of the package. The advantage is a higher density of connections compared to traditional leaded packages, enabling more compact designs and better electrical performance.The difference between PGA and BGA is the shape.

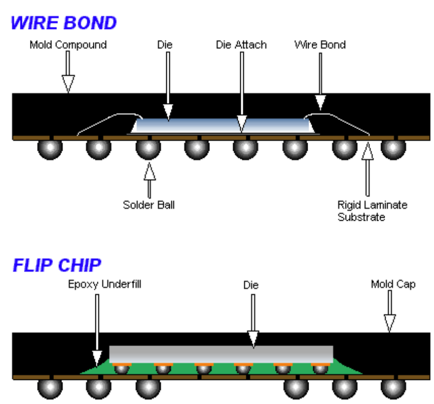
- **PGA (Pin Grid Array)** or Column Package: Pin array where contacts are pins.
- **BGA (Ball Grid Array)**: Pin array where contacts are solder balls.



9.5 Advanced Interconnection and Integration

9.5.1 Flip-Chip Bonding

The flip-chip method is an alternative to wire bonding where the chip is inverted (flipped) and connected directly to the substrate using solder bumps.
In this method, instead of using long gold wires, small pin contacts are utilized to establish electrical connections between the chip and the substrate. These pin contacts are directly bonded to the solder bumps on the chip, providing a compact and efficient interconnection. This approach reduces parasitic inductance and resistance, improving the overall electrical performance of the package.



The Pb : Sn alloy system is critical for many solder processes. The phase diagram (Figure 6.3) shows a eutectic composition at 61.9% Sn melting at 183°C (362°F).

9.5.2 Multi-Chip Modules (MCMs)

The need for high density, high performance, and cost-effectiveness has sped up the development and use of Multi-Chip Modules (MCMs). MCMs enable the integration of different chips with varied functions into a mini substrate, instead of inserting individual packages onto a large PCB (also known as second-level package).

Definitions

A Multi-Chip Module (MCM) can be defined in a couple of ways:

- A structure consisting of two or more integrated circuits electrically connected to a common circuit base and interconnected by conductors in that base.
- A structure in which a **packaging efficiency** of greater than 30% is achieved. **Packaging efficiency** is determined by the ratio of silicon die area to printed circuit board area. This definition implies a particular technology that allows chips to be packed closely together.

Goals

The basic idea behind developing MCM technology is to decrease the average spacing between ICs in an electronic system. The goals are:

- Higher performance resulting from reduced signal delays between chips.
- Improved signal quality between chips.
- Reduced overall size.
- Reduced number of external components.

Structure

An MCM substrate can be composed of different layers, and their number depends on the MCM technology used. These layers provide all the interconnections between the different mounted ICs.

9.5.3 3D Packaging

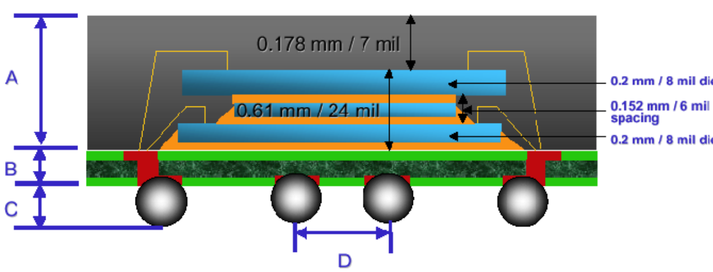
As a result, 3D packaging technology has evolved as a natural progression from the 2D packaging technology (MCMs).
3D packaging is achieved by stacking multiple chips vertically, one on top of the other. This stacking is facilitated by Through Silicon Vias (TSVs), which provide vertical electrical connections between the layers. The stacked configuration minimizes the footprint on the PCB while enabling high-density integration and improved performance. This approach is particularly useful in applications requiring compact designs, such as mobile devices and high-performance computing systems. As the requirements for high performance systems continually increase, even MCM technology is often not enough.

Advantages

- Footprint used by the 3D device is much smaller than the footprint used by every chip in a 2D MCM layout.
- **Through Via Connection (TSV)**: Provides short interconnect length, high density interconnect, and a low profile (without a spacer).

Through Silicon Vias (TSV)

TSV via diameters typically range from 5 μm to several tens of μm. They provide vertical electrical connections through the chip.



Package Outline				
Total (Thickness)	Mold Cap Thickness (A)	Substrate Thickness (B)	Ball Height (C)	Ball Pitch (D)
1.5 mm Max.	0.8 mm	0.38 mm	0.35 mm	0.8 mm

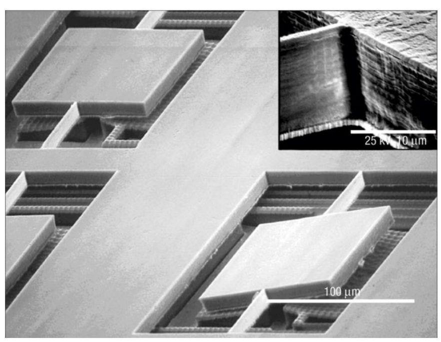
Chapter 10

Complementary Technologies

10.1 Bulk Micromachining

Bulk micromachining is a fundamental technique in microfabrication that involves the selective removal of the "bulk" substrate to create suspended three-dimensional structures. This method is widely used due to its simplicity and effectiveness, especially for applications requiring relatively large masses. Below are the key characteristics and techniques associated with bulk micromachining:

- **Selective Removal:** The process focuses on removing specific portions of the substrate while leaving other areas intact to form the desired microstructures.
- **Historical Significance:** Bulk micromachining is one of the earliest micromachining techniques, with its first applications dating back to the 1970s for the fabrication of micro pressure sensors.
- **Advantages:** It allows for the creation of structures with significant mass, which is not feasible with surface micromachining.
- **Limitations:** Compared to surface micromachining, bulk micromachining offers less versatility and lower resolution.



10.1.1 Bulk Micromachining Etching Techniques

Bulk micromachining relies on various etching techniques to achieve the desired structures. These techniques include:

Wet Etching

Wet etching is a chemical process that involves the use of liquid etchants to remove material from the substrate. The key steps in wet etching are as follows:

- **Masking:** The substrate is coated with a protective material (mask) to define the areas to be etched.
- **Submersion:** The substrate is submerged in an acid bath or other chemical solution that acts as the etchant.
- **Etching Process:** The exposed areas of the substrate react with the etchant and are dissolved.
- **Undercutting:** Due to the isotropic nature of wet etching, slight undercutting may occur beneath the mask, which can affect the precision of the etched structures.

Vapor Etching

Vapor etching is a dry etching process that uses vapor-phase reactants to remove material. It is highly selective and isotropic. The chemical reactions involved include:

- **Reaction with Silicon:** Xenon difluoride (XeF_2) reacts with silicon to form xenon gas and silicon tetrafluoride:

$$2XeF_2 + Si \rightarrow 2Xe + SiF_4$$

- **Reaction with Silicon Dioxide:** Hydrofluoric acid (HF) reacts with silicon dioxide to form silicon tetrafluoride and water:

$$SiO_2 + 4\ HF \rightarrow SiF_4 + 2\ H_2O$$

- **Process Characteristics:** Vapor etching is particularly useful for applications requiring high selectivity and minimal damage to surrounding materials.

Plasma Etching Process

Plasma etching is a versatile technique that uses ionized gases to remove material from the substrate. The process involves:

- **Electrode Configuration:** The etching chamber is equipped with top and bottom electrodes to generate the plasma.
- **Gas Introduction:** Specific etching gases are introduced into the chamber to react with the target material.
- **Vacuum Environment:** A vacuum pump maintains a low-pressure environment to sustain the plasma.
- **RF Power Source:** A radio frequency (RF) power source ionizes the gases, creating a plasma that facilitates the etching process.
- **Etching Techniques:**
  - **Reactive Ion Etching (RIE):** Combines chemical and physical etching mechanisms to achieve anisotropic profiles.
  - **Deep Reactive Ion Etching (DRIE):** A specialized technique for creating deep, high-aspect-ratio structures, commonly used in MEMS fabrication.
- **Etch Depths:** Plasma etching can achieve depths ranging from a few micrometers to several hundred micrometers, depending on the process parameters.

10.1.2 Etch Stop technique

P++ Etch Stop

In silicon, heavily boron-doped regions exhibit significantly slower etch rates, making them effective etch stops. Key characteristics include:

- **Selective Etching:** EDP (ethylene diamine pyrocatechol) is the most selective etchant for boron-doped silicon. The etch rate decreases by a factor of 100 as the boron concentration increases from 10^{19} atoms/cm^3 to 10^{20} atoms/cm^3, which is near the solid solubility limit of boron in silicon.
- **Applications:** This technique is widely used to create thin membranes and other structures with well-defined thicknesses.

Electrochemical Etch Stop

Electrochemical etch-stop techniques utilize the anodic passivation properties of silicon to achieve selective etching. The process involves:

- **Passivation:** At certain potentials, silicon forms an anodic oxide layer that stops etching, effectively passivating the surface.
- **Reverse Biased p-n Junction:** The technique combines anodic passivation with a reverse-biased p-n junction to achieve high selectivity between p-type and n-type silicon.
- **Electrical Configuration:** A positive voltage is applied to the n-type silicon via an ohmic contact, while the p-type silicon is connected to the etch solution through a counter electrode (usually platinum).

- **Etch Stop:** Once the p-type silicon is completely removed, the n-type silicon becomes exposed to the etch solution. Under anodic bias, the n-type silicon passivates, halting the etching process.

The electrochemical etch-stop process is one of the most cost-effective and precise techniques for fabricating thin, monocrystalline silicon membranes. It requires:

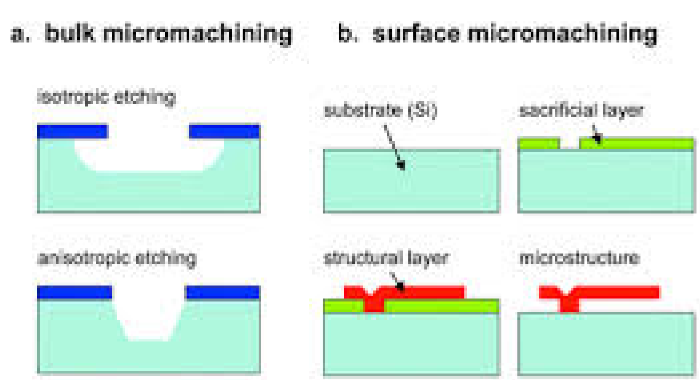
- A reliable wafer holder to protect the n-type backside of the wafer and provide electrical contact.
- A potentiostat to maintain the reverse bias and monitor the current over time.

10.2 Surface Micromachining

Surface micromachining is a critical technique in microfabrication that focuses on the deposition and etching of thin films to create intricate microstructures. This method is particularly advantageous for applications requiring high resolution and the ability to fabricate complex geometries. Unlike bulk micromachining, which involves removing large portions of the substrate, surface micromachining relies on the precise manipulation of thin films deposited on the substrate.

- Surface micromachining utilizes planar technologies to fabricate suspended microstructures. This involves depositing thin films on the bulk substrate and selectively etching specific areas to create the desired structures.
- While the technology is more complex compared to bulk micromachining, it offers greater versatility and flexibility in design.
- The process introduces the concepts of a **structural layer** (the layer that forms the functional part of the device) and a **sacrificial layer** (a temporary layer that is removed to release the structural layer).
- Surface micromachining typically allows for better resolution and more efficient use of the substrate surface compared to bulk micromachining.
- However, the thickness of devices and cavities created using this technique is limited to the micron scale. As a result, surface micromachining is not suitable for fabricating high structures or devices requiring large masses.
- A significant challenge in surface micromachining is the **stiction problem**, where microstructures stick to the substrate due to capillary forces during the drying process.

10.2.1 Surface vs. Bulk Micromachining



Surface micromachining differs from bulk micromachining in several key aspects:

- **Bulk Micromachining:** This technique involves the removal of large portions of the substrate to create structures. It is suitable for applications requiring high masses or thick structures.
- **Surface Micromachining:** This method focuses on the deposition and etching of thin films to create suspended microstructures. It is more versatile and allows for higher resolution but is limited in terms of structure thickness and mass.

10.2.2 Typical Devices Realized by Surface Micromachining

Surface micromachining is commonly used to fabricate devices with suspended structures, such as cantilevers, membranes, and beams. These devices often require the removal of the sacrificial layer to release the structural layer. To facilitate this process:

- **Etch Holes:** Etch holes are incorporated into the design to reduce the time required for removing the sacrificial layer underneath large-area structures. These holes allow the etchant to access and dissolve the sacrificial layer more efficiently.

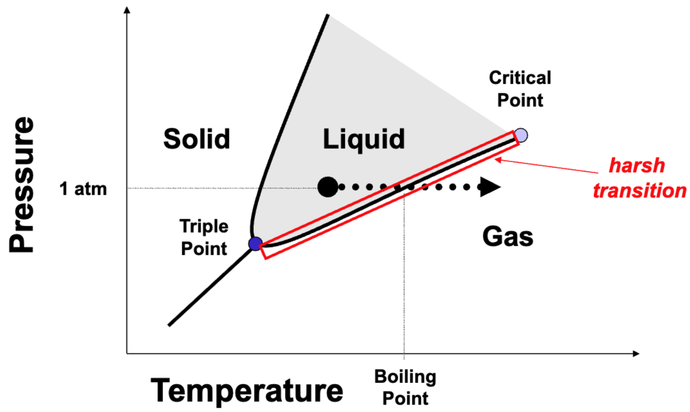
10.3 Criteria for selecting materials and etching solutions

When selecting materials and etching solutions for microfabrication processes, several critical criteria must be considered to ensure optimal performance and compatibility. These criteria include:

- **Selectivity:**
  - The etching solution must exhibit a high selectivity, meaning the etch rate on the sacrificial layer should be significantly higher than the etch rate on the structural layer. This ensures that the structural layer remains intact and undamaged during the etching process.
- **Etch Rate:**
  - The etching solution should have a rapid etch rate on the sacrificial layer. A faster etch rate reduces the overall etching time, improving process efficiency and minimizing the risk of prolonged exposure to the etchant, which could lead to unintended damage or degradation.
- **Deposition Temperature:**
  - In certain applications, it is essential to maintain a low overall processing temperature. This is particularly important for processes that involve integration with temperature-sensitive components, such as CMOS circuits or biological materials, where high temperatures could cause damage or alter material properties.
- **Intrinsic Stress of the Structural Layer:**
  - The structural layer must exhibit low intrinsic stress to ensure that it remains flat and stable after the release process. High stress in the structural layer can lead to warping, deformation, or mechanical failure, which would compromise the functionality of the final device.
- **Surface Smoothness:**
  - Surface smoothness is a critical factor for applications that require high optical quality or precision. A smooth surface minimizes scattering and ensures optimal performance in optical and other high-precision applications.

10.3.1 Evaporation Drying

The transition of a liquid to a gas phase, known as evaporation drying, presents significant challenges at the micro-scale. During this process, capillary forces can exert substantial stress on microstructures, leading to deformation or adhesion to the substrate. This phenomenon, often referred to as stiction, is particularly problematic for delicate microstructures with high aspect ratios. The capillary forces arise due to the surface tension of the liquid as it evaporates, creating a pulling effect that can cause adjacent structures to come into contact. Once in contact, these structures may adhere to each other or to the substrate, resulting in permanent deformation or failure of the microdevice.



10.3.2 Release Stiction

Stiction, or the adhesion of microstructures to the substrate or to each other, is a critical issue during the drying phase of microfabrication processes. This adhesion is primarily caused by capillary forces exerted by the drying rinse solution. As the liquid evaporates, the surface tension pulls the microstructures downward, leading to potential stiction. The specific adhesion mechanisms depend on the surface properties of the materials involved:

- **Van der Waals Forces:** These forces are the primary cause of stiction for hydrophobic surfaces. They arise due to the interaction of instantaneous dipoles between molecules.
- **Hydrogen Bonding:** For hydrophilic surfaces, hydrogen bonding is the dominant adhesion mechanism. This occurs when hydrogen atoms form bonds with electronegative atoms like oxygen or nitrogen on the surface.

The critical question during the drying process is whether the microdevices will successfully release from the substrate or remain adhered due to stiction.

Strategies to Mitigate Stiction

To address the challenges posed by stiction, several strategies can be employed to reduce its occurrence and impact:

- **Reduce Adhesion Area:** By minimizing the contact area between the microstructure and the substrate, the likelihood of stiction can be significantly reduced. This can be achieved by incorporating dimples or bumps into the design of the microstructures.
- **Integrate Supporting Microstructures:** Adding supporting features can increase the tolerance of microstructures to capillary forces. Examples include microfuses or sacrificial supporting layers made of photoresist, which provide temporary support during the drying process.
- **Sublimation Drying:** This technique involves freezing the liquid and then transitioning it directly from the solid phase to the gas phase, bypassing the liquid phase. This eliminates capillary forces and reduces the risk of stiction.
- **Supercritical CO_2 Drying:** In this method, the liquid is replaced with supercritical carbon dioxide, which has no surface tension. This prevents the formation of capillary forces during drying.
- **Vapor Phase Etching:** This process uses vapor-phase chemicals to etch away the sacrificial layer, avoiding the use of liquid solutions and thereby eliminating capillary forces.

10.4 Wafer Bonding

Wafer bonding is a fundamental technology in the field of microfabrication, enabling the joining of two or more wafers to create intricate and complex microstructures and systems. This technique is indispensable for applications that require the integration of thick or multi-level systems, such as microfluidic circuits and advanced mechanical systems like sensors and actuators. Additionally, wafer bonding plays a crucial role in the packaging of sensors and micromechanical structures at the wafer level, ensuring their protection and functionality.

- Facilitates the creation of thick or multi-level systems (multi-wafer stacks), which are essential for:
  - Microfluidic circuits and systems.

- Complex mechanical systems, including sensors, motors, and other devices.

- Enables the packaging of sensors and micromechanical structures at the wafer level, providing a robust and reliable solution for integration.

10.4.1 Classifying Wafer Bonding

Wafer bonding can be categorized based on the mechanisms and techniques used to achieve the bond. The primary classifications include:

- **Heat-Assisted Bonding:**
  - Includes methods such as Fusion Bonding, Eutectic Bonding, and Glass Frit Bonding.
- **Electrical Field-Assisted Bonding:**
  - Anodic Bonding is a prominent example of this category.
- **Chemistry-Assisted Bonding:**
  - Adhesive Bonding falls under this category.

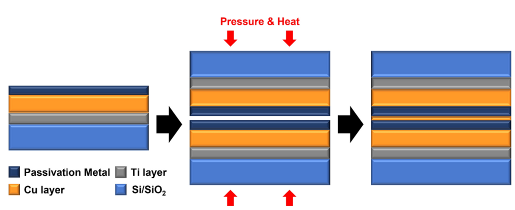
The general process of wafer bonding typically involves three main steps:

- **Surface Preparation:** This step is critical as the quality of the bond is highly dependent on the surface conditions. Ensuring that the surfaces are clean, flat, and free from contamination or particles is essential. For instance, a 1 μm particle can cause a void as large as 1 cm in diameter when bonding 8 in. wafers.
- **Contacting:** The wafers are brought into intimate contact under controlled conditions to initiate the bonding process.
- **Annealing:** The bonded wafers are subjected to thermal treatment to strengthen and stabilize the bond.

10.4.2 Fusion (or Direct) Bonding

Fusion bonding, also known as direct bonding, is a technique that involves the joining of polished surfaces of wafers without the use of intermediate layers. Key characteristics include:

- Applicable to Si/Si, Si/SiO_2, or SiO_2/SiO_2 bonding.
- The process begins by bringing cleaned and polished wafer surfaces into intimate contact at room temperature.
- A stable and permanent bond is formed by applying a large external force and/or subjecting the wafers to high temperatures.
- The result is a nearly perfect, high-strength bond that is suitable for demanding applications.

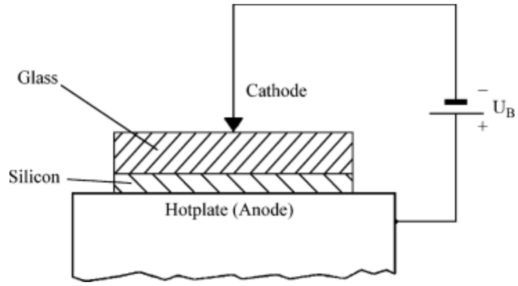


10.4.3 Anodic Bonding

Anodic bonding, also known as field-assisted glass-silicon sealing, is a process that enables the bonding of silicon to glass at relatively low temperatures. The key steps and characteristics of anodic bonding are as follows:

- The surfaces to be bonded must have a surface roughness of less than 0.1 μm to ensure close mating.
- The wafers are assembled and heated on a hot plate to a temperature between 400 and 500°C, which is significantly lower than the temperatures required for fusion bonding.
- A DC voltage, typically in the range of several hundred volts, is applied across the assembly, with the silicon wafer being positive relative to the glass.
- The bonding process begins under the negative electrode and spreads across the surface. No external force is required.

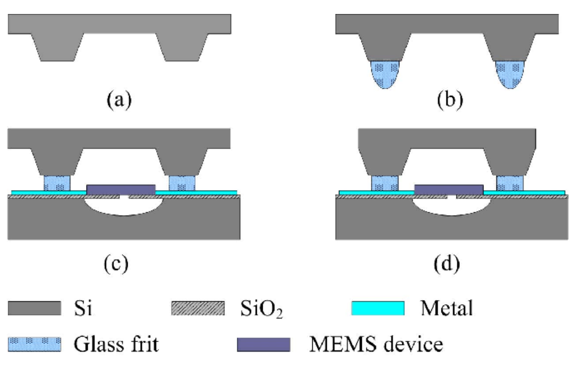
- The mechanism involves the migration of mobile ions (e.g., Na^+ or K^+) in the glass. These ions are attracted to the negative electrode, leaving behind a space charge layer that creates an electric field. This field pulls the surfaces into intimate contact, forming a strong bond.
- The bonding process is irreversible, and the resulting structures are held together by a chemical bond.



10.4.4 Glass Frit Bonding

Glass frit bonding is a versatile technique that uses a low-melting-point glass layer as an intermediate bonding layer. This method is particularly useful for applications where high temperatures or voltages are not feasible. Key features include:

- Enables bonding or sealing at relatively low temperatures (as low as 160°C) without the need for high voltages, metals, or adhesives.
- The glass frit material can be patterned to achieve localized and selective bonding.
- Inorganic glass frits require higher temperatures (300-600°C) but provide better sealing properties.
- The process can accommodate slight surface curvatures, roughness, and even surface topologies, such as conductive traces or interconnects, by reflowing the glass.
- The resulting bond is weaker compared to fusion or anodic bonding but is sufficient for many applications.
- Glass frit bonding is compatible with various material combinations, including Si-Si, Si-SiO_2, Pyrex-Quartz, and others.
- The sealing achieved can be hermetic, making it suitable for applications requiring airtight enclosures.



10.5 Replication Techniques

10.5.1 Micro-Injection Molding

Micro-injection molding is a replication technique that offers significantly higher throughput compared to hot embossing. This method is widely used for the mass production of microstructures due to its efficiency and scalability.

10.5.2 Micro-Injection Moulding

Micro-injection moulding is a replication technique used to reproduce many pieces from a mould insert.

Thermoplastics, such as PMMA, PVC, and ABS, are common materials used in micro-injection. Metal powders and ceramic powders mixed in polymers can also be injected and sintered to form metallic or ceramic parts.

Aspect ratios up to 100 and lateral precision down to 0.5 μm can be achieved. Surface roughness depends on the quality of the mould insert, but can be as small as 50 nm.

Micro-injection moulding is a variation of conventional injection moulding, adapted to produce pieces of extremely small feature size.

To be able to fill features with aspect ratio above 10, the temperature of the mould insert and holder must be particularly well controlled.

The polymer is injected with a typical pressure ranging from 500 to 2000 bars into the mould cavity, which is previously evacuated through a vacuum pump.

Demoulding often requires special handling or the addition of mould release compounds into the polymer mix.

10.5.3 Hot-Embossing

Hot-embossing is a replication process that involves the use of heat and pressure to transfer microstructures onto a substrate. Common materials used in hot-embossing include:

- PMMA (Polymethylmethacrylate): A transparent thermoplastic with excellent optical properties.
- POM (Polyoxymethylene Acetal): Known for its high stiffness and low friction.
- PC (Polycarbonate): A durable material with high impact resistance and optical clarity.
- PVDF (Polyvinylidene Fluoride): A highly non-reactive and pure thermoplastic fluoropolymer.
- PSU (Polyethersulfone): A high-performance thermoplastic with excellent thermal and mechanical properties.

10.5.4 (in-situ) Casting

The (in-situ) casting technique involves the following steps:

- A high-resolution mold is created using thick resist materials such as SU-8 or through processes like LIGA.
- Polydimethylsiloxane (PDMS), a soft and flexible silicone material, is poured onto the mold.
  - PDMS is known for its ability to conform to surface features with high fidelity.
  - The material undergoes heat curing, which can range from room temperature (RT) to over 150°C. The curing time varies from 24 hours to as little as 10 minutes depending on the conditions.
- Once cured, the PDMS is simply peeled off from the mold.
  - This process allows for the reproduction of sub-micron features at a low cost.

PDMS Properties:

PDMS exhibits a range of properties that make it highly suitable for microfabrication:

- It is a silicone elastomer available in a variety of viscosities.
- PDMS is flexible, with a Young's modulus of approximately 1 MPa, making it easy to mold.
- As an elastomer, it conforms to surfaces over large areas.
- It is chemically inert and optically transparent in the 300-1100 nm wavelength range.
- PDMS has low surface energy, allowing it to bond reversibly or permanently to other surfaces.
- It seals effectively to flat and clean surfaces, making it ideal for microfluidic applications.
- The material is durable, reusable, and exhibits low thermal expansion.
- It is biocompatible, even suitable for use as a food additive.
- PDMS is permeable to gases and solvents but impermeable to liquid water.
- It does not produce particulates, is environmentally safe, and is highly transparent.
- PDMS is commercially available under the Dow Corning brand.

10.6 Global Planarization-Chemical Mechanical Polishing

- Chemical Mechanical Polishing (CMP) is an essential process in the field of microfabrication, primarily aimed at achieving global planarization of substrates. This process is critical for ensuring the uniformity and precision required in advanced semiconductor manufacturing and other microfabrication applications.
- CMP operates by combining both mechanical and chemical forces in a synergistic manner. The mechanical aspect involves the physical abrasion of the wafer surface using a polishing pad and abrasive particles, while the chemical aspect involves the use of reactive chemicals in the slurry to selectively weaken and remove specific materials.
- During the CMP process, the wafer is mounted on a carrier and pressed against a rotating polishing pad. A controlled load force is applied to the back of the wafer to ensure consistent contact with the pad. Both the polishing pad and the wafer are counter-rotated to enhance the uniformity of the material removal process.
- A slurry, which is a mixture of abrasive particles and reactive chemicals, is continuously introduced between the wafer and the polishing pad. The abrasive particles in the slurry facilitate the mechanical removal of material, while the reactive chemicals selectively etch the surface, enhancing the overall planarization efficiency.
- The primary goal of CMP is to achieve a highly uniform and planar surface across the entire wafer. This uniformity is crucial for subsequent fabrication steps, such as lithography and etching, where surface irregularities can lead to defects and reduced device performance.
- CMP is widely used for planarizing various materials, including oxides, metals, and dielectrics, and is a key enabler for the fabrication of advanced integrated circuits, multi-level interconnects, and other high-precision microstructures.